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Estimation Theory

Year 2021-22

July 23, 2026

Universidad Carlos III de Madrid



Chapter 1
Statistical Estimation Theory

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1.1 Estimation Theory

Estimation theory plays a central role in many data science applications by providing foundational methods for inferring unknown parameters from observed data. Estimation techniques are used to design algorithms that learn from data and make predictions about unseen samples. In signal processing, it helps recover signals from noisy measurements, while in statistical inference, it forms the basis for hypothesis testing and parameter estimation.

1.1.1 The estimation problem

In its most general form, the estimation problem consists of designing a function, known as an *estimator*, that maps an input vector of observations to an estimate of a target variable. This target may represent an unknown physical quantity, a model parameter, or a latent variable in a probabilistic model.

Mathematically, we denote the observed data as a random vector $\mathbf{X}$, drawn from some observation space $\mathscr{X}$, and the target variable to be estimated as S. We will generally assume that the target variable is a scalar, although we will analyze some multidimensional scenarios.[1]

The goal of estimation is to construct a function $f : \mathscr{X} \to \mathbb{R}$, called the *estimator*, that produces an estimate $\hat{S} = f(\mathbf{X})$ of the unknown S, named the *estimation* or *prediction*. The estimator is a deterministic function, meaning that for a given input $\mathbf{x}$, it will consistently produce the same output. However, when the input $\mathbf{X}$ is random vector, the prediction $\hat{S}$ is also a random variable.

Figure 1.1 provides a schematic representation of the estimation problem

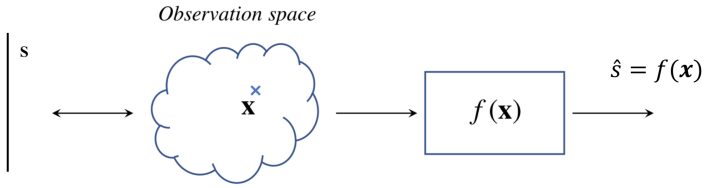


Fig. 1.1 Diagram block of estimation problems.

The estimator's performance is evaluated by how well $\hat{S}$ approximates S, typically using a predefined cost function (or, alternatively, a reward function) that quantifies the estimation error. A well-designed estimator seeks to minimize the expected cost (or to maximize the expected reward). There are two primary tasks in estimation problems:

- Analysis: given an estimator, assess its performance based on statistical criteria.
- Design: find an estimator $f()$ that optimizes a predefined performance goal.

[1] Note that we use capital letters to model the random variables, and we will use lowercase letters to denote an arbitrary realization of them.

1.1.2 Probability model

The statistical relation between the observations and the target variable is described by the **joint** probability density function (pdf), $p_{\mathbf{X},S}(\mathbf{x},s)$, or some distribution related to it. The joint pdf can be factorized as a product of conditional and marginal pdfs:

$$p_{\mathbf{X},S}(\mathbf{x},s) = p_{\mathbf{X}|S}(\mathbf{x}|s) \cdot p_S(s) = p_{S|\mathbf{X}}(s|\mathbf{x}) \cdot p_{\mathbf{X}}(\mathbf{x}) \tag{1.1}$$

In the context of estimation theory, these factors receive specific names:

- The **likelihood** of $S = s$ for observation $\mathbf{x}$, $p_{\mathbf{X}|S}(\mathbf{x}|s)$: it characterizes the generation of observations for each value of the target variable.
- The **prior (or *a priori*) distribution** of S, $p_S(s)$: it describes how much is known (or unknown) about the target variable before observing $\mathbf{X}$.
- The **posterior (or *a posteriori*) distribution** of S given $\mathbf{X} = \mathbf{x}$, $p_{S|\mathbf{X}}(s|\mathbf{x})$: it describes the knowledge (or the uncertainty) about S after observing $\mathbf{X}$.
- The **evidence** or marginal distribution of $\mathbf{X}$, $p_{\mathbf{X}}(\mathbf{x})$.

The information available to design the estimator may depend on the application. In some cases, the likelihood function is known, because it can be related to the physical generative process of the observations. If additionally, a prior distribution is available, the design can be grounded on the posterior distribution $p_{S|\mathbf{X}}(s|\mathbf{x})$, which can be calculated by means of Bayes' Theorem,

$$p_{S|\mathbf{X}}(s|\mathbf{x}) = \frac{p_{\mathbf{X},S}(\mathbf{x},s)}{p_{\mathbf{X}}(\mathbf{x})} = \frac{p_{\mathbf{X}|S}(\mathbf{x}|s)p_S(s)}{\int p_{\mathbf{X}|S}(\mathbf{x}|s')p_S(s')ds'} \tag{1.2}$$

1.1.3 Cost functions for estimation problems

The evaluation and design of an estimator require some objective criteria. In some cases, we will consider that this criterion materializes in the form of a **cost function** whose value we seek to minimize.

A cost function $c(s,\hat{s})$ is any measure of the discrepancy between the target variable and the estimation. It is generally non-negative, $c(s,\hat{s}) \geq 0$, with equality for $s = \hat{s}$. Some cost functions can be expressed as a function of the estimation error $e = s - \hat{s}$ and we will write[2] $c(s,\hat{s}) = c(s-\hat{s}) = c(e)$. Some examples are:

- Square error: $c(e) = e^2$.
- Absolute error: $c(e) = |e|$.
- Relative square error: $c(s,\hat{s}) = \frac{(s-\hat{s})^2}{s^2}$
- Cross Entropy: $c(s,\hat{s}) = -s\ln\hat{s} - (1-s)\ln(1-\hat{s})$, for $s,\hat{s} \in [0,1]$

[2] Note that the cost function is denoted with a lowercase letter, c, because it is a deterministic function, i.e., for fixed values of s and $\hat{s}$ the cost always takes the same value. However, as with the estimation function, the application of that function to random variables will result in another random variable, i.e., $C = c(S,\hat{S})$.

Since the target variable is unknown, the prediction cannot be computed by directly minimizing the cost $c(s,\hat{s})$, and we have to work with expectations. The expected value of the cost is usually referred as the **risk** of an estimator $\hat{s} = f(\mathbf{x})$:

$$R_f = \mathbb{E}\{c(S,\hat{S})\} = \int_{\mathbf{x}}\int_{s} c(s,f(\mathbf{x}))p_{S,\mathbf{X}}(s,\mathbf{x})dsd\mathbf{x} \tag{1.3}$$

By the *Law of Large Numbers*, this is the average cost we can expect from a given estimator, after a large number of predictions.

The **conditional risk** is the conditional mean for a given observation

$$R(\hat{s},\mathbf{x}) = \mathbb{E}\{c(S,\hat{s})|\mathbf{x}\} = \int_{s} c(s,\hat{s})p_{S|\mathbf{X}}(s|\mathbf{x})ds \tag{1.4}$$

Example 1.1 (Evaluation of estimators 1) Given the joint distribution

$$p_{S,X}(s,x) = \begin{cases} \frac{1}{x}, & 0 \le s \le x \text{ and } 0 < x \le 1 \\ 0, & \text{otherwise} \end{cases}, \tag{1.5}$$

consider the estimators $\hat{S}_1 = \frac{1}{2}X$ and $\hat{S}_2 = X$. Which is the best estimator from the point of view of the square error? To find out, we'll calculate the risk (i.e. the mean square error) for both estimators. Knowing that, for any w,

$$\begin{aligned} R &= \mathbb{E}\{(S-wX)^2\} = \int_0^1\int_0^x (s-wx)^2 p_{S,X}(s,x)dsdx = \int_0^1\int_0^x (s-wx)^2\frac{1}{x}dsdx \\ &= \int_0^1 \left(\frac{1}{3}-w+w^2\right)x^2dx = \frac{1}{3}\left(\frac{1}{3}-w+w^2\right) \end{aligned} \tag{1.6}$$

Taking $w = 1/2$ and $w = 1$ we get, respectively, the risks

$$R_1 = \mathbb{E}\left\{(S-\hat{S}_1)^2\right\} = \mathbb{E}\left\{\left(S-\frac{1}{2}X\right)^2\right\} = \frac{1}{3}\left(\frac{1}{3}-\frac{1}{2}+\frac{1}{4}\right) = \frac{1}{36} \tag{1.7}$$

$$R_2 = \mathbb{E}\{(S-\hat{S}_2)^2\} = \mathbb{E}\{(S-X)^2\} = \frac{1}{3}\left(\frac{1}{3}-1+1\right) = \frac{1}{9} \tag{1.8}$$

Therefore, from the point of view of the square error, $\hat{S}_1$ is a better estimator than $\hat{S}_2$.

Example 1.2 (Evaluation of estimators 2) Assume that S is a random variable of mean 0 and variance 1, and X is a noisy observation of S,

$$X = S+T \tag{1.9}$$

where T is a Gaussian random variable, independent of S, with mean 0 variance v. We will compute the risk for the estimator $\hat{S} = X$ and different cost functions. For the square error:

$$\mathbb{E}\{(S-\hat{S})^2\} = \mathbb{E}\{(S-X)^2\} = \mathbb{E}\{T^2\} = v \tag{1.10}$$

For the absolute error

$$\mathbb{E}\{|S-\hat{S}|\} = \mathbb{E}\{|T|\} = \int_{-\infty}^{\infty} |t| \frac{1}{\sqrt{2\pi v}} \exp\left(-\frac{t^2}{2v}\right) dt$$
$$= 2\int_{0}^{\infty} t \frac{1}{\sqrt{2\pi v}} \exp\left(-\frac{t^2}{2v}\right) dt = \sqrt{\frac{2v}{\pi}} \tag{1.11}$$

1.1.4 Bias and Variance

The performance of an estimator $\hat{S} = f(\mathbf{X})$ is often evaluated using two key quantities: *bias* and *variance*. These can be defined in both *conditional* and *unconditional* forms, depending on whether they are measured for a specific target variable or not.

Definition 1.1 (Conditional Bias) The *conditional bias* of an estimator $\hat{S}$ given that $S = s$ is defined as

$$\text{bias}\{\hat{S}|S=s\} = \mathbb{E}\{\hat{S}|S=s\} - s. \tag{1.12}$$

The conditional bias quantifies the expected deviation of the estimate from the true value of the target variable, s.

Theorem 1.1 (Bias-Variance decomposition) *For any estimator, $\hat{S}$ of random variable S, the conditional mean square error can be decomposed as*

$$\mathbb{E}\{(\hat{S}-S)^2|S=s\} = \text{bias}^2\{\hat{S}|S=s\} + \text{var}\{\hat{S}|S=s\} \tag{1.13}$$

Proof Since, for any random variable, V, $\mathbb{E}\{V^2\} = \mathbb{E}\{V\}^2 + \text{var}\{V\}$, taking $V = S - \hat{S}$, and conditioning over $S = s$, we get

$$\mathbb{E}\{(\hat{S}-S)^2|S=s\} = \mathbb{E}\{\hat{S}-S|S=s\}^2 + \text{var}\{\hat{S}-S|S=s\}$$
$$= \text{bias}^2\{\hat{S}|S=s\} + \text{var}\{\hat{S}|S=s\} \tag{1.14}$$

The bias-variance decomposition highlights a major tradeoff in estimation:

- Low-bias estimators tend to have high variance, leading to unstable predictions.
- Low-variance estimators may introduce **bias** to improve stability.

y Note that, in the above expressions, the bias, variance and mean square error are conditioned over a specific value of the target variable. We can extend these definitions to measures defined from unconditional means:

Definition 1.2 (Unconditional Bias) The *unconditional bias* of an estimator $\hat{S}$ of a random variable S is

$$\text{bias}\{\hat{S}\} = \mathbb{E}\{\hat{S}\} - \mathbb{E}\{S\}. \tag{1.15}$$

A similar unconditional bias-variance decomposition can obtained, that is given by

$$\begin{aligned}\mathbb{E}\{(\hat{S}-S)^2\} &= \mathbb{E}\{\hat{S}-S\}^2 + \mathrm{var}\{\hat{S}-S|S=s\} \\ &= \mathrm{bias}^2\{\hat{S}\} + \mathrm{var}\{\hat{S}-S\} \end{aligned} \tag{1.16}$$

If $\mathrm{bias}\{\hat{S}|S=s\}=0$ for all s, the estimator is said to be *conditionally unbiased*. If $\mathrm{bias}\{\hat{S}\}=0$, it is *unbiased in the Bayesian sense*, meaning that its expected value matches the prior mean of S.

1.2 Design of estimators

1.2.1 Maximum Likelihood (ML) estimation

The maximum likelihood estimator (ML) uses the likelihood as the reward function to be maximized:

$$\hat{s}_{\mathrm{ML}} = \underset{s}{\mathrm{argmax}}\, p_{\mathbf{X}|S}(\mathbf{x}|s) = \underset{s}{\mathrm{argmax}} \ln(p_{\mathbf{X}|S}(\mathbf{x}|s)) \tag{1.17}$$

The ML estimator selects the value of the parameter s that maximizes the likelihood of observing $\mathbf{x}$ when $S=s$. Loosely speaking, observing $\mathbf{x}$ when $S=\hat{s}_{\mathrm{ML}}$ is less unexpected than if S takes any other value. Note that $p_{\mathbf{X}|S}(\mathbf{x}|s)$, which is a density function over random variable $\mathbf{X}$, is not maximized with respect to $\mathbf{x}$, but s.

Example 1.3 (ML Estimation)

We want to compute the ML estimate of a random variable S given observation X. The likelihood function is given by

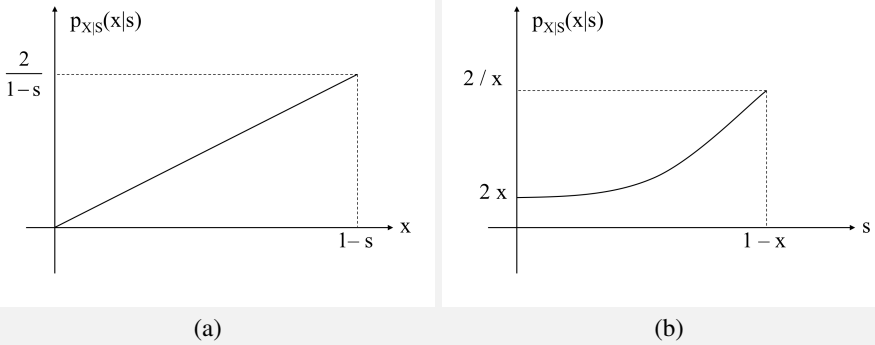
$$p_{X|S}(x|s) = \frac{2x}{(1-s)^2}, \qquad 0 \le x \le 1-s, \quad 0 \le s \le 1 \tag{1.18}$$

and it is represented in Figure 1.2(a) as a function of x, for a specific value of s. Note that the likelihood function is a conditional density function of the random variable X. However, since we aim to estimate S, it may be more useful to represent the likelihood as a function of s, for a given observation x, as shown in Fig.1.2(b),[a]. This representation suggests that the ML estimator is

$$\hat{s}_{\mathrm{ML}} = 1-x$$

which can be corroborated by inspecting (1.18), which shows that the likelihood increases with s. If we apply the estimation function to random variable X instead of a specific value, we have

$$\hat{S}_{\mathrm{ML}} = 1-X$$



(a) (b)

Fig. 1.2 Representation of the likelihood distribution of the example 1.3 as a function of x and s.

[a] Note that the integral with respect to s of $p_{X|S}(x|s)$ will not generally be equal to one, since the likelihood function is not a probability density function of the target variable S.

Note that the second equality in (1.17) states that the maximization of the likelihood is equivalent to the maximization of its logarithm (the **log-likelihood** function). Since the logarithm function is strictly increasing, $p_{\mathbf{X}|S}(\mathbf{x}|s_1) > p_{\mathbf{X}|S}(\mathbf{x}|s_2)$ implies $\ln(p_{\mathbf{X}|S}(\mathbf{x}|s_1)) > \ln(p_{\mathbf{X}|S}(\mathbf{x}|s_2)))$ and, thus, the logarithm does not alter the outcome of the maximization. The logarithm is useful to simplify the optimization when the likelihood is a product of several factors or an exponential function, as it will transform products into sums and an exponential into the exponents.

Note that the maximum likelihood does not need any probability model about the target variable, S, which is treated as a deterministic parameter.

1.2.2 Maximum a posteriori (MAP) estimation

We define the maximum a posteriori (MAP) estimator as the mode of the posterior distribution, that is

$$\hat{s}_{\text{MAP}} = \underset{s}{\text{argmax}}\, p_{S|\mathbf{X}}(s|\mathbf{x}) = \underset{s}{\text{argmax}} \ln(p_{S|\mathbf{X}}(s|\mathbf{x})) \tag{1.19}$$

Using the definition of conditional pdf and Bayes' rule, it is easy to see that the MAP estimator can be altenatively computed as

$$\hat{s}_{\text{MAP}} = \underset{s}{\text{argmax}}\, p_{S,\mathbf{X}}(s,\mathbf{x}) \tag{1.20}$$

$$= \underset{s}{\text{argmax}}\{p_{\mathbf{X}|S}(\mathbf{x}|s)p_S(s)\} \tag{1.21}$$

Eq. (1.21) demonstrates that the MAP estimator seeks to maximize the likelihood function, modulated by the prior distribution. This integration of the prior distribution allows the MAP estimator to incorporate existing knowledge or assumptions about the target S before observing the data, $\mathbf{x}$. When the prior distribution is uniform across the domain of S, the distinction between the MAP and ML estimators vanishes, as the MAP estimator effectively reduces to the ML estimator.

However, when the prior distribution is not uniform, the MAP estimator is biased towards values of the target variable with higher prior probabilities. This shift illustrates the MAP estimator's sensitivity to prior knowledge.

Beyond their mathematical formulations, the MAP and ML estimators embody fundamentally different inference philosophies. The ML estimator optimizes the likelihood function, which models the probability of the observed data under various values of the target, treating s as a fixed but unknown parameter. This approach aligns with the **frequentist paradigm**, which interprets probability as the long-run frequency of events and does not incorporate prior information about S.

Conversely, the MAP estimator embraces a **Bayesian framework**, treating the target as a random variable. This perspective allows the incorporation of prior knowledge or beliefs about S through the prior distribution, $p_S(s)$, and the posterior distribution, $p_{S|\mathbf{X}}(s|\mathbf{x})$, updates this knowledge based on new evidence from the data. The Bayesian approach, therefore, provides a probabilistic framework for updating beliefs about uncertain parameters in light of new data.

Example 1.4 (MAP estimation) Given the posterior density

$$p_{S|X}(s|x) = \frac{1}{x^2} s \exp\left(-\frac{s}{x}\right), \qquad x \geq 0, \quad s \geq 0 \tag{1.22}$$

the MAP estimator can be computed by maximizing

$$\ln(p_{S|X}(s|x)) = -2\ln(x) + \ln(s) - \frac{s}{x}, \qquad x \geq 0, \quad s \geq 0, \tag{1.23}$$

Since $\ln(p_{S|X}(s|x))$ tends to $-\infty$ around $s = 0$ and $s = \infty$, its maximum must be at some intermediate point with zero derivative:

$$\frac{\partial}{\partial s} \ln p_{S|X}(s|x)\Big|_{s=\hat{s}_{\text{MAP}}} = \frac{1}{\hat{s}_{\text{MAP}}} - \frac{1}{x} = 0, \qquad x \geq 0, \quad s \geq 0. \tag{1.24}$$

Therefore,

$$\hat{s}_{\text{MAP}} = x \tag{1.25}$$

1.2.3 Minimum risk estimators

On major drawback of the MAP estimator is that a high posterior density at a given point does not necessarily mean that the estimated value is likely to be close to the true parameter, as the example in Fig. 1.3 shows, where the MAP estimate lies in a region with low probability. This motivates the application of other Bayesian estimators that take into account the cost of large deviations from the true parameter.

The Bayesian estimator for a cost function $c(S,\hat{S})$ is the one minimizing the risk (1.3). Applying the total expectation theorem, the risk of estimator $\hat{s} = f(\mathbf{x})$ can be expressed as

$$R_f = \mathbb{E}\{c(S,\hat{S})\} = \int_{\mathbf{x}} \mathbb{E}\{c(S,\hat{s})|\mathbf{X} = \mathbf{x}\} p_{\mathbf{X}}(\mathbf{x}) \mathbf{dx} \tag{1.26}$$

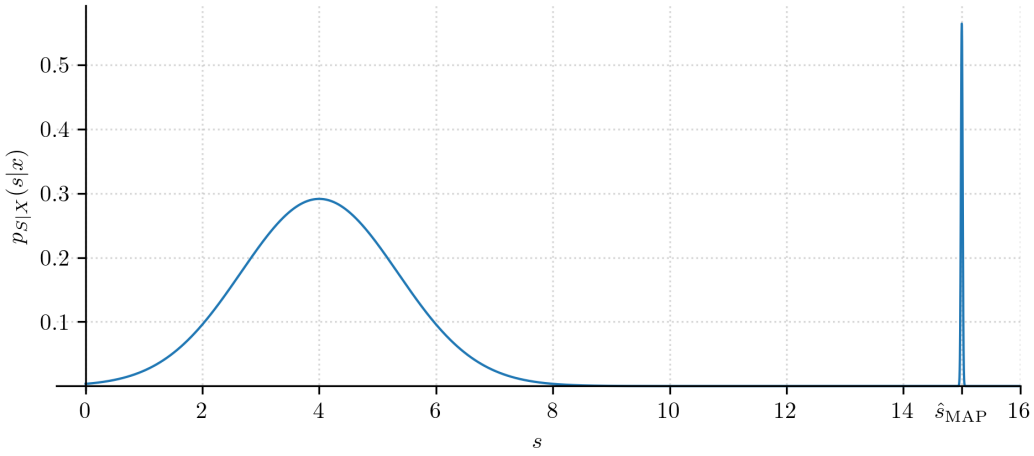


Fig. 1.3 A visual example of a posterior distribution, where the MAP estimate can be far from the true value of the target parameter with high probability.

where $\hat{s} = f(\mathbf{x})$. That is, the risk is the integral of the conditional risk, and, thus, the estimator minimizing R_f should minimize the conditional risk for each observation $\mathbf{x}$,

$$\hat{s}^* = \underset{\hat{s}}{\operatorname{argmin}}\ \mathbb{E}\{c(S,\hat{s})|\mathbf{X}=\mathbf{x}\} \tag{1.27}$$

Example 1.5 (Calculation of a minimum mean square error estimator) Following the example 1.1, we can calculate the posterior distribution of S through

$$p_{S|X}(s|x) = \frac{p_{S,X}(s,x)}{p_X(x)}. \tag{1.28}$$

Knowing that

$$p_X(x) = \int_0^1 p_{S,X}(s,x)ds = \int_0^x \frac{1}{x}ds = 1, \qquad 0 \le x \le 1 \tag{1.29}$$

we obtain

$$p_{S|X}(s|x) = \begin{cases} \frac{1}{x}, & 0 < s < x < 1 \\ 0, & \text{otherwise} \end{cases} \tag{1.30}$$

The conditional risk will be given by

$$\begin{aligned} \mathbb{E}\{c(S,\hat{s})|X=x\} &= \mathbb{E}\{(S-\hat{s})^2|X=x\} \\ &= \int_0^1 (s-\hat{s})^2 p_{S|X}(s|x)ds \\ &= \frac{1}{x}\int_0^x (s-\hat{s})^2 ds = \frac{1}{x}\left(\frac{(x-\hat{s})^3}{3} + \frac{\hat{s}^3}{3}\right) \\ &= \frac{1}{3}x^2 - \hat{s}x + \hat{s}^2. \end{aligned} \tag{1.31}$$

As a function of $\hat{s}$, conditional risk is a second-degree polynomial, whose minimum can be calculated through differentiation. Since

$$\frac{d}{d\hat{s}}\mathbb{E}\{c(S,\hat{s})|X=x\} = -x+2\hat{s}, \tag{1.32}$$

the Bayesian estimator associated with the square error is

$$\hat{s}^* = \frac{1}{2}x, \tag{1.33}$$

which matches the estimator $\hat{S}_1$ from the example 1.1. Therefore, $\hat{S}_1$ is the best possible estimator from the point of view of the mean square error.

1.3 Common Bayesian estimators

In this section we derive some of the most widely used Bayesian estimators, starting from different choices of the loss function.

1.3.1 Minimum Mean Squared Error estimator (MSE)

The minimum mean squared error (MSE) estimator is the Bayesian estimator associated with the cost function $c(e) = e^2 = (s-\hat{s})^2$, and therefore is given by

$$\begin{aligned}\hat{s}_{\text{MSE}} &= \underset{\hat{s}}{\operatorname{argmin}}\ \mathbb{E}\{c(S,\hat{s})|\mathbf{X}=\mathbf{x}\} \\ &= \underset{\hat{s}}{\operatorname{argmin}}\ \mathbb{E}\{(S-\hat{s})^2|\mathbf{X}=\mathbf{x}\}\end{aligned} \tag{1.34}$$

Figure 1.4 illustrates the minimum MSE estimation problem. The risk can be obtained by integrating (with respect to s) the product of the square error and the posterior pdf of S. The argument for minimization is $\hat{s}$, which allows shifting the graph corresponding to the cost function (represented with discontinuous stroke) so that the result of that integral is minimal.

For the square error, the conditional risk in (1.34) becomes

$$\mathbb{E}\{(S-\hat{s})^2|\mathbf{X}=\mathbf{x}\} = \mathbb{E}\{S^2|\mathbf{X}=\mathbf{x}\} - 2\mathbb{E}\{S|\mathbf{X}=\mathbf{x}\}\hat{s}+\hat{s}^2 \tag{1.35}$$

This is a second-degree polynomial that can be minimized by differentiation to get

$$\hat{s}_{\text{MSE}} = \mathbb{E}\{S|\mathbf{X}=\mathbf{x}\} = \int s\cdot p_{S|\mathbf{X}}(s|x)ds \tag{1.36}$$

In other words, the minimum MSE estimator of S is the posterior mean of S given $\mathbf{X}$.

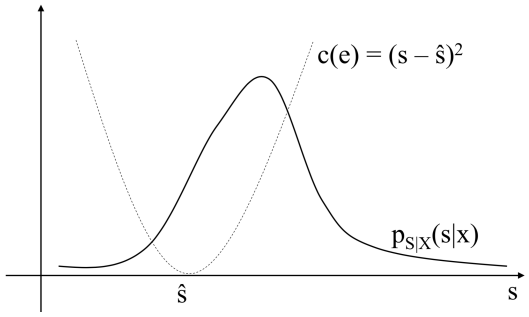


Fig. 1.4 Graphical representation of the process of calculating the posterior mean for a generic value $\hat{s}$.

Example 1.6 (MSE estimator) According to (1.36), minimum mean squared error estimator obtained in 1.1 can alternatively be derived as follows

$$\hat{s}_{\text{MSE}} = \int_0^1 s p_{S|X}(s|x) ds = \int_0^x \frac{s}{x} ds = \frac{1}{2} x \tag{1.37}$$

which is consistent with (1.33).

1.3.2 Minimum Mean Absolute Deviation Estimator (MAD)

When the cost is the absolute deviation, $c(e) = |e| = |s - \hat{s}|$, the minimizer of the conditional risk (which we will refer to as the Mean Absolute Deviation, or MAD, estimator), is given by

$$\hat{s}_{\text{MAD}} = \underset{\hat{s}}{\text{argmin}}\ \mathbb{E}\{|S - \hat{s}|\ |\mathbf{X} = \mathbf{x}\} = \tag{1.38}$$

$$= \underset{\hat{s}}{\text{argmin}} \int_s |s - \hat{s}|\ p_{S|\mathbf{X}}(s|\mathbf{x}) ds \tag{1.39}$$

The computation of the conditional risk is illustrated in Fig. 1.5. This representation also suggests the convenience of splitting the integral into two parts corresponding to the two slopes of the cost function:

$$\mathbb{E}\{|S - \hat{s}|\ |\mathbf{X} = \mathbf{x}\} = \int_{-\infty}^{\hat{s}} (\hat{s} - s)\ p_{S|\mathbf{X}}(s|\mathbf{x}) ds + \int_{\hat{s}}^{\infty} (s - \hat{s})\ p_{S|\mathbf{X}}(s|\mathbf{x}) ds \tag{1.40}$$

$$= \hat{s} \left[\int_{-\infty}^{\hat{s}} p_{S|\mathbf{X}}(s|\mathbf{x}) ds - \int_{\hat{s}}^{\infty} p_{S|\mathbf{X}}(s|\mathbf{x}) ds \right] + \int_{\hat{s}}^{\infty} s\ p_{S|\mathbf{X}}(s|\mathbf{x}) ds - \int_{-\infty}^{\hat{s}} s\ p_{S|\mathbf{X}}(s|\mathbf{x}) ds \tag{1.41}$$

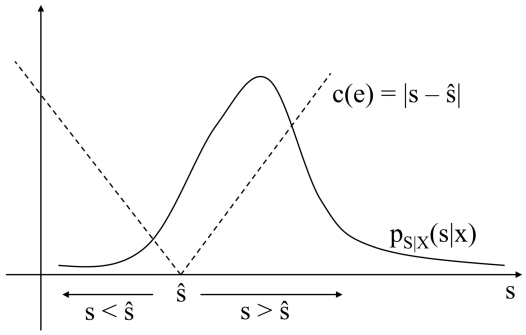


Fig. 1.5 Calculation of the posterior mean absolute error for a generic value $\hat{s}$.

Applying the fundamental theorem of calculus[3] we can write[4]

$$\frac{d\mathbb{E}\{|S-\hat{s}|\ |\mathbf{X}=\mathbf{x}\}}{d\hat{s}} = 2F_{S|\mathbf{X}}(\hat{s}|\mathbf{x}) - 1 \tag{1.42}$$

where $F_{S|\mathbf{X}}(s|\mathbf{x})$ is the posterior cumulative distribution function of S given $\mathbf{X}$. Since this derivative must vanish at the minimum, we get $F_{S|\mathbf{X}}(\hat{s}_{\text{MAD}}|\mathbf{x}) = 1/2$. In other words, the minimum MAD estimator is given by the median of $p_{S|\mathbf{X}}(s|\mathbf{x})$:

$$\hat{s}_{\text{MAD}} = \text{median}\{S|\mathbf{X}=\mathbf{x}\} \tag{1.43}$$

Remember that the median of a distribution is the point that splits its domain into two sets with the same probability, so the minimum MAD estimator satisfies

$$P\{S > \hat{s}_{\text{MAD}}|\mathbf{x}\} = P\{S < \hat{s}_{\text{MAD}}|\mathbf{x}\} \tag{1.44}$$

In practice, this can be computed as the solution of

$$\int_{-\infty}^{\hat{s}_{\text{MAD}}} p_{S|\mathbf{X}}(s|\mathbf{x})ds = \frac{1}{2} \tag{1.45}$$

Example 1.7 (MAD Estimator)
In the scenario of the example 1.1, the posterior distribution of S given X is uniform between 0 and x, the median of which is $x/2$. Thus,

$$\hat{s}_{\text{MAD}} = \frac{1}{2}x \tag{1.46}$$

[3] $\frac{d}{dx}\int_{t_0}^{x} g(t)dt = g(x)$.

[4] Strictly speaking, it is assumed here that the posterior distribution is absolutely continuous, that is, $p_{S|X}(s|x)$ does not contain singularities (i.e. points with nonzero probability mass).

Note that, in this case, the MAD estimator matches the MSE obtained at (1.33). This is a consequence of the symmetry of the a posterior distribution.

1.4 Estimation with constrains

1.4.1 General principles

Occasionally, it might be beneficial to prescribe a specific parametric form for the estimator, denoted as $\hat{S} = f_{\mathbf{w}}(\mathbf{X})$, where $\mathbf{w}$ represents a vector of parameters. For instance, in scenarios involving two observations, $\mathbf{X} = [X_1, X_2]^T$, a design constraint might necessitate limiting the search for an estimator to the family of quadratic estimators, characterized by $\hat{S} = w_0 + w_1 X_1^2 + w_2 X_2^2$. In such situations, the task of designing an estimator consists of identifying the optimal parameter vector $\mathbf{w}^*$ that minimizes the risk, while adhering to the specified constraints on the estimator structure:

$$\begin{aligned}\mathbf{w}^* &= \underset{\mathbf{w}}{\operatorname{argmin}}\, \mathbb{E}\{c(S,\hat{S})\} = \underset{\mathbf{w}}{\operatorname{argmin}}\, \mathbb{E}\{c(S, f_{\mathbf{w}}(\mathbf{X}))\} \\ &= \underset{\mathbf{w}}{\operatorname{argmin}} \int_{\mathbf{x}}\int_{s} c(s, f_{\mathbf{w}}(\mathbf{x})) p_{S,\mathbf{X}}(s,\mathbf{x}) ds d\mathbf{x}. \end{aligned} \tag{1.47}$$

Restricting the estimator to a specific analytic form typically results in a higher risk than what could be achieved with a Bayesian estimator tailored to the same cost function. The exception to this rule occurs when the imposed constraints align with the optimal estimator form, essentially when the Bayesian estimator inherently fits within the specified constraints. Despite potentially incurring higher costs, practical considerations may justify opting for such constrained estimators, such as simplification in design or implementation. An exploration of this concept is presented in Section 1.4.2, focusing on linear estimators that achieve minimum MSE.

Example 1.8 (Estimator with Constraints)
Continuing the example 1.5, we want to calculate the minimum MSE estimator with the form $\hat{s} = wx^2$. Since the conditional risk is given by (1.31), the risk can be computed as

$$\begin{aligned}\mathbb{E}\{c(S,\hat{S})\} &= \int_x \mathbb{E}\{c(S,\hat{s})|X=x\}\; p_X(x)dx \\ &= \int_x \left(\frac{1}{3}x^2 - \hat{s}x + \hat{s}^2\right) p_X(x)dx \end{aligned} \tag{1.48}$$

Forcing $\hat{s} = wx^2$ and taking into account that $p_X(x) = 1$ for $0 < x < 1$, we get the MSE as a function of w.

$$\mathbb{E}\{c(S,wX^2)\} = \int_0^1 \left(\frac{1}{3}x^2 - wx^3 + w^2x^4\right) dx \tag{1.49}$$

$$= \frac{1}{9} - \frac{1}{4}w + \frac{1}{5}w^2 \tag{1.50}$$

Since the risk is a quadratic funtion of w, it can be optimized by differentiation, to get

$$w^* = \frac{5}{8}, \tag{1.51}$$

and therefore the estimator is $\hat{s} = \frac{5}{8}x^2$.

1.4.2 Linear (in the parameters) estimation of minimum MSE

In this section we will focus on the study of estimators that are a linear combination of variables related to the observations, using the minimization of the MSE as a design criterion. More specifically, we will consider estimators given by the general expression

$$\hat{S} = \mathbf{w}^\intercal \mathbf{Z} \tag{1.52}$$

where

$$\mathbf{Z} = \phi(\mathbf{X}) \tag{1.53}$$

is some known transformation of the observations. The nature of this transformation may depend on the application, but there are some cases of particular interest:

- **Linear estimation**: in this case, ϕ is the identity function, so that $\mathbf{Z} = \mathbf{X}$, and the estimator a linear function of the observations

$$\hat{S} = w_0 X_0 + w_1 X_1 + \cdots + w_{N-1} X_{N-1} \tag{1.54}$$

- **Linear estimation with independent term**: in this case, $\mathbf{Z} = (1, \mathbf{X}^\intercal)^\intercal$ and

$$\hat{S} = w_0 + w_1 X_0 + \cdots + w_N X_{N-1} \tag{1.55}$$

- **Polynomial estimation**: the components of $\mathbf{Z}$ are monomials of the observational variables. For instance, for a scalar observation $X \in \mathbb{R}$, we can take $\mathbf{Z} = (1, X, X^2, \ldots X^M)$, for some $M > 1$, and the estimation becomes an M-degree polynomial of the observation.

$$\hat{S} = w_0 + w_1 X + w_2 X^2 \cdots + w_M X^M \tag{1.56}$$

Note that in general, $\mathbf{X}$ and $\mathbf{Z}$ may have different dimensions. Note, also, that, despite the estimator may be a non-linear function of the observation, all estimators given by (1.52) are linear functions of the parameters. For this reason, we will refer to them as estimators that are *linear in the parameters*.

By imposing a restriction on the analytic form of the estimator, linear-in-the-parameter estimators will generally obtain lower performance than the optimal Bayesian estimator. However, the interest in linear estimators is justified by their simplicity and ease of design. As we shall see, the linear estimator of minimum MSE depends exclusively on first and second-order statistical moments (means and covariances) associated with the target variable and the transformed observation, $\mathbf{Z}$.

1.4.2.1 Minimization of the mean squared error.

We will consider as design criteria the squared error, $c(e) = (s - \hat{s})^2$, so the optimal weight vector will be the one that minimizes the MSE risk:

$$\mathbf{w}^* = \underset{\mathbf{w}}{\operatorname{argmin}}\ \mathbb{E}\{(S - \hat{S})^2\} = \underset{\mathbf{w}}{\operatorname{argmin}}\ \mathbb{E}\{(S - \mathbf{w}^\intercal \mathbf{Z})^2\} \tag{1.57}$$

and we will refer to the estimator associated with the weight vector as $\hat{S}_{\text{LMSE}}$:

$$\hat{S}_{\text{LMSE}} = \mathbf{w}^{*T}\mathbf{Z}$$

The MSE can be expanded as

$$\begin{aligned} MSE &= \mathbb{E}\{(S - \hat{S})^2\} = \mathbb{E}\{(S - \mathbf{w}^\intercal \mathbf{Z})^2\} \\ &= \mathbb{E}\{S^2\} - 2\mathbb{E}\{\mathbf{w}^\intercal \mathbf{Z} S\} + \mathbb{E}\{(\mathbf{w}^\intercal \mathbf{Z})^2\} \\ &= \mathbb{E}\{S^2\} - 2\mathbb{E}\{S\mathbf{Z}\}^\intercal \mathbf{w} + \mathbf{w}^\intercal \mathbb{E}\{\mathbf{Z}\mathbf{Z}^\intercal\}\mathbf{w} \\ &= \mathbb{E}\{S^2\} - 2\mathbf{r}_{S\mathbf{Z}}^\intercal \mathbf{w} + \mathbf{w}^\intercal \mathbf{R}_{\mathbf{Z}}\mathbf{w} \end{aligned} \tag{1.58}$$

where

- $\mathbf{r}_{S\mathbf{Z}} = \mathbb{E}\{S\mathbf{Z}\}$ is the **cross-correlation vector** between S and $\mathbf{Z}$
- $\mathbf{R}_{\mathbf{Z}} = \mathbb{E}\{\mathbf{Z}\mathbf{Z}^\intercal\}$ is the **autocorrelation matrix** of $\mathbf{Z}$.

Thus, the MSE is a second-degree polynomial in $\mathbf{w}$. This is illustrated in Figure 1.6, which depicts the error surface in a case with two observations. Being the function to minimize quadratic in weights (minimization argument), the error surface will take the form of a N dimensional paraboloid.

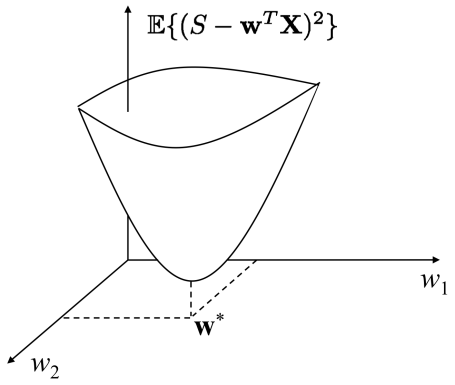


Fig. 1.6 Surface of the MSE for the linear estimator with $\mathbf{Z} = \mathbf{X}$, as a function of the parameters.

Since the MSE is non-negative, it is guaranteed to be a convex function of $\mathbf{w}$ and, thus, its minimum must be located at a point with zero gradient[5] (with respect to $\mathbf{w}$):

$$\nabla_{\mathbf{w}} MSE|_{\mathbf{w}=\mathbf{w}^*} = -2\mathbf{r}_{S\mathbf{Z}} + 2\mathbf{R}_{\mathbf{Z}}\mathbf{w}^* = \mathbf{0} \tag{1.59}$$

therefore, the optimal weight vector is any solution of

$$\mathbf{R}_{\mathbf{Z}}\mathbf{w}^* = \mathbf{r}_{S\mathbf{Z}} \tag{1.60}$$

If the autocorrelation matrix is invertible, we get

$$\boxed{\mathbf{w}^* = \mathbf{R}_{\mathbf{Z}}^{-1}\mathbf{r}_{S\mathbf{Z}}} \tag{1.61}$$

1.4.2.2 Alternative expression for the linear case

We can obtain an alternative expression for the optimal weights in the case of the linear estimator with the independent term in (1.55), that can be written as

$$\hat{S} = w_0 + \mathbf{w}_{1:}^{\intercal}\mathbf{X} \tag{1.62}$$

where $\mathbf{w}_{1:}$ is the weight vector after removing the first component, w_0. To do so, we can re-express (1.60) as the equivalent pair of equations

$$\mathbb{E}\{w_0^* + \mathbf{w}_{1:}^{*\intercal}\mathbf{X}\} = \mathbb{E}\{S\} \tag{1.63}$$

$$\mathbb{E}\{\mathbf{X}\}w_0^* + \mathbb{E}\{\mathbf{X}\mathbf{X}^{\intercal}\}\mathbf{w}_{1:}^* = \mathbb{E}\{S\mathbf{X}\} \tag{1.64}$$

that is

$$w_0^* = m_S - \mathbf{w}_{1:}^{*\intercal}\mathbf{m}_{\mathbf{x}} \tag{1.65}$$

$$\mathbb{E}\{\mathbf{X}\mathbf{X}^{\intercal}\}\mathbf{w}_{1:}^* = \mathbb{E}\{S\mathbf{X}\} - w_0^*\mathbf{m}_{\mathbf{X}} \tag{1.66}$$

where $m_S = \mathbb{E}\{S\}$, $m_{\mathbf{X}} = \mathbb{E}\{\mathbf{X}\}$. Now using the expressions that relate the correlation and covariance of two variables:

$$\mathbb{E}\{S\mathbf{X}\} = \mathbf{v}_{S\mathbf{X}} + m_S\mathbf{m}_{\mathbf{X}} \tag{1.67}$$

$$\mathbb{E}\{\mathbf{X}\mathbf{X}^{\intercal}\} = \mathbf{V}_{\mathbf{X}} + \mathbf{m}_{\mathbf{X}}\mathbf{m}_{\mathbf{X}}^{\intercal} \tag{1.68}$$

we get

$$w_0^* = m_S - \mathbf{w}_{1:}^{*\intercal}\mathbf{m}_{\mathbf{x}} \tag{1.69}$$

$$\mathbf{v}_{S\mathbf{X}} + m_S\mathbf{m}_{\mathbf{X}} = (\mathbf{V}_{\mathbf{X}} + \mathbf{m}_{\mathbf{X}}\mathbf{m}_{\mathbf{X}}^{T})\mathbf{w}_{1:}^* + w_0^*\mathbf{m}_{\mathbf{X}} \tag{1.70}$$

Solving these equations for w_0^* and $w_{1:}^*$ we get

[5] The gradient of a function scale $f(\mathbf{w})$ with respect to the vector $\mathbf{w}$ is defined as a vector formed by the derivatives of the function with respect to each one of the components of $\mathbf{w}$: $\nabla_{\mathbf{w}} f(\mathbf{w}) = \left[\frac{\partial f}{\partial w_1}, \ldots \frac{\partial f}{\partial w_N}\right]^T$.

$$w_0^* = m_S - \mathbf{w}_{1:}^{*\,T}\mathbf{m}_{\mathbf{x}} \tag{1.71}$$

$$\mathbf{w}_{1:}^* = \mathbf{V}_{\mathbf{X}}^{-1}\,\mathbf{v}_{S,\mathbf{X}} \tag{1.72}$$

The optimal estimator is, thus,

$$\hat{S}_{\text{LMSE}} = m_S + \mathbf{v}_{S\mathbf{X}}^{\intercal}\mathbf{V}_{\mathbf{X}}(\mathbf{X} - \mathbf{m}_{\mathbf{X}}) \tag{1.73}$$

The bias term w_0 compensates for differences between the means of the target variable and the observations. Therefore, when all the variables involved have zero mean, $w_0^* = 0$, and the estimator is purely linear in the observations.

1.4.2.3 Minimum squared mean error

The minimum MSE can be computed by replacing the optimal weights (1.61) into (1.58)

$$\begin{aligned}\text{MSE}^* &= \mathbb{E}\{S^2\} - 2\mathbf{r}_{S\mathbf{Z}}^{\intercal}\mathbf{R}_{\mathbf{Z}}^{-1}\mathbf{r}_{S\mathbf{Z}} + (\mathbf{R}_{\mathbf{Z}}^{-1}\mathbf{r}_{S\mathbf{Z}})^{\intercal}\mathbf{R}_{\mathbf{Z}}\mathbf{R}_{\mathbf{Z}}^{-1}\mathbf{r}_{S\mathbf{Z}} \\ &= \mathbb{E}\{S^2\} - \mathbf{r}_{S\mathbf{Z}}^{\intercal}\mathbf{R}_{\mathbf{Z}}^{-1}\mathbf{r}_{S\mathbf{Z}}\end{aligned} \tag{1.74}$$

For the linear estimator (1.73)

$$\text{MSE}^* = v_S - \mathbf{w}^{*\,T}\mathbf{v}_{S\mathbf{X}} \tag{1.75}$$

1.5 Estimation with Gaussian distributions

In this section, we delve into the estimation of random variables within the context where the combined distribution of all involved variables (the target variable along with the observational variables) is a multidimensional Gaussian. This scenario is particularly significant due to the prevalent occurrence of these distributions in signal processing, communications and various other fields.

When the joint distribution $p_{S,\mathbf{X}}(s,\mathbf{x})$ is Gaussian, all marginal and conditional distributions retain a Gaussian form. In particular, the posterior distribution, $p_{S|\mathbf{X}}(s|\mathbf{x})$ is Gaussian. Since the mean, median, and mode of the Gaussian distribution align, $\hat{s}_{\text{MSE}} = \hat{s}_{\text{MAD}} = \hat{s}_{\text{MAP}}$. Thus, our discussion in this section will primarily concentrate on deriving the estimator that minimizes the MSE.

Besides, we will demonstrate that the minimum MSE estimator and, consequently, the MAP and MAD estimators are linear, which will allow us to use the results shown in the previous section for minimum MSE estimation.

1.5.1 One dimensional variables

Assume that the joint distribution of X and S is Gaussian:

$$p_{S,X}(s,x) \sim G\left(\begin{bmatrix} s \\ x \end{bmatrix}, \begin{bmatrix} v_X & \rho \\ \rho & v_S \end{bmatrix}\right) \tag{1.76}$$

where ρ is the covariance between the two random variables. From the joint distribution, we can obtain any other distribution involving the variables S and X; specifically, the posterior distribution can be obtained as:

$$p_{S|X}(s|x) = \frac{p_{S,X}(s,x)}{p_X(x)} = \frac{\frac{1}{2\pi\sqrt{v_X v_S - \rho^2}} \exp\left[-\frac{1}{2(v_X v_S - \rho^2)} \begin{bmatrix} s \\ x \end{bmatrix}^{\mathsf{T}} \begin{bmatrix} v_X & -\rho \\ -\rho & v_S \end{bmatrix} \begin{bmatrix} s \\ x \end{bmatrix}\right]}{\frac{1}{\sqrt{2\pi v_X}} \exp\left[-\frac{x^2}{2v_X}\right]} \tag{1.77}$$

where it has been necessary to calculate the inverse of the covariance matrix of S and X.

Noting that, as a function of s, $p_{S|X}(s|x)$ differs from $p_{S,X}(s,x)$ in the scale factor $p_X(x)$, which does not depend on s, $p_{S|X}(s|x)$ should be a Gaussian pdf too. Therefore, it must be writable as

$$p_{S|X}(s|x) = \frac{1}{\sqrt{2\pi v_{S|X}}} \exp\left[-\frac{(s - m_{S|X})^2}{2v_{S|X}}\right] \tag{1.78}$$

where $m_{S|X}$ and $v_{S|X}$ are the posterior mean and variance, respectively, to be determined.

Joining (1.77) and (1.78), we can write

$$\frac{2\pi\sqrt{v_X v_S - \rho^2}}{\sqrt{2\pi v_{S|X}}\sqrt{2\pi v_X}} \exp\left[-\frac{(s - m_{S|X})^2}{2v_{S|X}} + \frac{1}{2(v_X v_S - \rho^2)} \begin{bmatrix} s \\ x \end{bmatrix}^{T} \begin{bmatrix} v_X & -\rho \\ -\rho & v_S \end{bmatrix} \begin{bmatrix} s \\ x \end{bmatrix} - \frac{x^2}{2v_X}\right] = 1 \tag{1.79}$$

which can be simplified to

$$\frac{\sqrt{v_X v_S - \rho^2}}{\sqrt{v_{S|X} v_X}} \exp\left[-\frac{s^2 - 2m_{S|X}s + m_{S|X}^2}{2v_{S|X}} + \frac{v_X s^2 - 2\rho x s + v_S x^2}{2(v_X v_S - \rho^2)} - \frac{x^2}{2v_X}\right] = 1 \tag{1.80}$$

Note that the equation above must be satisfied for any $s \in \mathbb{R}$. Since the right-hand side is constant, and the exponent on the left-hand side is a polynomial function of s, the coefficients multiplying s^2 and s must be zero. Thus

$$\frac{m_{S|X}}{v_{S|X}} = \frac{\rho x}{v_X v_S - \rho^2} \tag{1.81}$$

$$\frac{1}{v_{S|X}} = \frac{v_X}{v_X v_S - \rho^2} \tag{1.82}$$

From (1.82) we get the posterior variance

$$v_{S|X} = v_S - \frac{\rho^2}{v_X} \tag{1.83}$$

and, replacing (1.83) into (1.81) we get the posterior mean, which is the minimum MSE estimate

$$\hat{s}_{\text{MSE}} = \hat{s}_{\text{MAD}} = \hat{s}_{\text{MAP}} = m_{S|X} = \frac{\rho}{v_X} x \tag{1.84}$$

Note that the estimator is a linear function of the observation.

Exercise 1.1 Generalize the above result for the case where the variables S and X have (non-zero) means m_S and m_X, respectively. Demonstrate that in such a case, the estimator is

$$\hat{s}_{\text{MSE}} = m_S + \frac{\rho}{v_X}(x - m_X) \tag{1.85}$$

Example 1.9 (Estimation of a Gaussian signal contaminated by Gaussian noise)

In this example, we will consider the case in which the observation is the sum of the target variable and an independent noise component: $X = S + R$. Both the target and the noise are zero-mean Gaussian random variables with variances v_S and v_R, respectively.

Figure (1.7) represents the situation described for a case with $v_S < v_R$.

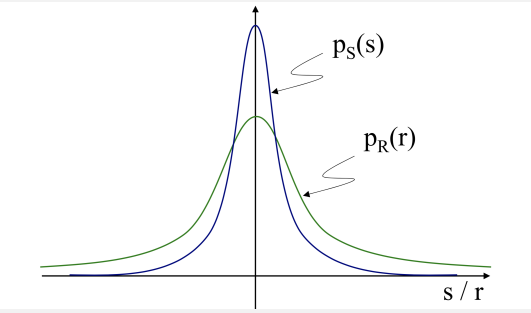


Fig. 1.7 Estimation of Gaussian random variable S contaminated by Gaussian noise R.

According to (1.84), for the resolution of the problem, we must find the variance of X and the covariance between S and X (ρ). The variance v_X is obtained simply as the sum of v_S and v_R because both are independent variables. For the covariance calculation,

$$\rho = \mathbb{E}\{(X - m_X)(S - m_S)\} = \mathbb{E}\{X\,S\} = \mathbb{E}\{(S+R)S\} = \mathbb{E}\{S^2\} + \mathbb{E}\{S\,R\} = v_S \tag{1.86}$$

where independence of S and R has been used, and the fact that all variables (including X) have zero means.

Replacing these results in (1.84) we get

$$\hat{s}_{\text{MSE}} = \frac{v_S}{v_S + v_R} x \tag{1.87}$$

This result can be interpreted quite intuitively: when the variance of the noise is much lower than that of the signal (high Signal to Noise Ratio (SNR), $v_S \gg v_R$) we get $\hat{s}_{\text{MSE}} \to x$, which makes sense since the effect of the noise component, in this case, is not very significant; on the contrary, when the SNR is very small ($v_S \ll v_R$), the observation barely provides information about the S value in each experiment, so the estimator keeps the mean value of the signal component, $\hat{s}_{\text{MSE}} \to 0$.

1.5.2 Multidimensional variables

In a general multidimensional case, $\mathbf{S}$ and $\mathbf{X}$ can be random vectors of dimensions N and M, respectively, with joint Gaussian distribution.

$$p_{\mathbf{S},\mathbf{X}}(\mathbf{s},\mathbf{x}) \sim G\left(\begin{bmatrix}\mathbf{m_S}\\ \mathbf{m_X}\end{bmatrix}, \begin{bmatrix}\mathbf{V_S} & \mathbf{V_{SX}}\\ \mathbf{V_{SX}}^T & \mathbf{V_X}\end{bmatrix}\right) \tag{1.88}$$

being $\mathbf{m_S}$ and $\mathbf{m_X}$ the means of $\mathbf{S}$ and $\mathbf{X}$, respectively, $\mathbf{V_S}$ and $\mathbf{V_X}$ the covariance matrix of $\mathbf{S}$ and $\mathbf{X}$, respectively, and $\mathbf{V_{SX}}$ the matrix of crossed covariances of $\mathbf{S}$ and $\mathbf{X}$, that is,

$$\mathbf{V_S} = \mathbb{E}\{(\mathbf{S}-\mathbf{m_S})(\mathbf{S}-\mathbf{m_S})^\intercal\} \tag{1.89}$$
$$\mathbf{V_X} = \mathbb{E}\{(\mathbf{X}-\mathbf{m_X})(\mathbf{X}-\mathbf{m_X})^\intercal\} \tag{1.90}$$
$$\mathbf{V_{SX}} = \mathbb{E}\{(\mathbf{S}-\mathbf{m_S})(\mathbf{X}-\mathbf{m_X})^\intercal\} \tag{1.91}$$

The calculation of the posterior distribution of $\mathbf{S}$ given $\mathbf{X}$ is more complex than in the one-dimensional case, but it follows a similar procedure, which we will omit here. It can be shown that the posterior distribution is Gaussian with mean

$$\mathbf{m_{S|X}} = \mathbf{m_S} + \mathbf{V_{SX}}\mathbf{V_X}^{-1}(\mathbf{x}-\mathbf{m_X}) \tag{1.92}$$

and covariance

$$\mathbf{V_{S|X}} = \mathbf{V_S} - \mathbf{V_{SX}}\mathbf{V_X}^{-1}\mathbf{V_{SX}}^T \tag{1.93}$$

Since the minimum MSE estimator of $\mathbf{S}$ given $\mathbf{X}$ is precisely the posterior mean, we can write

$$\hat{\mathbf{s}}_{\text{MSE}} = \hat{\mathbf{s}}_{\text{MAD}} = \hat{\mathbf{s}}_{\text{MAP}} = \mathbf{m_{S|X}} = \mathbf{m_S} + \mathbf{V_{SX}}\mathbf{V_X}^{-1}(\mathbf{x}-\mathbf{m_X}) \tag{1.94}$$

1.5.3 Linear estimation and Gaussian estimation

Note that, if the target variable is scalar, (1.94) is identical to (1.73). This is not coincidental: if the minimum MSE estimator in the Gaussian case is linear, it must be equal to the best linear estimator, which is given by (1.94).

1.6 ML estimation of probability distributions parameters

Sometimes we may be interested in estimating the parameters of a probability distribution, such as the mean or variance of a Gaussian distribution, the decay parameter that characterizes an exponential distribution, or values a and b delimiting the interval in which a uniform distribution is defined.

In these cases, the prior distribution of these variables is not usually known, even more, in many cases, these parameters are said to be deterministic and are not treated as random parameters. However, if a set of observations generated from these distributions is available, we can obtain the likelihood of these variables and estimate their values with maximum likelihood criteria.

Example 1.10 (ML estimate of the mean and variance of a one-dimensional Gaussian distribution)
The weight of individuals in a family of molluscs is known to obey a Gaussian distribution, but the mean and variance are unknown. The weight of l individuals taken independently, $\{X^{(k)}\}_{k=1}^{l}$, is available.

The likelihood of the mean and the variance for a single observation x, in this case, is given by:

$$p_X(x) = p_{X|m,v}(x|m,v) = \frac{1}{\sqrt{2\pi v}} \exp\left[-\frac{(x-m)^2}{2v}\right] \tag{1.95}$$

Since we must construct the estimator based on the joint observation of l observations, we will need to calculate the joint distribution of all of them which, being independent observations, is obtained as the product of individual observations:

$$\begin{aligned} p_{\{X^{(k)}\}|m,v}(\{x^{(k)}\}|m,v) &= \prod_{k=1}^{l} p_{X|m,v}(x^{(k)}|m,v) \\ &= \frac{1}{(2\pi v)^{l/2}} \prod_{k=1}^{l} \exp\left[-\frac{(x^{(k)}-m)^2}{2v}\right] \end{aligned} \tag{1.96}$$

The ML estimators of m and v will be the values maximizing (1.96). The analytical form of this function suggests the use of the logarithm to simplify the maximization:

$$L = \ln\left[p_{\{X^{(k)}\}|m,v}(\{x^{(k)}\}|m,v)\right] = -\frac{l}{2}\ln(2\pi v) - \frac{1}{2v}\sum_{k=1}^{l}(x^{(k)}-m)^2 \tag{1.97}$$

Differentiating (1.97) with respect to m and v, we get the system of equations to solve

$$\begin{aligned} \left.\frac{d\,L}{d\,m}\right|_{\substack{m=\hat{m}_{\mathrm{ML}} \\ v=\hat{v}_{\mathrm{ML}}}} &= -\frac{1}{v}\sum_{k=1}^{l}(x^{(k)}-m)\Bigg|_{\substack{m=\hat{m}_{\mathrm{ML}} \\ v=\hat{v}_{\mathrm{ML}}}} = 0 \\ \left.\frac{d\,L}{d\,v}\right|_{\substack{m=\hat{m}_{\mathrm{ML}} \\ v=\hat{v}_{\mathrm{ML}}}} &= -\frac{l}{2v} + \frac{1}{2v^2}\sum_{k=1}^{l}(x^{(k)}-m)^2\Bigg|_{\substack{m=\hat{m}_{\mathrm{ML}} \\ v=\hat{v}_{\mathrm{ML}}}} = 0 \end{aligned} \tag{1.98}$$

Solving for m the first equation we get

$$\hat{m}_{\mathrm{ML}} = \frac{1}{l}\sum_{k=1}^{l} x^{(k)} \tag{1.99}$$

which is the sample average of the observations. On the other hand, we can solve the second equation for v to get

$$\hat{v}_{\mathrm{ML}} = \frac{1}{l}\sum_{k=1}^{l} (x^{(k)} - \hat{m}_{\mathrm{ML}})^2 \tag{1.100}$$

which is the sample variance. Note that, if instead of applying the estimation function (of m or v) on some specific observations we did it on generic values $\{X^{(k)}\}$, the estimators could be treated as random variables, i.e.,

$$\hat{M}_{\mathrm{ML}} = \frac{1}{l}\sum_{k=1}^{l} X^{(k)} \tag{1.101}$$

$$\hat{V}_{\mathrm{ML}} = \frac{1}{l}\sum_{k=1}^{l} [X^{(k)} - \hat{M}_{\mathrm{ML}}]^2 \tag{1.102}$$

1.7 Some key results on Estimation Theory

In estimation theory, an estimator should ideally satisfy certain desirable properties to ensure its reliability and optimality. This section introduces three fundamental concepts: *consistency*, the *Cramér-Rao lower bound*, and *efficiency*.

Throughout this section, we assume that the parameter S is **deterministic** (frequentist setting), meaning it is an unknown but fixed quantity that does not vary across different samples.

1.7.1 Consistency

An estimator is consistent if it converges to the true parameter as the sample size increases. This property ensures that, with enough data, the estimator provides an accurate estimate.

Definition 1.3 (Mean Square Consistency) An estimator $\hat{S}_n$ of a target variable S, based on a set on n i.i.d. observational variables $X_0, \ldots, X_{n-1}$ is **mean square consistent** (or, simply, consistent) if

$$\lim_{n\to\infty} \mathbb{E}\{(\hat{S}_n - S)^2 | S = s\} = 0. \tag{1.103}$$

Using the bias-variance decomposition in (1.13), we can conclude that mean square consistency is equivalent to require the variance to vanish as $n \to \infty$, and the estimator to be asymptotically unbiased, that is:

$$\lim_{n\to\infty} \text{var}\{\hat{S}_n|S=s\} = 0, \tag{1.104}$$

$$\lim_{n\to\infty} \text{bias}\{\hat{S}_n|S=s\} = 0. \tag{1.105}$$

1.7.2 Cramér-Rao Lower Bound

The Cramér-Rao lower bound (CRLB) provides a fundamental limit on the variance of any unbiased estimator, establishing the best possible precision an estimator can achieve.

Theorem 1.2 (Cramér-Rao Lower Bound) *Let $\hat{S}$ be an unbiased estimator of a parameter S based on a sample $X_0, X_1, \ldots, X_{n-1}$ with likelihood function $p_{X|S}(x|s)$. The conditional variance of any unbiased estimator satisfies:*

$$var\{\hat{S}|S=s\} \geq \frac{1}{I(s)}, \tag{1.106}$$

*where $I(s)$ is the **Fisher information**, given by:*

$$I(s) = \mathbb{E}\left\{\left(\frac{\partial}{\partial s}\log p_{X|S}(x|s)\right)^2\right\}. \tag{1.107}$$

This result establishes a theoretical lower bound on the variance of an unbiased estimator, providing a benchmark for evaluating estimator performance.

1.7.3 Efficiency

An estimator is efficient if it attains the Cramér-Rao lower bound, meaning it achieves the minimum possible variance among all unbiased estimators.

Definition 1.4 (Efficiency) An unbiased estimator $\hat{S}$ of S is said to be **efficient** if it attains the Cramér-Rao lower bound:

$$\text{var}\{\hat{S}|S=s\} = \frac{1}{I(s)}. \tag{1.108}$$

An estimator is **asymptotically efficient** if its variance asymptotically approaches the CRLB, i.e.,

$$\lim_{n\to\infty} n \cdot \text{Var}(\hat{S}_n) = \frac{1}{I(S)}. \tag{1.109}$$

Efficiency is a desirable property since it ensures that an estimator makes optimal use of the available information, achieving the lowest possible variance.

Theorem 1.3 (Asymptotic Efficiency of Maximum Likelihood Estimation) *Let $\hat{S}_{ML}$ be the maximum likelihood estimator of a deterministic parameter S, based on an i.i.d. sample $X_1, X_2, \ldots, X_n$ with likelihood function $p_{X|S}(x|s)$. Under regularity[a] conditions, the maximum likelihood estimate is asymptotically efficient.*

[a] The regularity conditions are assumptions that ensure the MLE behaves well as the sample size increases. This refers to some conditions on the differentiability of the log-likelihood function, and others.

1.8 Problems

1.1 The posterior distribution of S given X is

$$p_{S|X}(s|x) = x^2 \exp(-x^2 s), \qquad s \geq 0$$

Compute estimators $\hat{S}_{\text{MSE}}$, $\hat{S}_{\text{MAD}}$ y $\hat{S}_{\text{MAP}}$.

1.2 Consider an estimation problem given by the following posterior distribution:

$$p_{S|X}(s|x) = x \exp(-xs), \;\; s > 0 \tag{1.110}$$

Compute estimators $\hat{S}_{\text{MSE}}$, $\hat{S}_{\text{MAD}}$ y $\hat{S}_{\text{MAP}}$.

1.3 A r.v. S must be estimated from the observation of another r.v. X by means of a linear mean square error estimator given by:

$$\hat{S}_{\text{LMSE}} = w_0 + w_1 X$$

Knowing that $\mathbb{E}\{X\} = 1$, $\mathbb{E}\{S\} = 0$, $\mathbb{E}\{X^2\} = 2$, $\mathbb{E}\{S^2\} = 1$ y $\mathbb{E}\{SX\} = 1/2$, compute:

a) The values for w_0 y w_1.

b) The mean square error of the estimator, $\mathbb{E}\left\{\left(S - \hat{S}_{\text{LMSE}}\right)^2\right\}$.

1.4 (Linear estimation of minimum mean squared error) We want to construct a linear estimator of minimum mean squared error that will allow us to estimate the random variable S from the random variables X_1 and X_2. Knowing that

$$\begin{array}{lll} \mathbb{E}\{S\} = 1/2 & \mathbb{E}\{X_1\} = 1 & \mathbb{E}\{X_2\} = 0 \\ \mathbb{E}\{S^2\} = 4 & \mathbb{E}\{X_1^2\} = 3/2 & \mathbb{E}\{X_2^2\} = 2 \\ \mathbb{E}\{SX_1\} = 1 & \mathbb{E}\{SX_2\} = 2 & \mathbb{E}\{X_1 X_2\} = 1/2 \end{array}$$

get the weights from the desired estimator and calculate its squared mean error. Calculate the estimated value for the observation vector: $[X_1, X_2] = [3, 1]$.

1.5 Let X and S be two random variables with joint pdf

$$p_{X,S}(x,s) \begin{cases} 2 \; 0 < x < 1, 0 < s < x \\ 0 \text{ resto} \end{cases}$$

a) Compute the minimum mean square error estimate of S given X, $\hat{S}_{\text{MSE}}$.
b) Compute the risk of estimator $\hat{S}_{\text{MSE}}$.

1.6 A digitized image of dimensions $8x8$ is available, whose luminance values are statistically independent and evenly distributed between 0 (white) and 1 (black); the image has been modified by applying a transformation of the form $Y = X^r$ on each pixel; $r > 0$, where X is the r.v. associated with the pixels of the original image and Y is associated with the transformed image. Obtain the expression that allows estimating r by maximum likelihood given the 64 pixel values of the transformed image $\{y^{(k)}\}_{k=1}^{64}$, without knowing the original image.

1.7 For the design of a communication system, it is desired to estimate the signal attenuation between the transmitter and the receiver, as well as the noise power introduced by the channel when this noise is Gaussian of zero mean and independent of the transmitted signal. For this, the transmitter sends a signal with a constant amplitude of 1 and the receiver collects a set of K observations available at its input.

a) Estimate the channel attenuation, α, and the noise variance, v_r, by maximum likelihood, when the available observations on the receiver are

$$\{0.55, 0.68, 0.27, 0.58, 0.53, 0.37, 0.45, 0.53, 0.86, 0.78\}.$$

b) If the system is to be used for the transmission of digital signals with unipolar coding (a A signal level is used to transmit a bit 1 and the signal level is maintained at 0 for the transmission of bit 0), considering equiprobability between symbols, indicate the minimum level of signal that should be used in the coding, $A_{\min}$, to guarantee a SNR level in the receiver of 3 dB.

1.8 Company *Like2Call* offers hosting services for call centers. In order to dimension the staff of operators the company is designing a statistical model to characterize the activity in the hosted call centers. One of the components of such a model relies on the well-known fact that the times between incoming calls follow an exponential distribution

$$p_{X|S}(x|s) = s\ \exp(-s\,x), \qquad x > 0$$

where random variable X represents the time before a new call arrives, and S is a parameter that depends on the time of the day and each particular call-center service (e.g., attention to the clients of an insurance company, customers of an online bank, etc).

For random variable S, the following *a priori* model is assumed:

$$p_S(s) = \exp(-s), \qquad s > 0.$$

With this information, we would like to design an estimator of S that is based on the first K incoming calls for each implemented service and time interval, i.e., represented by a vector of i.i.d. observations $\mathbf{X} = [X_0, X_1, \cdots, X_{K-1}]$,.

a) Obtain the maximum likelihood estimator or S based on the observation vector $\mathbf{X}$ and verify that it depends just on the sum of all observations, $z = \sum_{k=0}^{K-1} x_k$.
b) Compute the posterior distribution of S given $\mathbf{X}$, $p_{S|\mathbf{X}}(s|\mathbf{x})$.

c) Obtain the maximum *a posteriori* estimator of S given $\mathbf{X}$, $\hat{s}_{\text{MAP}}$.
d) Obtain the minimum mean square error estimator of S given $\mathbf{X}$, $\hat{s}_{\text{MSE}}$.
e) Calculate the mean square error given $\mathbf{X}$ of a generic estimator $\hat{S}$, and particularize the result for estimators of the following analytical shape $\hat{s}_c = \frac{c}{z+1}$.
f) Find expressions for $p_{Z|S}(z|s)$, $p_{Z,S}(z,s)$, and $p_Z(z)$.
g) Calculate the mean square error of a generic estimator $\hat{s}_c = \frac{c}{z+1}$. Study how the result changes with c and K.

You can use the following results:

i.
$$\int_0^\infty x^N \exp(-x)dx = N!$$

ii. If $f(x) = a\, exp(-a\, x)$, $x > 0$ then

$$\underbrace{f(x) * f(x) * \cdots * f(x)}_{N \text{ times}} = \frac{a^N\, x^{N-1}}{(N-1)!} exp(-a\, x),\ x > 0$$

iii. For K an integer
$$\int_0^\infty \frac{K\, x^{K-1}}{(x+1)^{K+3}} dx = \frac{2}{(K+2)(K+1)}$$

Solution 1.1 (Solution to problem 2.8)

a) Since $p_{\mathbf{X}|S}(\mathbf{x}|s) = s^K\ \exp(-s\, z)$, with $z > 0$, we have $\ln p_{\mathbf{X}|S}(\mathbf{x}|s) = K \ln s - s\, z$ and, thus,

$$\frac{d}{ds} \ln p_{\mathbf{X}|S}(\mathbf{x}|s) = \frac{K}{s} - z$$

Therefore,

$$\hat{s}_{\text{ML}} = \frac{K}{z}$$

b) The joint distribution of S and X is

$$p_{\mathbf{X},S}(\mathbf{x},s) = p_{\mathbf{X}|S}(\mathbf{x}|s) p_S(s) = s^K\ \exp[-s(z+1)]$$

therefore

$$p_{\mathbf{X}}(\mathbf{x}) = \int p_{\mathbf{X},S}(\mathbf{x},s)\, ds = \int_0^\infty s^K\ \exp[-s(z+1)]\, ds$$

With the change of variable $s' = s(z+1)$ the integral can be simplified using expression (i), and we get

$$p_{S|\mathbf{X}}(s|\mathbf{x}) = \frac{p_{\mathbf{X},S}(\mathbf{x},s)}{p_{\mathbf{X}}(\mathbf{x})} = \frac{(z+1)^{K+1}\, p_{\mathbf{X},S}(\mathbf{x},s)}{K!} = \frac{s^K (z+1)^{K+1}\ \exp[-s(z+1)]}{K!}$$

c) The MAP estimate is

$$\begin{aligned}
&\hat{s}_{\text{MAP}} = \arg\max_s p_{S|\mathbf{X}}(s|\mathbf{x}) = \arg\max_s p_{\mathbf{X},S}(\mathbf{x},s) \\
&\ln p_{\mathbf{X},S}(\mathbf{x},s) = K \ln s - s\,(z+1) \\
&\frac{d}{ds} \ln p_{\mathbf{X},S}(\mathbf{x},s) = \frac{K}{s} - (z+1) \\
&\hat{s}_{\text{MAP}} = \frac{K}{z+1}
\end{aligned}$$

d) The minimum MSE estimate is

$$\hat{s}_{\text{MSE}} = \mathbb{E}\{S|\mathbf{x}\} = \int s\, p_{S|\mathbf{X}}(s|\mathbf{x})\, ds = \frac{(z+1)^{K+1}}{K!} \int_0^\infty s^{K+1}\, \exp[-s(z+1)]\, ds$$

Replacing again $s' = s(z+1)$ and using expression (i), we get

$$\hat{s}_{\text{MSE}} = \frac{K+1}{z+1}$$

e) The calculation is somehow tedious, but can be summarized as follows:

$$\begin{aligned}
\mathbb{E}\left\{(S-\hat{s})^2|X\right\} &= \int_0^\infty (s-\hat{s})^2\, p_{S|X}(s|x) ds \\
&= \frac{(z+1)^{K+1}}{K!}\left[\frac{(K+2)!}{(z+1)^{K+3}} + \hat{s}^2 \frac{K!}{(z+1)^{K+1}} - 2\hat{s}\frac{(K+1)!}{(z+1)^{K+2}}\right] \\
&= \frac{(K+2)(K+1)+c^2-2c(K+1)}{(z+1)^2}
\end{aligned}$$

For the MAP and MSE estimators, the expressions are substantially simplified:

$$\begin{aligned}
\mathbb{E}\left\{(S-\hat{s}_{MAP})^2|z\right\} &= \frac{K+2}{(z+1)^2} \\
E\left\{(S-\hat{s}_{MSE})^2|z\right\} &= \frac{K+1}{(z+1)^2}
\end{aligned}$$

f) Using the fact that Z is the sum of K i.i.d. variables (given S):

$$p_{Z|S}(z|s) = \underbrace{[s\, \exp(-s\, z)] * \cdots * [s\, \exp(-s\, z)]}_{K \text{ times}} = \frac{s^K\, z^{K-1}}{(K-1)!} \exp(-s\, z), \qquad z > 0$$

The joint pdf of Z and S can now be obtained as

$$p_{Z,S}(z,s) = p_{Z|S}(z|s) p_S(s) = \frac{s^K\, z^{K-1}}{(K-1)!} \exp[-s\,(z+1)], \qquad s,z > 0$$

Finally, integrating s out, we have

$$p_Z(z) = \int p_{Z,S}(z,s) ds = \frac{z^{K-1}}{(K-1)!} \int_0^\infty s^K \exp[-s\,(z+1)] = \frac{K\, z^{K-1}}{(z+1)^{K+1}},\ \ z > 0$$

g)

$$\mathbb{E}\left\{(S-\hat{S}_c)^2\right\} = \int \mathbb{E}\left\{(S-\hat{s}_c)^2|z\right\}\, p_Z(z) dz$$

Using the results from the previous two sections we can obtain

$$\mathbb{E}\left\{(S-\hat{S}_c)^2\right\} = \left[(K+2)(K+1)+c^2-2c(K+1)\right] \int_0^\infty \frac{K\, z^{K-1}}{(z+1)^{K+3}} dz$$

The value of the integral is given in (iii). Simplifying also for the MAP and MSE estimators:

$$\begin{aligned}
\mathbb{E}\left\{(S-\hat{S}_{MAP})^2\right\} &= \frac{2}{K+1} \\
E\left\{(S-\hat{S}_{MSE})^2\right\} &= \frac{2}{K+2}
\end{aligned}$$

Chapter 2
Statistical Decision Theory

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2.1 Introduction to Decision Theory

In this section we provide a formal presentation of decision theory.

Decision theory (also named **detection theory** or **hypothesis testing**) is a mathematical framework used to make optimal choices under conditions of uncertainty. It employs models and statistical analysis to evaluate and compare the outcomes of different decisions, aiming to identify the most advantageous option based on established criteria. This field utilizes concepts from statistics and economics to aid in strategic planning and risk management by calculating the probabilities and impacts of potential scenarios. Decision theory focuses on maximizing benefits and minimizing costs or risks, providing a structured approach to rational decision-making. Through precise quantitative methods, it guides individuals and organizations in policy formulation and decision implementation.

2.1.1 Hypotheses-based problems

In this course, we will only cover a particular class of detection or classification problems to which we will refer as *hypotheses-based problems*. The goal is to infer the true hypothesis, which cannot be directly observed, from a set of measurements or observations. Thus, we consider a scenario with M hypotheses, and denote the random variable that identifies the hypothesis as H. This is depicted in Fig. 2.1, where $H \in \{0,1,\ldots,M-1\}$. We also assume that we have access to an observation vector $\mathbf{x}$, which can be considered as the realization of a random variable $\mathbf{X}$ lying in the observation space $\mathscr{X}$. We assume also that there is a certain statistical relationship between H and $\mathbf{X}$. Otherwise, i.e., if H and $\mathbf{X}$ were independent, it would make no sense to use $\mathbf{x}$ to make an informed inference about the value of H.

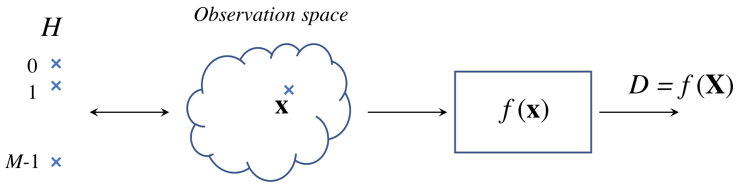


Fig. 2.1 Diagram block of hypothesis testing problems.

In this context, a **detector** is a function that, for any input $\mathbf{x} \in \mathscr{X}$, outputs a *decision* $d = f(\mathbf{x}) \in \{0,1,\ldots M-1\}$, i.e., it makes a guess on the value of the true hypothesis. Depending on the application scenario, the detector receives another names, like **decision-maker** or **classifier**. In this chapter we will take these terms as synonymous.

We should make a few considerations about the functions $f(\mathbf{x})$ that we admit as valid detectors in this course:

- Funcion $f(\cdot)$ is *deterministic*. This implies that if the same vector is presented several times, the function will output the same value each time. Irrespective of this, when the input is a random vector, $D = f(\mathbf{X})$ is a random variable.
- The function is surjective, that is, every input $\mathbf{x}$ generates one and only one output, but different inputs can generate the same output. Hence, the function divides the observation space into M non-overlapping regions, $\mathscr{X}_d$, $d = 0,1,\ldots M-1$, i.e., one region per hypotheses. The boundaries between regions are known as *decision boundaries*.

Example 2.1 The detector $f(x) = u(x^2-1)$, where $u(\cdot)$ is the step function, is defined for any x on the real line, and is characterized by the following decision regions:

$$\mathscr{X}_0 = \{x \in \Re | x^2 - 1 < 0\} = (-1,1),$$
$$\mathscr{X}_1 = \{x \in \Re | x^2 - 1 \geq 0\} = (-\infty,-1] \cup [1,\infty).$$

where we have assumed $u(0) = 1$. In this example, the regions are connected and non-empty.

Example 2.2 The detector $f(\mathbf{x}) = \arg\min_i y_i(\mathbf{x})$ defined over $\mathscr{X} = [0,1]^2$, with

$$\begin{aligned} y_0(\mathbf{x}) &= \|\mathbf{x}\|^2, \\ y_1(\mathbf{x}) &= x_1 - x_0 + 1, \\ y_2(\mathbf{x}) &= x_0 - x_1 + 1, \end{aligned}$$

is characterized by the decision regions depicted in Fig. 2.2.

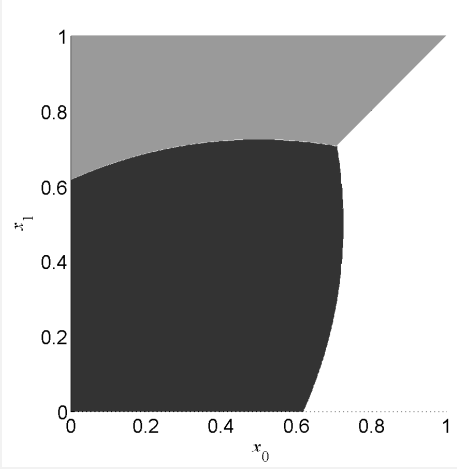


Fig. 2.2 Decision regions for the detector given in Example 2.2: $\mathscr{X}_0$ (black), $\mathscr{X}_1$ (grey), and $\mathscr{X}_2$ (white).

2.1.2 Modeling uncertainty

We review now the main distributions that will be employed in detection problems:

- *A priori* probability distribution of the hypotheses: This is a discrete distribution that quantifies the probability of each hypothesis independently of the observations. If we did not have access to any observation, the decision-maker should rely entirely on these probabilities.
$$P_H(h), \qquad \text{for} \quad h = 0, 1, \ldots, M-1.$$
- Likelihood of the hypotheses: the conditional distribution of the observation, given the hypothesis. For a continuous observation variable, it is given by a probability density function,
$$p_{\mathbf{X}|H}(\mathbf{x}|h) \qquad \text{for} \quad \mathbf{x} \in \mathscr{X},$$
for each hypothesis $h = 0, 1, \ldots, M-1$. Note that the domain of each distribution may depend on the hypothesis.
- *A posteriori* distribution of the hypotheses:
$$P_{H|\mathbf{X}}(h|\mathbf{x}), \qquad \text{for} \quad h = 0, 1, \ldots, M-1.$$

It is a probability measure of the uncertainty about the hypothesis after the observation $\mathbf{X} = \mathbf{x}$. This distribution plays a fundamental role for the statistical design of detectors.

A priori and *a posteriori* probabilities are related by Bayes' Theorem, which states

$$P_{H|\mathbf{X}}(h|\mathbf{x}) = \frac{p_{\mathbf{X}|H}(\mathbf{x}|h)P_H(h)}{p_{\mathbf{X}}(\mathbf{x})}.$$

Bayes' Theorem shows how to update the probabilities of the hypotheses in light of the observation $\mathbf{X} = \mathbf{x}$. Note that if we know both the *a priori* probabilities of the hypothesis and their likelihoods, the joint distribution of $\mathbf{X}$ and H can be calculated. The joint distribution is the most complete characterization of the random variables, and from it any other probability function can be calculated as well.

In the following, we consider two different kinds of problems involving M-ary hypothesis testing problems:

- Analysis of detectors: Here, the detector is given, and the objective is to analyze its performance with respect to certain performance metrics.
- Detector design: The goal is to build a function $f(\mathbf{x})$ to optimize a desired performance metric.

2.2 Performance metrics

The first problem that we consider is the evaluation of the performance of a given detector. In this section, we review different metrics that can be used to assess performance. In all cases, we consider first the multiple hypothesis test scenario, and afterwards we specialize it to the binary case.

2.2.1 Probability of error

The probability of error is the probability of a wrong decision, i.e., the output of the detector is not equal to the true hypothesis. Under a frequentist approach, it can be interpreted as the average number of decision errors after a infinite sequence of independent observations. It can be computed from the likelihood functions and the prior probabilities as

$$\begin{aligned} P_e = \{D \neq H\} &= 1 - P\{D = H\} \\ &= 1 - \sum_{h=0}^{M-1} P\{D = h, H = h\} \\ &= 1 - \sum_{h=0}^{M-1} P\{D = h | H = h\} P_H(h) \\ &= 1 - \sum_{h=0}^{M-1} P_H(h) \int_{\mathscr{X}_h} p_{\mathbf{X}|H}(\mathbf{x}|h) d\mathbf{x}, \end{aligned}$$

where we have exploited that the probability of error is one minus the probability of correct decision, or **accuracy**. This is, in most cases, more convenient since the number of combinations where D and H are equal is (much) smaller than the number of combinations where they differ. Moreover, the last line of the previous expression follows from

$$P(D=h|H=h) = P(\mathbf{x} \in \mathscr{X}_d|H=h) = \int_{\mathscr{X}_h} p_{\mathbf{X}|H}(\mathbf{x}|h)d\mathbf{x},$$

which states that, conditioned on $H=h$, the probability of $D=h$ is precisely the integral of the likelihood of that hypothesis in the region where the given detector decides in favor of hypothesis h, i.e., $\mathscr{X}_h$.

The *conditional* probability of error for a particular observation vector $\mathbf{x} \in \mathscr{X}_d$ can be computed as

$$P\{H \neq d|\mathbf{x}\} = 1 - P\{H=d|\mathbf{x}\} = 1 - P_{H|\mathbf{X}}(d|\mathbf{x}) = \sum_{\substack{l=0 \\ l\neq d}}^{M-1} P_{H|\mathbf{X}}(l|\mathbf{x}) \tag{2.1}$$

In other words, the probability of error at a particular $\mathbf{x} \in \mathscr{X}_d$ is the sum of the *a posteriori* probabilities of hypothesis different from d conditioned on this particular observation.

2.2.1.1 Binary case: P_e, P_{FA}, P_{M} and P_{D}

For the binary case, contrary to the multiple hypotheses test, computing the probability of error involves as many terms as the probability of a correct decision since

$$\begin{aligned} P_e &= P\{D=0,H=1\} + P\{D=1,H=0\} \\ &= P\{D=0|H=1\}P_H(1) + P\{D=1|H=0\}P_H(0). \end{aligned}$$

In the expression above we find two terms that are normally referred to as the **probability of false alarm** (also known as probability of *Type I error* or *significance level*) and the **probability of missing** (or probability of *Type II* error):

$$\begin{aligned} P_{\mathrm{FA}} &= P\{D=1|H=0\} = \int_{\mathscr{X}_1} p_{\mathbf{X}|H}(\mathbf{x}|0)d\mathbf{x}, \\ P_{\mathrm{M}} &= P\{D=0|H=1\} = \int_{\mathscr{X}_0} p_{\mathbf{X}|H}(\mathbf{x}|1)d\mathbf{x}. \end{aligned}$$

Similarly, the **probability of detection** (*power* or *sensitivity*) is defined as

$$P_{\mathrm{D}} = P\{D=1|H=1\} = 1 - P_{\mathrm{M}},$$

and

$$P\{D=0|H=0\} = 1 - P_{\mathrm{FA}},$$

is the **specificity**. Using these definitions, the probability of error can now be expressed more compactly as

$$P_e = P_{\mathrm{M}}P_H(1) + P_{\mathrm{FA}}P_H(0).$$

Interestingly, for the computation of P_{FA} and P_{M}, only likelihoods are required. However, in order to compute the overall probability of error, we also need to know the *a priori* probabilities of the hypothesis.

2.2.2 Receiver Operating Characteristic (ROC)

We also introduce here an important concept for the analysis of binary hypothesis tests: the receiver operating characteristic (ROC) curve. The ROC curve plots the probability of false alarm, P_{FA}, against the probability of detection, P_{D} for different values of some parameter. Figure 2.3 shows the ROC curves of two different detectors, Detector 1 and Detector 2. As can be seen in this figure, the performance of Detector 2 is clearly better than that of Detector 1, since for each P_{FA}, the P_{D} of Detector 2 is equal or larger than that of Detector 1. Moreover, both detectors perform better than a random decision whose ROC curve is also shown in the figure. One final comment is in order. For almost all detectors it is not possible to increase the probability of detection without increasing the probability of false alarm.

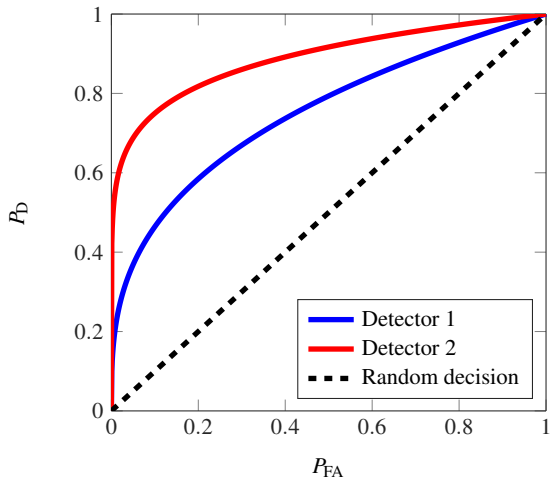


Fig. 2.3 ROC curves for two different detectors.

2.2.3 Risk

In scenarios where the consequences of each type of error are different, the probability of error is not an adequate performance measure. Imagine, for instance, a detector that discriminates whether there are or not suspicious tumor masses in a medical image. Such detector is used as a pre-diagnosis system, so that patients that can have a tumor are then explored with more accurate (but also invasive) techniques. In this case, there is a clear

asymmetry between both kinds of errors: The incorrect decision that tumor masses are present would result in an unnecessary biopsy and inconvenience for the patient, but the opposite error could delay the diagnosis until a time when the process is irreversible.

To assign a penalty to different kinds of errors, we can define a cost function $c(d,h)$, or

$$c_{dh}, \qquad d,h = 0,\ldots,M-1,$$

whose value is the cost of deciding $D = d$ when the true hypothesis is $H = h$. In general, we will assume that the cost for each particular d and h is deterministic. However, the cost c_{DH}, as a function of the random variables D and H, is a random variable, too. Given a detector $D = \phi(\mathbf{X})$, we define the **risk** R_ϕ as the expected value of the cost,

$$R_\phi = \mathbb{E}\{c_{DH}\} = \sum_{h=0}^{M-1}\sum_{d=0}^{M-1} c_{dh} P_H(h) P_{D|H}(d|h). \tag{2.2}$$

Since $\phi(\mathbf{x}) = d$ when the observation belongs to the decision region of $D = d$, that is, $\mathbf{x} \in \mathcal{X}_d$, the conditional probabilities $P_{D|H}(d|h)$ can be calculated as

$$P_{D|H}(d|h) = P\{X \in \mathcal{X}_d | H = h\} = \int_{\mathcal{X}_d} p_{\mathbf{X}|H}(\mathbf{x}|h) d\mathbf{x} \tag{2.3}$$

therefore

$$R_\phi = \sum_{h=0}^{M-1} P_H(h) \sum_{d=0}^{M-1} c_{dh} \int_{\mathbf{x}\in\mathcal{X}_d} p_{\mathbf{X}|H}(\mathbf{x}|h) d\mathbf{x}. \tag{2.4}$$

Example 2.3 Consider a multiclass decision problem with three hypotheses whose likelihoods are:

$$\begin{array}{ll} p_{X|H}(x|0) = 1 & 0 < x < 1 \\ p_{X|H}(x|1) = 2(1-x) & 0 < x < 1 \\ p_{X|H}(x|2) = 2x & 0 < x < 1 \end{array}$$

knowing that the prior probabilities of the hypotheses are: $P_H(0) = 0.4$ and $P_H(1) = P_H(2) = 0.3$, and the cost policy is given by $c_{hh} = 0$, $h = 0,1,2$ and $c_{hd} = 1$, $h \neq d$. Obtain the risk of the decision-maker:

$$\phi(x) = \begin{cases} 1, & x < 0.5 \\ 2, & x > 0.5 \end{cases}$$

Applying the expression (2.2) to this problem we have:

$$\begin{aligned} R_\phi = {} & c_{10} P_H(0) P_{D|H}(1|0) + c_{20} P_H(0) P_{D|H}(2|0) + c_{01} P_H(1) P_{D|H}(0|1) \\ & + c_{21} P_H(1) P_{D|H}(2|1) + c_{02} P_H(2) P_{D|H}(0|2) + c_{12} P_H(2) P_{D|H}(1|2) \end{aligned}$$

where the terms $P_{D|H}(d|h)$ can be calculated using (2.3)

$$
\begin{aligned}
P_{D|H}(0|1) &= P_{D|H}(0|2) = 0 \\
P_{D|H}(1|0) &= \int_{\mathscr{X}_1} p_{X|H}(x|0)\,dx = \int_0^{0.5} 1\,dx = 0.5 \\
P_{D|H}(2|0) &= \int_{\mathscr{X}_2} p_{X|H}(x|0)\,dx = \int_{0.5}^{1} 1\,dx = 0.5 \\
P_{D|H}(1|2) &= \int_{\mathscr{X}_1} p_{X|H}(x|2)\,dx = \int_0^{0.5} 2x\,dx = 0.25 \\
P_{D|H}(2|1) &= \int_{\mathscr{X}_2} p_{X|H}(x|1)\,dx = \int_{0.5}^{1} 2(1-x)\,dx = 0.25
\end{aligned}
$$

and substituting these, we arrive at

$$R_\phi = 0.4\cdot 0.5 + 0.4\cdot 0.5 + 0.3\cdot 0.25 + 0.3\cdot 0.25 = 0.55$$

Finally, we define the **conditional risk** as the expected cost conditioned on a given value of $\mathbf{x}$, $\mathbb{E}\{c_{dH}|\mathbf{x}\}$. Taking into account that, for a given $\mathbf{x}$ and a given detector, the decision value is fixed, it is only required to take expectations with respect to such hypothesis. The conditional risk for a given observation $\mathbf{x} \in \mathscr{X}_d$ is given by

$$\mathbb{E}\{c_{dH}|\mathbf{x}\} = \sum_{h=0}^{M-1} c_{dh} P_{H|X}(h|\mathbf{x}). \tag{2.5}$$

Example 2.4 Continuing with Example 2.3, the conditional risk of each decision can be calculated as:

$$\mathbb{E}\{c(d,H)|x\} = c_{d0}P_{H|X}(0|x) + c_{d1}P_{H|X}(1|x) + c_{d2}P_{H|X}(2|x)$$

where the posterior distributions can be obtained by applying Bayes' Theorem:

$$
\begin{aligned}
P_{H|X}(0|x) &= \frac{p_{X|H}(x|0)P_H(0)}{\sum_{h=0}^{2} p_{X|H}(x|h)P_H(h)} = \frac{1\cdot 0.4}{1\cdot 0.4 + 2(1-x)\cdot 0.3 + 2x\cdot 0.3} = 0.4 \\
P_{H|X}(1|x) &= \frac{p_{X|H}(x|1)P_H(1)}{\sum_{h=0}^{2} p_{X|H}(x|h)P_H(h)} = \frac{2(1-x)\cdot 0.3}{1} = 0.6(1-x) \\
P_{H|X}(2|x) &= \frac{p_{X|H}(x|2)P_H(2)}{\sum_{h=0}^{2} p_{X|H}(x|h)P_H(h)} = \frac{2x\cdot 0.3}{1} = 0.6x
\end{aligned}
$$

This leads to:

- if $d = 0$:
$$
\begin{aligned}
\mathbb{E}\{c_{0,H}|x\} &= c_{00}P_{H|X}(0|x) + c_{01}P_{H|X}(1|x) + c_{02}P_{H|X}(2|x) \\
&= 0\cdot 0.4 + 1\cdot 0.6(1-x) + 1\cdot 0.6x = 0.6
\end{aligned}
$$
- if $d = 1$:
$$
\begin{aligned}
\mathbb{E}\{c_{1,H}|x\} &= c_{10}P_{H|X}(0|x) + c_{11}P_{H|X}(1|x) + c_{12}P_{H|X}(2|x) \\
&= 1\cdot 0.4 + 0\cdot 0.6(1-x) + 1\cdot 0.6x = 0.4 + 0.6x
\end{aligned}
$$
- if $d = 2$:
$$
\begin{aligned}
\mathbb{E}\{c_{2,H}|x\} &= c_{20}P_{H|X}(0|x) + c_{21}P_{H|X}(1|x) + c_{22}P_{H|X}(2|x) \\
&= 1\cdot 0.4 + 1\cdot 0.6(1-x) + 0\cdot 0.6x = 1 - 0.6x
\end{aligned}
$$

2.2.3.1 Binary case: risk

For the binary case, a simpler expression can be obtained in terms of P_{FA}, P_{M}, and P_{D} as follows

$$\begin{aligned} R_\phi &= c_{00}P(D=0,H=0)+c_{01}P(D=0,H=1) \\ &= c_{00}P(D=0|H=0)P_H(0)+c_{01}P_{\mathrm{M}}P_H(1)+c_{10}P_{\mathrm{FA}}P_H(0)+c_{11}P_{\mathrm{D}}P_H(1). \\ &= c_{00}(1-P_{\mathrm{FA}})P_H(0)+c_{01}P_{\mathrm{M}}P_H(1)+c_{10}P_{\mathrm{FA}}P_H(0)+c_{11}(1-P_{\mathrm{M}})P_H(1). \\ &= (c_{00}P_H(0)+c_{11}P_H(1))+(c_{01}-c_{11})P_{\mathrm{M}}P_H(1)+(c_{10}-c_{00})P_{\mathrm{FA}}P_H(0) \end{aligned} \tag{2.6}$$

The previous expression shows that the risk of a decision-maker is the sum of three components:

- $(c_{00}P_H(0)+c_{11}P_H(1))$ is the minimum risk of the ideal decision-maker, the one with $P_{\mathrm{M}}=0$ and $P_{\mathrm{FA}}=0$ who succeeds with probability 1.
- $(c_{01}-c_{11})P_H(1)P_{\mathrm{M}}$ is the increase in risk caused by miss errors.
- $(c_{10}-c_{00})P_H(0)P_{\mathrm{FA}}$ is the increase in risk caused by false alarms.

Note that the ideal decision-maker is, in general, unachievable, because if the likelihoods of the hypotheses overlap, it is not possible to avoid errors. The optimal decision-maker will be the one who finds a good compromise between miss errors and false alarms, such that the risk in (2.6) is minimized.

2.3 Detector design

Once we have studied different ways of analyzing the performance of a given detector, we turn our attention to the problem of designing detectors that maximize one of these performance metrics.

2.3.1 Maximum likelihood and maximum *a posteriori* detectors

A first possibility would be to rely directly on the maximization of the available probability density functions:

- The detector that maximizes the likelihood is known as the *maximum likelihood* (ML) detector:
$$d_{ML} = \arg\max_h p_{\mathbf{X}|H}(\mathbf{x}|h).$$
- The detector that selects the hypothesis with maximum *a posteriori* probability is known as the maximum *a posteriori* (MAP) detector:
$$d_{MAP} = \arg\max_h P_{H|\mathbf{X}}(h|\mathbf{x}).$$

Finally, there are two properties that are worth considering with respect to these detectors:

1. When the *a priori* probabilities of the hypothesis are the same, i.e., $P_H(h)=1/M$, the ML and MAP detectors are identical. This can be shown from the Bayes' Theorem, since in this case

$$d_{MAP} = \arg\max_h P_{H|\mathbf{X}}(h|\mathbf{x}) = \arg\max_h \frac{p_{\mathbf{X}|H}(\mathbf{x}|h)P_H(h)}{p_{\mathbf{X}}(\mathbf{x})} = \arg\max_h p_{\mathbf{X}|H}(\mathbf{x}|h) = d_{ML}.$$

2. The MAP detector minimizes the probability of error. Note that according to (2.1) the probability of error for a given $\mathbf{x}$ can be expressed as

$$P(D \neq H|\mathbf{x}) = 1 - P_{H|\mathbf{X}}(h|\mathbf{x}).$$

Since the MAP detector selects, for every $\mathbf{x}$, the hypothesis maximizing $P_{H|\mathbf{X}}(h|\mathbf{x})$, it minimizes the conditional error probability $P(D \neq H|\mathbf{x})$. Since the overall error probability is an integral over the conditional error probabilities, i.e.,

$$P(D \neq H) = \int_{\mathscr{X}} P(D \neq H|\mathbf{x}) p_{\mathbf{X}}(\mathbf{x}) \mathbf{dx},$$

the MAP detector minimizes the overal probability of error, too.

2.3.1.1 Binary case: ML and MAP detectors

The expressions of the ML and MAP detectors become fairly simple for the binary case:

- Maximum likelihood detector:

$$p_{\mathbf{X}|H}(\mathbf{x}|1) \underset{D=0}{\overset{D=1}{\gtrless}} p_{\mathbf{X}|H}(\mathbf{x}|0),$$

which can be expressed as a *likelihood ratio test* (LRT)

$$\frac{p_{\mathbf{X}|H}(\mathbf{x}|1)}{p_{\mathbf{X}|H}(\mathbf{x}|0)} \underset{D=0}{\overset{D=1}{\gtrless}} 1,$$

where we have taken into account that the likelihoods are non-negative. Sometimes, it will be more convenient to work with the *log-likelihood ratio test* (LLRT)

$$\log\left[\frac{p_{\mathbf{X}|H}(\mathbf{x}|1)}{p_{\mathbf{X}|H}(\mathbf{x}|0)}\right] = \log p_{\mathbf{X}|H}(\mathbf{x}|1) - \log p_{\mathbf{X}|H}(\mathbf{x}|0) \underset{D=0}{\overset{D=1}{\gtrless}} 0, \tag{2.7}$$

which can be done because the logarithm is a monotonically increasing function.

- Maximum *a posteriori* detector:

$$p_{H|\mathbf{X}}(1|\mathbf{x}) \underset{D=0}{\overset{D=1}{\gtrless}} p_{H|\mathbf{X}}(0|\mathbf{x}),$$

which can also be expressed as a LRT as

$$\frac{p_{\mathbf{X}|H}(\mathbf{x}|1)}{p_{\mathbf{X}|H}(\mathbf{x}|0)} \underset{D=0}{\overset{D=1}{\gtrless}} \frac{P_H(0)}{P_H(1)}. \tag{2.8}$$

As in the general case with M hypothesis, the MAP detector minimizes the probability of error and the ML and MAP detectors are the same if $P_H(0) = P_H(1) = 0.5$. Moreover,

we can see that both detectors can be expressed as a LRT

$$\frac{p_{\mathbf{X}|H}(\mathbf{x}|1)}{p_{\mathbf{X}|H}(\mathbf{x}|0)} \underset{D=0}{\overset{D=1}{\gtrless}} \eta, \tag{2.9}$$

where η is a threshold. When this threshold is 1, the LRT is the ML detector and for $\eta = P_H(0)/P_H(1)$, the LRT becomes the MAP detector, that is, minimum P_e detector. Hence, we get two different points in the ROC curve. Actually, sweeping the value of the threshold generates the complete ROC curves in Figure 2.3.[1]

2.3.2 Bayesian decision-making: the minimum risk detector

As we have already studied, sometimes it makes more sense to measure the performance of a detector in terms of the expected cost. Therefore, it is important to tackle the problem of designing a detector that is optimum with respect to the expected cost.

Remember that the expected cost of a detector deciding d for an observation $\mathbf{x}$ is given by Equation (2.5), which we reproduce here for convenience:

$$\mathbb{E}\{c_{dH}|\mathbf{x}\} = \sum_{h=0}^{M-1} c_{dh} P_{H|\mathbf{X}}(h|\mathbf{x}). \tag{2.10}$$

Minimizing the expected cost over the whole observation space requires that decisions for each observation minimize the conditional expected cost. That is, for each $\mathbf{x}$ the above expression should be minimized, and the expression of the minimum mean cost detector can be stated as follows:

$$d^\star = \arg\min_d \sum_{h=0}^{M-1} c_{dh} P_{H|\mathbf{X}}(h|\mathbf{x}).$$

Hence, when designing the detector, we need to evaluate the cost of the different decisions for each observation vector, and select the decision for which the expected cost is minimized.

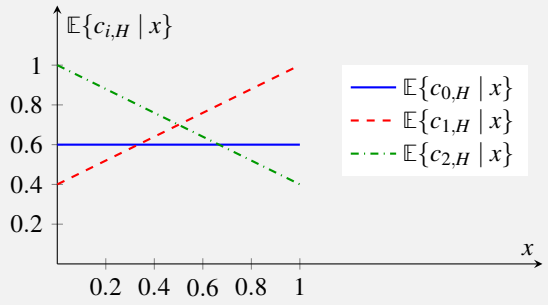
Example 2.5 Continuing with Example 2.4, since

$$\begin{aligned}\mathbb{E}\{c_{0,H}|x\} &= 0.6\\ \mathbb{E}\{c_{1,H}|x\} &= 0.4 + 0.6x\\ \mathbb{E}\{c_{2,H}|x\} &= 1 - 0.6x\end{aligned}$$

(as shown in the figure), the Bayesian detector is

[1] This actually applies to all detectors that can be written as $\phi(\mathbf{x}) \underset{D=0}{\overset{D=1}{\gtrless}} \eta$. That is, comparing a function of the observations with a threshold achieves a given $(P_{\mathrm{FA}}, P_{\mathrm{D}})$ point in the ROC curve. These detectors are known as threshold detectors.

$$D = \begin{cases} 1 & 0 \le X \le \frac{1}{3} \\ 0 & \frac{1}{3} \le X \le \frac{2}{3} \\ 2 & \frac{2}{3} \le X \le 1 \end{cases}$$



It is interesting to point out that when the cost function penalizes equally all kinds of errors, i.e.,

$$c_{dh} = \begin{cases} 0, & d = h \\ c, & d \neq h \end{cases}$$

the detector with minimum risk is MAP. This is easily proved by replacing these costs into the expression for the minimum expected cost detector

$$\begin{aligned} d^\star &= \arg\min_d \sum_{h=0}^{M-1} c_{dh} P_{H|\mathbf{X}}(h|\mathbf{x}) \\ &= \arg\min_d \; c \sum_{h \neq d} P_{H|\mathbf{X}}(h|\mathbf{x}) \\ &= \arg\min_d \left\{1 - P_{H|\mathbf{X}}(d|\mathbf{x})\right\} \\ &= \arg\max_d P_{H|\mathbf{X}}(d|\mathbf{x}) \\ &= d_{MAP}. \end{aligned}$$

2.3.2.1 Binary case: Minimum risk detector

In the binary case, we can also express the optimum detector with respect to a cost function as a LRT. Let us start by particularizing (2.10) for $d = 0$ and $d = 1$, and then follow the criterion of deciding in favor of the minimum cost, i.e.,

$$\mathbb{E}\{c_{0H}|\mathbf{x}\} \underset{D=0}{\overset{D=1}{\gtrless}} \mathbb{E}\{c_{1H}|\mathbf{x}\}.$$

Now, using the definition of expectation, the criterion becomes

$$c_{00}P_{H|\mathbf{X}}(0|\mathbf{x}) + c_{01}P_{H|\mathbf{X}}(1|\mathbf{x}) \underset{D=0}{\overset{D=1}{\gtrless}} c_{10}P_{H|\mathbf{X}}(0|\mathbf{x}) + c_{11}P_{H|\mathbf{X}}(1|\mathbf{x}),$$

which after some algebra can be rewritten as

$$\frac{P_{H|\mathbf{X}}(1|\mathbf{x})}{P_{H|\mathbf{X}}(0|\mathbf{x})} \underset{D=0}{\overset{D=1}{\gtrless}} \frac{c_{10}-c_{00}}{c_{01}-c_{11}}.$$

Finally, using Bayes' Theorem, we may rewrite the *a posteriori* probabilities in terms of the likelihoods and the *a priori* probabilities, which finally yields

$$\frac{P_{\mathbf{X}|H}(\mathbf{x}|1)}{P_{\mathbf{X}|H}(\mathbf{x}|0)} \underset{D=0}{\overset{D=1}{\gtrless}} \frac{c_{10}-c_{00}}{c_{01}-c_{11}} \frac{P_H(0)}{P_H(1)},$$

and corresponds to yet another point of the ROC curve of the LRT.

2.3.3 Non-Bayesian detectors

Non-Bayesian detectors are those that do not ground on a probability model for the hypothesis. Their design depends on the likelihood functions only. This is the case, for instance, of the ML detector. Other non-Bayesian detectors, in the binary case can be expressed as the LRT in (2.9) for different valures of η. The Neyman-Pearson detector is a classical example.

2.3.3.1 Binary case: Neyman-Pearson detector

The Neyman-Pearson (NP) detector is a well known detector for binary problems which maximizes the probability of detection while it provides a bound on the probability of false alarm. Before proceeding with the derivation, let us recall the definitions of probability of false alarm and detection

$$P_{\mathrm{FA}} = \int_{\mathscr{X}_1} p_{\mathbf{X}|H}(\mathbf{x}|0)d\mathbf{x},$$
$$P_{\mathrm{D}} = \int_{\mathscr{X}_1} p_{\mathbf{X}|H}(\mathbf{x}|1)d\mathbf{x}.$$

The NP detector can be derived as the solution of the following constrained optimization problem:

$$\text{maximize } P_{\mathrm{D}}, \quad \text{subject to } P_{\mathrm{FA}} \leq \alpha,$$

The solution to this kind of problems is obtained from the Lagrangian, $\mathscr{L}(\mathscr{X}_1, \eta)$, defined as:

$$\begin{aligned}\mathscr{L}(\mathscr{X}_1, \eta) &= P_{\mathrm{D}} - \eta(P_{\mathrm{FA}} - \alpha) \\ &= \int_{\mathscr{X}_1} p_{\mathbf{X}|H}(\mathbf{x}|1)d\mathbf{x} - \eta\left(\int_{\mathscr{X}_1} p_{\mathbf{X}|H}(\mathbf{x}|0)d\mathbf{x} - \alpha\right) \\ &= \int_{\mathscr{X}_1} \left(p_{\mathbf{X}|H}(\mathbf{x}|1) - \eta\, p_{\mathbf{X}|H}(\mathbf{x}|0)\right) d\mathbf{x} + \eta\,\alpha. \end{aligned} \tag{2.11}$$

Note that the optimization variable is the decision region $\mathscr{X}_1$ (the set of observations where we decide $D = 1$). To maximize the Lagrangian, we must choose $\mathscr{X}_1$ such that we include all points $\mathbf{x}$ where the integrand is non-negative:

$$\mathscr{X}_1 = \{\mathbf{x} \mid p_{\mathbf{X}|H}(\mathbf{x}|1) - \eta\, p_{\mathbf{X}|H}(\mathbf{x}|0) \geq 0\}.$$

This result leads directly to the Likelihood Ratio Test (LRT):

$$\Lambda(\mathbf{x}) = \frac{p_{\mathbf{X}|H}(\mathbf{x}|1)}{p_{\mathbf{X}|H}(\mathbf{x}|0)} \underset{D=0}{\overset{D=1}{\gtrless}} \eta,$$

where the threshold η is selected to achieve the desired probability of false alarm α.

In practice, when the ROC of the LRT is a continuous function, the NP detector can be computed in two steps:

1. Express P_{FA} as a function of η for the LRT
2. Find η such that $P_{\text{FA}}(\eta) = \alpha$.

2.3.3.2 Minimax classifiers

The decision rule of the MAP detector depends on the prior probability $P_H(1)$. In many practical scenarios, however, this prior is unknown or may vary, leading to a mismatch between the design prior and the true operating conditions. The minimax classifier provides a robust strategy by minimizing the maximum possible error probability.

Let $P = P_H(1)$ be the true (unknown) prior probability, and let Q be the design prior used to construct the MAP detector. The resulting probability of error, $P_{\text{e}}(P,Q)$, is a function of both. The minimax detector is found by solving:

$$Q^* = \underset{Q}{\operatorname{argmin}} \max_{P} \left[P \cdot P_{\text{M}}(Q) + (1-P) \cdot P_{\text{FA}}(Q)\right]. \tag{2.12}$$

This formulation implies that we seek a design prior Q such that the detector performs well even for the "least favorable" true prior P. Since the error probability $P_{\text{e}}(P,Q)$ is a linear function of P, the maximum error occurs at the boundaries ($P = 0$ or $P = 1$) unless the slope of the error function is zero.

Therefore, the minimax solution is achieved when the error probability is independent of the true prior P. For binary problems, this occurs when the Likelihood Ratio Test (LRT) threshold, η is chosen such that:

$$\boxed{P_{\text{FA}}(\eta) = P_{\text{M}}(\eta)} \tag{2.13}$$

In this state, the classifier is robust: no matter how the true prior P fluctuates, the error probability remains constant and equal to P_{FA}.

2.4 Gaussian models

In this section, we will derive the likelihood ratio test for Gaussian observations under several assumptions. Then, depending on the threshold, we would obtain the different detectors: NP, ML, MAP, and minimum cost.

Before proceeding, we introduce the multivariate real Gaussian probability density function (PDF), which is given by

$$P_{\mathbf{X}}(\mathbf{x}) = \frac{1}{(2\pi)^{N/2}|\mathbf{V}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x}-\mathbf{m})^T\mathbf{V}^{-1}(\mathbf{x}-\mathbf{m})\right),$$

where $\mathbf{x}$ is an N-dimensional vector, $\mathbf{m}$ is the mean vector, and $\mathbf{V}$ is the cross-covariance matrix. Then, under hypothesis $h=0$, the likelihood is

$$P_{\mathbf{X}|H}(\mathbf{x}|0) = \frac{1}{(2\pi)^{N/2}|\mathbf{V}_0|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x}-\mathbf{m}_0)^T\mathbf{V}_0^{-1}(\mathbf{x}-\mathbf{m}_0)\right),$$

whereas it is

$$P_{\mathbf{X}|H}(\mathbf{x}|1) = \frac{1}{(2\pi)^{N/2}|\mathbf{V}_1|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x}-\mathbf{m}_1)^T\mathbf{V}_1^{-1}(\mathbf{x}-\mathbf{m}_1)\right),$$

under hypothesis $h=1$. For this hypothesis test, the LLRT in (2.7) becomes

$$-\frac{1}{2}\log|\mathbf{V}_1| - \frac{1}{2}(\mathbf{x}-\mathbf{m}_1)^T\mathbf{V}_1^{-1}(\mathbf{x}-\mathbf{m}_1) + \frac{1}{2}\log|\mathbf{V}_0| + \frac{1}{2}(\mathbf{x}-\mathbf{m}_0)^T\mathbf{V}_0^{-1}(\mathbf{x}-\mathbf{m}_0) \underset{D=0}{\overset{D=1}{\gtrless}} \log(\eta)$$

or, equivalently,

$$(\mathbf{x}-\mathbf{m}_0)^T\mathbf{V}_0^{-1}(\mathbf{x}-\mathbf{m}_0) - (\mathbf{x}-\mathbf{m}_1)^T\mathbf{V}_1^{-1}(\mathbf{x}-\mathbf{m}_1) \underset{D=0}{\overset{D=1}{\gtrless}} \mu \tag{2.14}$$

where

$$\mu = 2\log(\eta) + \log|\mathbf{V}_1| - \log|\mathbf{V}_0|,$$

with η being a threshold selected according to the performance criterion.

After a careful look at (2.14), it can be shown that the optimal detector in the Gaussian case is given by a second-order polynomial function. Hence, the decision boundaries[2] are quadratic surfaces. For instance, for 2D problems ($N=2$), these boundaries are hyperbolas, parabolas, ellipses or straight lines.

In the following sections, we consider a few particular cases, and we conclude this section with two examples.

[2] We obtain the decision boundaries for the equality in (2.14).

Example 2.6 Figure 2.4 shows the decision boundaries for the ML detector ($\eta = 1$ in (2.14)), for a detection problem with 2D Gaussian observations with the following means and cross-covariance matrices:

$$\mathbf{m}_0 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \mathbf{V}_0 = \begin{pmatrix} 1.2 & 0.43 \\ 0.43 & 1.75 \end{pmatrix},$$

and

$$\mathbf{m}_1 = \begin{pmatrix} 3 \\ 3 \end{pmatrix}, \mathbf{V}_1 = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}.$$

In this figure, the gray color gradient represents the value of the likelihoods $P_{\mathbf{X}|H}(\mathbf{x}|0)$ and $P_{\mathbf{X}|H}(\mathbf{x}|1)$, where darker colors denote larger values. Moreover, the white curves are the iso-probability lines and the black curve is the decision boundary, which in this case is a hyperbola (the symmetric part is not shown in this figure).

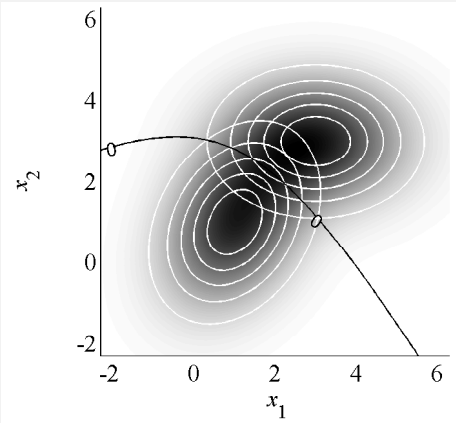


Fig. 2.4 Hyperbolic decision boundary of the ML detector and likelihoods for a Gaussian detection problem with 2D observations.

Example 2.7 Figure 2.5 shows an equivalent figure to that of the previous example, but for a problem with the following means and cross-covariance matrices:

$$\mathbf{m}_0 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \mathbf{V}_0 = \begin{pmatrix} 0.7 & 0 \\ 0 & 0.7 \end{pmatrix},$$

and

$$\mathbf{m}_1 = \begin{pmatrix} 0.2 \\ 0.4 \end{pmatrix}, \mathbf{V}_1 = \begin{pmatrix} 0.5 & 0 \\ 0 & 0.2 \end{pmatrix}.$$

In this case, the decision boundary is an ellipse.

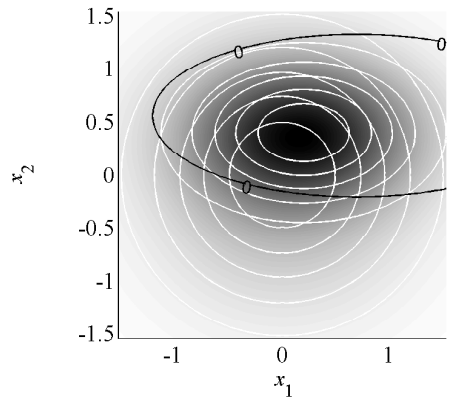


Fig. 2.5 Elliptic decision boundary of the ML detectors and likelihoods for a Gaussian detection problem with 2D observations.

2.4.1 Identical cross-covariance matrices

This section considers the case of $\mathbf{V}_1 = \mathbf{V}_0 = \mathbf{V}$. Then, the LLRT becomes

$$(\mathbf{x}-\mathbf{m}_0)^T\mathbf{V}^{-1}(\mathbf{x}-\mathbf{m}_0) - (\mathbf{x}-\mathbf{m}_1)^T\mathbf{V}^{-1}(\mathbf{x}-\mathbf{m}_1) \underset{D=0}{\overset{D=1}{\gtrless}} \mu.$$

Now, expanding the quadratic forms, the above expression simplifies to

$$(\mathbf{m}_1-\mathbf{m}_0)^T\mathbf{V}^{-1}\mathbf{x} \underset{D=0}{\overset{D=1}{\gtrless}} \tilde{\mu}, \tag{2.15}$$

where $\tilde{\mu} = \mu/2 + \mathbf{m}_1^T\mathbf{V}^{-1}\mathbf{m}_1/2 - \mathbf{m}_0^T\mathbf{V}^{-1}\mathbf{m}_0/2$. In this particular case, the LLRT in (2.15) is a linear function of the observation vector $\mathbf{x}$.

Example 2.8 Figure 2.6 shows three decision boundaries for an example with

$$\mathbf{m}_0 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \mathbf{V}_0 = \begin{pmatrix} 0.44 & 0.32 \\ 0.32 & 0.81 \end{pmatrix}$$

and

$$\mathbf{m}_1 = \begin{pmatrix} 3 \\ 3 \end{pmatrix}, \mathbf{V}_1 = \begin{pmatrix} 0.44 & 0.32 \\ 0.32 & 0.81 \end{pmatrix}.$$

The label of each decision boundary is $\log(\eta)$. Then, $\log(\eta) = 0$ corresponds to the ML detector.

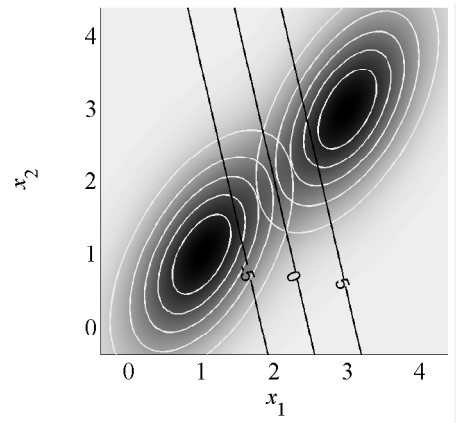


Fig. 2.6 Decision boundaries of the LLRT and likelihoods for a Gaussian detection problem with 2D observations and identical covariance matrices.

Example 2.9 (Matched filter) In this example, we derive one of the most well-known detectors, the matched filter (MF). The MF is the LLRT to the detection of a known signal contaminated by zero-mean Gaussian noise. Concretely, under hypothesis $h = 0$, the observations are given by noise only:

$$x[n] = w[n], \quad n = 0, \ldots, N-1,$$

and under hypothesis $h = 1$, the observations are

$$x[n] = s[n] + w[n], \quad n = 0, \ldots, N-1,$$

where $s[n]$ is a known signal and $w[n]$ is additive white Gaussian noise with zero mean and variance σ^2, i.e., $w[n] \sim \mathcal{N}(0, \sigma^2)$. To use the LLRT already derived in this section, we must first define the vector

$$\mathbf{x} = \big(x[0]\ x[1]\ \cdots\ x[N-1]\big)^T = \mathbf{s} + \mathbf{w},$$

with $\mathbf{s} = \big(s[0]\ s[1]\ \cdots\ s[N-1]\big)^T$ and $\mathbf{w} = \big(w[0]\ w[1]\ \cdots\ w[N-1]\big)^T$, and obtain the distributions of $\mathbf{x}$ under both hypothesis. Under hypothesis $h = 0$, the observation vector $\mathbf{x}$ collects samples of a Gaussian process, which makes it also Gaussian. Hence, only the mean and cross-covariance matrices are required:

$$\mathbf{m}_0 = \mathbb{E}\{\mathbf{x}|0\} = \mathbb{E}\{\mathbf{w}\} = \big(\mathbb{E}\{w[0]\}\ \mathbb{E}\{w[1]\}\ \cdots\ \mathbb{E}\{w[N-1]\}\big)^T = \mathbf{0},$$

and

$$\begin{aligned}
\mathbf{V}_0 &= \mathbb{E}\left\{(\mathbf{x}-\mathbf{m}_0)(\mathbf{x}-\mathbf{m}_0)^T|0\right\} = \mathbb{E}\left\{\mathbf{w}\mathbf{w}^T\right\} \\
&= \mathbb{E}\left\{\big(w[0]\ w[1]\ \cdots\ w[N-1]\big)^T \big(w[0]\ w[1]\ \cdots\ w[N-1]\big)\right\} \\
&= \begin{pmatrix}
\mathbb{E}\left\{w^2[0]\right\} & \mathbb{E}\{w[0]w[1]\} & \cdots\ \mathbb{E}\{w[0]w[N-1]\} \\
\mathbb{E}\{w[1]w[0]\} & \mathbb{E}\left\{w^2[1]\right\} & \cdots\ \mathbb{E}\{w[1]w[N-1]\} \\
\vdots & \vdots & \ddots\ \vdots \\
\mathbb{E}\{w[N-1]w[0]\} & \mathbb{E}\{w[N-1]w[1]\} & \cdots\ \mathbb{E}\left\{w^2[N-1]\right\}
\end{pmatrix}.
\end{aligned}$$

The cross-covariance matrix $\mathbf{V}_0$ can be simplified taking into account that the noise is white, i.e., $\mathbb{E}\{w[n]w[n-m]\} = \sigma^2\delta[m]$, which yields

$$\mathbf{V}_0 = \begin{pmatrix} \sigma^2 & 0 & \cdots & 0 \\ 0 & \sigma^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma^2 \end{pmatrix} = \sigma^2 \mathbf{I}.$$

Similarly, under hypothesis $h = 1$, the observations are Gaussian with mean

$$\mathbf{m}_1 = \mathbb{E}\{\mathbf{x}|1\} = \mathbb{E}\{\mathbf{s}+\mathbf{w}\} = \mathbb{E}\{\mathbf{s}\} + \mathbb{E}\{\mathbf{w}\} = \mathbf{s},$$

since $\mathbf{s}$ is deterministic, and cross-covariance matrix

$$\mathbf{V}_1 = \mathbb{E}\left\{(\mathbf{x}-\mathbf{m}_1)(\mathbf{x}-\mathbf{m}_1)^T|1\right\} = \mathbb{E}\left\{(\mathbf{s}+\mathbf{w}-\mathbf{s})(\mathbf{s}+\mathbf{w}-\mathbf{s})^T\right\} = \mathbb{E}\left\{\mathbf{w}\mathbf{w}^T\right\} = \sigma^2\mathbf{I}.$$

Hence, the detection problem is that of Gaussian observations with identical covariance matrices, for which the LLRT is

$$(\mathbf{m}_1 - \mathbf{m}_0)^T \mathbf{V}^{-1}\mathbf{x} = \frac{1}{\sigma^2}\mathbf{s}^T\mathbf{x} \underset{D=0}{\overset{D=1}{\gtrless}} \tilde{\mu} \Rightarrow \underbrace{\sum_{n=0}^{N-1} s[n]x[n]}_{MF} \underset{D=0}{\overset{D=1}{\gtrless}} \sigma^2\tilde{\mu}.$$

Alternatively, and the motivation for the term matched filter, is because the above detector can be rewritten as a filtering of the signal $x[n]$ with the filter $h[n] = s[N-1-n]$, followed by sampling every N samples. Finally, we also would like to point out that the matched filter is a filter that maximizes the signal-to-noise ratio.

2.4.2 Zero means

We consider now that $\mathbf{m}_0 = \mathbf{m}_1 = \mathbf{0}$, which yields

$$\mathbf{x}^T\left(\mathbf{V}_0^{-1} - \mathbf{V}_1^{-1}\right)\mathbf{x} \underset{D=0}{\overset{D=1}{\gtrless}} \mu.$$

Example 2.10 Figure 2.7 shows the ML decision boundary for 2D Gaussian observations with

$$\mathbf{m}_0 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \mathbf{V}_0 = \begin{pmatrix} 0.62 & -0.22 \\ -0.22 & 0.37 \end{pmatrix},$$

and

$$\mathbf{m}_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \mathbf{V}_1 = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}.$$

The region $\mathscr{X}_0$ is given by the interior of the ellipse. Moreover, since the variance of the observations in every direction is larger under hypothesis $h = 1$, points further away from the origin should be assigned $d = 1$.

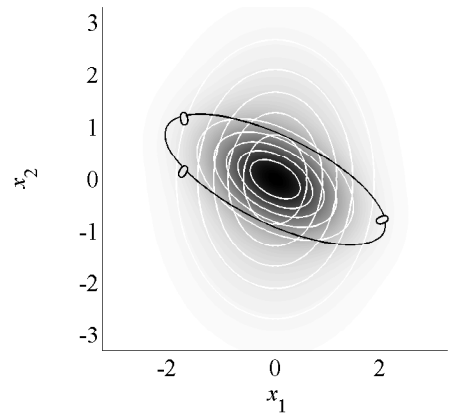


Fig. 2.7 Elliptic decision boundary for a 2D Gaussian problem with zero means.

Example 2.11 Figure 2.8 shows the ML decision boundary for 2D Gaussian observations with

$$\mathbf{m}_0 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \mathbf{V}_0 = \begin{pmatrix} 0.33 & 0.39 \\ 0.39 & 0.77 \end{pmatrix}$$

and

$$\mathbf{m}_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \mathbf{V}_1 = \begin{pmatrix} 0.39 & -0.19 \\ -0.19 & 0.16 \end{pmatrix}.$$

In this example, the variance under hypothesis $h = 1$ is larger only along dimension 1, whereas it is smaller along dimension 2. Hence, as a consequence, the boundary is a hyperbola.

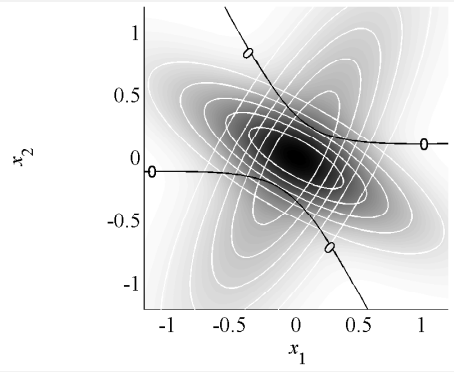


Fig. 2.8 Hyperbolic decision boundary for a 2D Gaussian problem with zero means.

2.5 Problems

2.1 Consider the decision problem with three hypotheses given by the observation $\mathbf{x} = (x_1, x_2) \in [0,1]^2$ and likelihoods:

$$p_{\mathbf{X}|H}(\mathbf{x}|0) = 2(1-x_1) \tag{2.16}$$

$$p_{\mathbf{X}|H}(\mathbf{x}|1) = 2x_1 \tag{2.17}$$

$$p_{\mathbf{X}|H}(\mathbf{x}|2) = 2x_2 \tag{2.18}$$

a) Determine the ML (Maximum Likelihood) classifier
b) Represent the decision regions.

2.2 Consider the decision problem given by the observation $x \in [0,1]$, likelihoods:

$$p_{X|H}(x|0) = 2(1-x) \tag{2.19}$$

$$p_{X|H}(x|1) = 1 \tag{2.20}$$

and prior probability $P_H(1) = 1/4$.

a) Determine the ML classifier.
b) Determine the MAP (Maximum A Posteriori) classifier.
c) Given that $c_{01} = 2$, $c_{10} = 1$, $c_{11} = c_{00} = 0$, determine the decision-maker of minimum risk.
d) Consider a threshold detector over x in the form:

$$x \underset{D=0}{\overset{D=1}{\gtrless}} \eta \tag{2.21}$$

Calculate the probabilities of false alarm, miss and error, as a function of η.
e) Apply the result to the previous three decision-makers. Verify that the MAP classifier obtains the minimum probability of error.
f) Determine the risk for the previous three decision-makers, using the cost parameters from part (c), and verify that the decision-maker obtained in said part achieves the lowest risk.

2.3 Consider a one-dimensional binary decision problem with equiprobable hypotheses, defined by the likelihoods:

$$p_{X|H}(x|0) = \frac{1}{6}, \quad |x| \leq 3,$$

$$p_{X|H}(x|1) = \frac{3}{2}x^2, \quad |x| \leq 1$$

a) Determine the ML (Maximum Likelihood) classifier.
b) Determine the values of P_{FA} (false alarm probability), P_{M} (miss probability), and P_{e} (error probability) for the above classifier.

2.4 Consider the decision problem defined by the observation $x \geq 0$ and likelihoods:

$$p_{X|H}(x|0) = \exp(-x); \tag{2.22}$$

$$p_{X|H}(x|1) = 2\exp(-2x); \tag{2.23}$$

a) Determine the LRT (Likelihood Ratio Test) decision rule.
b) Determine the ROC (Receiver Operating Characteristic).

2.5 Consider a binary decision problem where the likelihood for hypothesis $H = 0$ is uniform over the interval $0 < x < 1$, while $p_{X|H}(x|1) = 2x$, $0 < x < 1$.

a) Obtain the general expression for a Likelihood Ratio Test with parameter η. Graphically represent both likelihoods on the same axes, indicating the decision regions for the ML case.
b) Obtain the analytic expression of the ROC curve for the LRT. Plot this curve indicating the operating points for the ML classifier and the Neyman-Pearson detector with parameter $\alpha = 0.1$ (i.e., $P_{\text{FA}} \leq \alpha = 0.1$).

2.6 Company E manufactures 10,000 units of a product daily. It has been estimated that:

- The sale of a unit in good condition nets a profit of 3 Euros.
- Placing a defective unit on the market causes (on average) a loss of 81 Euros.
- The withdrawal of a unit (whether defective or not) results in a loss of 1 Euro.

An automatic inspection system is available that obtains, for each unit, an observation x_1. Defining $H = 0$ as the hypothesis "the unit is not defective" and $H = 1$ as "the unit is defective",

$$p_{X_1|H}(x_1|0) = \exp(-x_1)u(x_1) \tag{2.24}$$

$$p_{X_1|H}(x_1|1) = \lambda_1 \exp(-\lambda_1 x_1)u(x_1) \tag{2.25}$$

where $\lambda_1 = 1/2$. Additionally, the assembly line produces, on average, one defective unit for every 100 non-defective ones.

The aim is to incorporate an automatic mechanism for withdrawing defective units based on the observation of x_1.

a) Design the detector that provides Company E with the highest expected profit.
b) Determine the maximum expected daily profit that can be achieved.
c) A company offers Company E an innovative inspection device that provides, for each product, in addition to x_1, a new observation x_2, statistically independent from x_1, such that

$$p_{X_2|H}(x_2|0) = \exp(-x_2)u(x_2) \tag{2.26}$$

$$p_{X_2|H}(x_2|1) = \lambda_2 \exp(-\lambda_2 x_2)u(x_2) \tag{2.27}$$

where $\lambda_2 = 1/4$. The cost of this machine is 6000 Euros. Determine an expression for the average time it would take Company E to amortize this machine.

Chapter 3
Stochastic Processes

[1]

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[1] **Note**: Parts of this chapter compiles and adapts material from other sources. Secs 3.1-3.4 are an adaptation taken from [1] (Creative Commons license BY-NC-SA 4.0). The presentation of Secs. 3.5.6 and 3.5.7 follows [2] (see Sec. 3.3, in Spanish). This document is published under license BY-NC-SA 4.0

This chapter introduces the concept of stochastic process and analyzes some important examples.

3.1 Introduction

A stochastic process (SP) is a collection of random variables $\{X_n\}$ indexed by time, n. Depending on the nature of the time variable, there are two types of SP:

- **Continuous-time** SP: when the time variable is a real number ($n \in \mathbb{R}$)
- **Discrete-time** SP: when the time variable is an integer ($n \in \mathbb{Z}$).

This chapter discusses discrete-time processes only. Thus, unless contrary stated, all SP will be assumed to be discrete.

Depending on the range of the time variable, two types of SP will be considered:

- **One-sided** SP: $\{X_n, n \geq 0\}$. A one-sided SP is a sequence of random variables.
- **Two-sided** SP: $\{X_n, n \in \mathbb{Z}\}$, the SP is defined for all integer times.

Example 3.1 Consider the one-sided stochastic process X_n such that, for all n, $X_n \in \{-n, n\}$ with $P\{X_n = n\} = P\{X_n = -n\} = \frac{1}{2}$ and all variables in the collection are mutually independent.

A realization of all random variables in an SP defines a signal. For this reason, we can define stochastic processes in the following alternative but equivalent form:

Definition 3.1 (Stochastic Process)
A discrete-time stochastic process is a probability measure over a space of (one-sided or two-sided) discrete-time signals.

Example 3.2 Consider the one-sided stochastic process X_n such that

$$X_n = n, \quad \text{for all } n \geq 0 \tag{3.1}$$

or

$$X_n = -n, \quad \text{for all } n \geq 0 \tag{3.2}$$

with equal probabilities. Note that the universe set of this SP if finite, containing only two signals.

3.1.1 Characterization of a stochastic process

In a discrete-time SP, the time variable is discrete, but the random variables defining the SP can be continuous or discrete. In general, a SP is completely characterized when we can determine the joint distribution $P(X_{n_0}, X_{n_1}, \ldots, X_{n_{M-1}})$ for any subset of random variables from the process. However, for many SP, the computation of these distributions is often

unfeasible. We will pay attention to processes satisfying several properties that simplify the characterization.

3.2 Independent processes

Definition 3.2 (Independent process) A stochastic process X_n is statistically independent (or, simply, independent) if all its random variables are statistically independent, that is, for any integer $M > 0$ and any subset of M random variables $X_{n_0}, X_{n_1}, \ldots, X_{n_{M-1}}$ with $n_0 < n_1, \ldots < n_{M-1}$ we have

$$P(X_{n_0}, X_{n_1}, \ldots, X_{n_{M-1}}) = \prod_{k=0}^{M-1} P(X_{n_k}) \tag{3.3}$$

Note: A quick explanation on the mathematical notation. Eq. (3.3) should be interpreted as: the joint distribution of the set of M random variables $X_{n_0}, X_{n_1}, \ldots, X_{n_{M-1}}$ is the product of the marginal distribution of each variable. Thus, for instance, the cumulative distribution functions (CDFs) get factorized as follows:

$$F_{X_{n_0}, X_{n_1}, \ldots, X_{n_{M-1}}}(x_{n_0}, x_{n_1}, \ldots, x_{n_{M-1}}) = \prod_{k=0}^{M-1} F_{X_{n_k}}(x_{n_k}) \tag{3.4}$$

for any $x_{n_0}, x_{n_1}, \ldots, x_{n_{M-1}}$ in the respective domains of the variables. If all the random variables in the process are continuous, Eq. (3.4) is equivalent to the factorization of the density functions

$$p_{X_{n_0}, X_{n_1}, \ldots, X_{n_{M-1}}}(x_{n_0}, x_{n_1}, \ldots, x_{n_{M-1}}) = \prod_{k=0}^{M-1} p_{X_{n_k}}(x_{n_k}) \tag{3.5}$$

Also, if the random variables are discrete, this is equivalent to the factorization of the probability mass functions

$$P_{X_{n_0}, X_{n_1}, \ldots, X_{n_{M-1}}}(x_{n_0}, x_{n_1}, \ldots, x_{n_{M-1}}) = \prod_{k=0}^{M-1} P_{X_{n_k}}(x_{n_k}) \tag{3.6}$$

In general, we will use formulations like (3.3) instead of (3.4), (3.5), or (3.6) to express relationships between distributions, because it is simpler and encompasses all of them.

A particularly interesting case arises when all variables in the SP follow the same distribution.

Definition 3.3 (IID process)
A stochastic process X_n is independent and identically distributed, or IID, if it is independent and all variables in the process follow the same distribution, i.e. $P(X_n) = P(X_m)$. This implies that the CDFs re also identical, i.e., $F_{X_n}(x) = F_{X_m}(x)$, for all x and any n, m in the time domain of the process.

One of the simplest IID processes is the Bernoulli process.

Example 3.3 A Bernoulli process X_n with parameter p, denoted as $\mathscr{B}(p)$ is an IID process given by Bernoulli random variables:

$$P_{X_n}(x) = p^x(1-p)^{1-x}, \qquad x \in \{0,1\} \tag{3.7}$$

Note that Eq. (3.7) is equivalent to stating that $P\{X_n = 1\} = p$ and $P\{X_n = 0\} = 1-p$. Bernoulli and other IID processes are commonly used as building blocks of other, more complex, processes.

Example 3.4 Figure 3.1 shows 4 realizations of the independent stochastic process X_n, where each sample of the process follows a Gaussian distribution with mean and standard deviation equal to $4\sin(0.05\pi n)$.

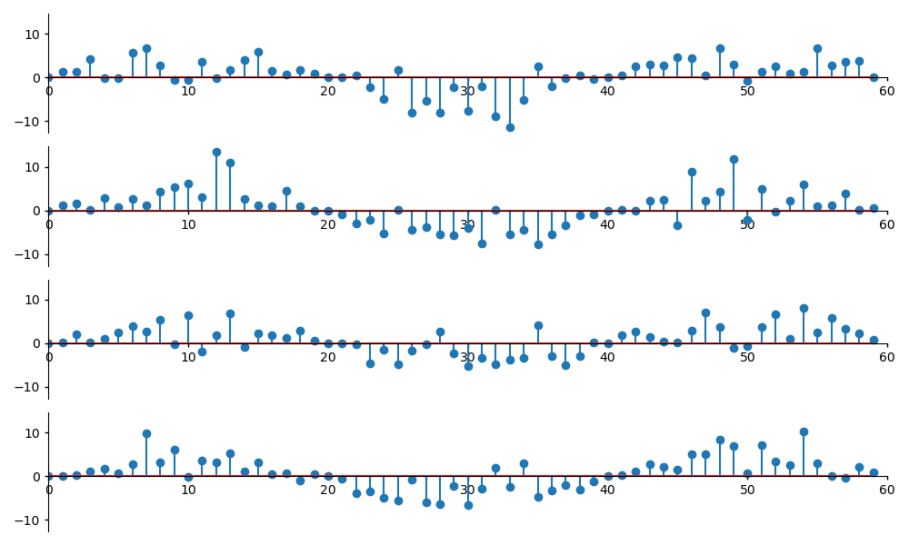


Fig. 3.1 4 realizations of process X_n

3.3 Random walks

Definition 3.4 (Simple random walk (SRW)) Let $B_n \in \{-1,1\}$ be a one-sided IID process with $P\{B_n = 1\} = \frac{1}{2}$. A simple random walk $\{X_n\}$ is defined by $X_0 = 0$ and

$$X_n = \sum_{k=1}^{n} B_k, \qquad n > 0 \tag{3.8}$$

Note that we can express the random variables in a SRW recursively as

$$X_n = \sum_{k=1}^{n-1} B_k + B_n = X_{n-1} + B_n, \qquad n > 0 \tag{3.9}$$

Thus, a SRW is not an independent process. It is not an identically distributed process either, since the domain of the random variables in the process grows with n.

$$-n \leq X_n \leq n \tag{3.10}$$

Fig. 3.2 shows several realizations of the SRW. Note that the realizations are quite far from the bounds. This is because the probability that X_n is close to n or $-n$ vanishes for large n.

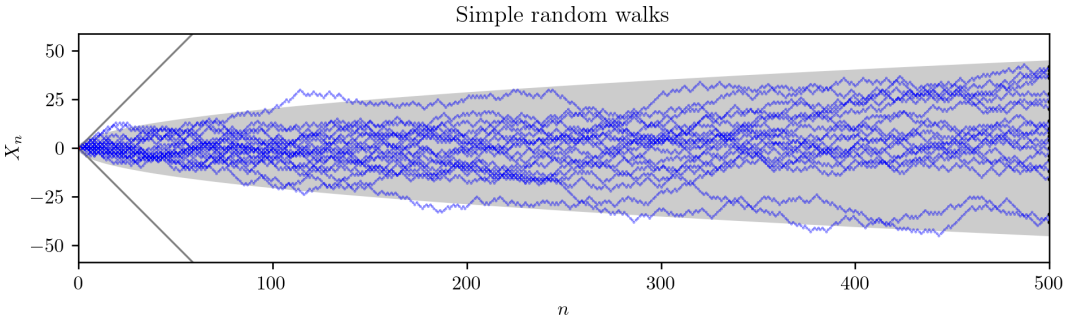


Fig. 3.2 20 realizations of a simple random walk. The straight lines represent the boundaries of the region containing all possible trajectories. The shaded area covers two standard deviations: all trajectories tend to be inside the gray area around 95% of the time.

The mean of the process is given by

$$\mathbb{E}\{X_n\} = \sum_{k=0}^{n} \mathbb{E}\{B_k\} = 0 \tag{3.11}$$

and, since B_n is IID, the variance is given by

$$\text{var}\{X_n\} = \sum_{k=1}^{n} \text{var}\{B_k\} = n \tag{3.12}$$

Therefore, the standard deviation is $\sqrt{n}$ and we can expect from the process to lie between the limits $-\sqrt{n}$ and $\sqrt{n}$ most of the time. This can be expressed more precisely by noting that, since X_n is the sum of n independent random variables with zero-mean and bounded variance and, as a consequence of the central limit theorem, X_n converges in distribution to a zero-mean Gaussian random variable with variance n. This implies that, after large n, we can expect that $|X_n| < 2\sqrt{n}$ approximately 95% of time.

3.3.1 Properties of random walks

- **Independent increment** For all $0 \leq n_0 \leq n_1 \leq n_2 \leq n_3$, the random variables $X_{n_1} - X_{n_0}$ and $X_{n_3} - X_{n_2}$ are mutually independent
- **Time invariance** For all $h \geq 1$, the SP $Z_n = X_{n+h} - X_h$ is a simple random walk.

3.3.2 Stopping times

The SRW can be used as a model of the total winning and losses of a fair game. Assume that we play a game based on repeated trials of flipping a coin. Every time the result is 'heads', we win 1 point while we loose 1 point otherwise. By taking $B_n = 1$ if the n-th toss is 'heads', process $X_n = X_{n-1} + B_n$ is an SRW that accounts for the evolution of the total score with time.

Since the SRW is zero mean, we cannot expect a positive gain after an arbitrary number of steps. One may wonder if there is a way to ensure a positive score by choosing the appropriate stopping time. Let's assume we set an upper and lower limits, $U > 0$ and $-L < 0$, respectively, such that, when $X_n = U$ or $X_n = -L$, we stop betting. By choosing L, we set the maximum loss we can tolerate, by choosing U, we set the gains that will make us stop playing.

We define *stopping time*, N as the first time where X_N reaches U or $-L$. Note that N is a random variable, because it is a function of the random variables in the SP:

$$N = \min\{n \mid X_n = U \text{ or } X_n = -L\} \tag{3.13}$$

Is there a way to select U and L in order to expect a positive gain? To answer this question, let us compute the probability of stopping at level U, that is $P\{X_N = U|X_0 = 0\}$. Thus, defining function

$$f(i) = P\{X_N = U|X_0 = i\}$$

our goal is to compute $f(0)$.

Applying the total probability theorem,

$$\begin{aligned} f(i) =& P\{X_N = U|X_0 = i\} \\ =& P\{X_1 = i+1|X_0 = i\}P\{X_N = U|X_0 = i, X_1 = i+1\} \\ &+ P\{X_1 = i-1|X_0 = i\}P\{X_N = U|X_0 = i, X_1 = i-1\} \\ =& \frac{1}{2}P\{X_N = U|X_1 = i+1\} + \frac{1}{2}P\{X_N = U|X_1 = i-1\} \\ =& \frac{1}{2}f(i+1) + \frac{1}{2}f(i-1) \end{aligned} \tag{3.14}$$

Noting, also, that $f(U) = 1$, and $f(-L) = 0$, we can apply standard methods for solving linear difference equations to find the unique solution

$$f(i) = \frac{L+i}{L+U} \tag{3.15}$$

so that

$$f(0) = P\{X_N = U|X_0 = 0\} = \frac{L}{L+U} \tag{3.16}$$

Thus, the gambler can increase the probability of scoring U points by increasing L. However, this implies a higher risk to suffer large temporal losses. The expected gain becomes:

$$\mathbb{E}\{X_N\} = Uf(0) - L(1-f(0)) = 0 \tag{3.17}$$

Thus, there is no way to set the bounds U and L in order to expect a positive gain.

3.4 Markov Chains

One important property of the simple random walk is that the effect of the past on the future is summarized only by the current state, rather than the whole history of the process. In other words, the distribution of X_{n+1} depends only on the value of X_n and not on the whole set of values of $X_0, X_1, \ldots, X_n$. A stochastic process with such property is called a **Markov chain**.

Definition 3.5 (Markov chain) Let X_n be a one-sided discrete-time SP that, for each n, takes values in some discrete set $\mathscr{S}$, which we call the **state space**. We say that X_n is a Markov chain if

$$P(X_{n+1}|X_n, X_{n-1}, \ldots, X_0) = P(X_{n+1}|X_n) \tag{3.18}$$

for all $n > 0$.

Note that this definition can be extended to two-sided SP, by conditioning over $\{X_k, k \leq n\}$ in the left side of (3.18).

We will say that an MC is **finite** if $\mathscr{S}$ is a finite set. In this case, we usually take the elements of $\mathscr{S}$ as consecutive integers. For instance, for a state space with m elements, we will take $\mathscr{S} = \{0, 1, \ldots, m-1\}$ or $\mathscr{S} = \{1, 2, \ldots, m\}$ under our convenience.

Note that the SRW is a one-sided MC, but it is not finite.

Any finite MC can be described in terms of the transition probabilities

$$p_{ij} = P\{X_{n+1} = j | X_n = i\} \tag{3.19}$$

All the elements of a MC model can be encoded in a **transition probability matrix**

$$\mathbf{P} = \begin{pmatrix} p_{00} & p_{01} & \cdots & p_{0,m-1} \\ p_{10} & p_{11} & \cdots & p_{1,m-1} \\ \vdots & \vdots & \ddots & \vdots \\ p_{m-1,0} & p_{m-1,1} & \cdots & p_{m-1,m-1} \end{pmatrix} \tag{3.20}$$

Using (3.19), it is easy to see that

$$\sum_{j \in \mathscr{S}} p_{ij} = 1 \tag{3.21}$$

thus, all rows in $\mathbf{P}$ sum up to one. For this reason, we say that $\mathbf{P}$ is a **right-stochastic** matrix. In matrix form, this property can be expressed as

$$\mathbf{P}\mathbb{1}_m = \mathbb{1}_m \tag{3.22}$$

where $\mathbb{1}_m$ is an all-ones vector with dimension m.

Note that, in general, the transition probabilities p_{ij} (and, thus, the transition matrix) may change with time. However, the case of constant probabilities is particularly interesting.

Definition 3.6 (Homogeneous Markov Chain) A markov chain X_n is homogeneous if the transition probabilities do not depend on time, that is.

$$P\{X_{n+1} = j | X_n = i\} = P\{X_{m+1} = j | X_m = i\}, \qquad i \in \mathscr{S}, \quad j \in \mathscr{S} \tag{3.23}$$

for any m and n in the time domain of the process.

Example 3.5 (Two-state machine) A machine can be either working or broken on a given day. If it is working, it will break down in the next day with probability 0.01, and will continue working with probability 0.99. If it breaks down on a given day, it will be repaired and be working in the next day with probability 0.8, and will continue to be broken down with probability 0.2. We can model this machine by a Markov chain with two states: 0 ("working"), and 1 ("broken down"). The transition probability matrix is given by

$$\begin{pmatrix} 0.99 & 0.01 \\ 0.8 & 0.2 \end{pmatrix}$$

Example 3.6 (SRW) A simple random walk is an MC. However, since its sample space is infinity, it cannot be described by a transition probability matrix.

3.4.1 Transition graph

The transition probability matrix of a given homogeneous MC can be represented by a directed graph, whose nodes are the states, and and edges from node i to node j represents a transition from state i to state j with nonzero probability. The weight of the edge is the transition probability.

Example 3.7 The transition graph of an MC given by state space $\mathscr{S} = \{0, 1, 2\}$ and transition probability matrix

$$\mathbf{P} = \begin{pmatrix} 0.4 & 0.2 & 0.4 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{3.24}$$

is shown in Figure 3.3.

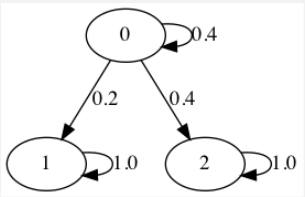


Fig. 3.3 The transition graph for the markov chain given by (3.24)

3.4.2 State probabilities

The transition probability matrix is useful to compute any probabilities about the state of the process at any given time. For instance, the n-th step transition probabilities, defined as $r_{ij}(n) = P\{X_n = j | X_0 = i\}$ can be computed recursively, by applying the total probability theorem, as

$$r_{ij}(n) = \sum_{k \in \mathscr{S}} P\{X_n = j | X_{n-1} = k, X_0 = i\} P\{X_{n-1} = k | X_0 = i\} = \sum_{k \in \mathscr{S}} r_{ik}(n-1) p_{kj}, \qquad n > 1 \tag{3.25}$$

Note that $r_{ij}(1) = p_{ij}$.

Defining the n-step transition probability matrix $\mathbf{R}_n$ as the matrix with components $r_{ij}(n)$, the recurrent relation (3.25) can be written in matrix form as

$$\mathbf{R}_n = \mathbf{R}_{n-1} \mathbf{P} \tag{3.26}$$

and, noting that $\mathbf{R}_1 = \mathbf{P}$, we get

$$\mathbf{R}_n = \mathbf{P}^n \tag{3.27}$$

Using these relations we can compute the state probabilities at any given time, given the initial state or the initial state probabilities.

For instance, let $\mathbf{q}_n$ be the state probability vector at time n, that is

$$q_{n,i} = P\{X_n = i\}, \quad n \geq 0, \quad i \in \mathscr{S} \tag{3.28}$$

then we can express the state probability vector at time n as a function of the initial state probabilities using (3.27) as

$$\begin{aligned} q_{n,j} = P\{X_n = j\} &= \sum_{i \in \mathscr{S}} P\{X_n = j | X_0 = i\} P\{X_0 = i\} \\ &= \sum_{i \in \mathscr{S}} r_{ij}(n) P\{X_0 = i\} \end{aligned} \tag{3.29}$$

which, in vector form, can be computed as

$$\mathbf{q}_n = \mathbf{R}_n^\intercal \mathbf{q}_0 = (\mathbf{P}^\intercal)^n \mathbf{q}_0 \tag{3.30}$$

3.4.3 Stationary distribution

Definition 3.7 (Stationary distribution) A stationary distribution of an MC is a probability distribution over the state space $\mathscr{S}$ (where $P\{X_0 = j\} = \pi_j$) such that

$$\mathbf{P}^\intercal \boldsymbol{\pi} = \boldsymbol{\pi} \tag{3.31}$$

Example 3.8 Let $\mathscr{S} = \mathbb{Z}_m$ and $X_0 = 0$. Consider the MC X_n such that $X_{n+1} = (X_n + 1) \mod m$ with probability $\frac{1}{2}$ and $X_{n+1} = (X_n - 1) \mod m$ with probability $\frac{1}{2}$.

The stationary distribution of this MC is $\pi_i = \frac{1}{m}$.

Note that the vector $\boldsymbol{\pi} = (\pi_1, \pi_2, \ldots, \pi_m)^\intercal$ is an eigenvector of $\mathbf{P}^\intercal$ with eigenvalue 1. Hence the following theorem can be deduced from the Perron-Frobenius theorem.

Theorem 3.1 *If $p_{ij} > 0$ for all $i, j \in \mathscr{S}$, then there exists a unique stationary distribution of the system.*

Moreover, if $\mathbf{r}_i(n)$ is the state conditional probability vector with components $r_{ij}(n)$, $j \in \mathscr{S}$ (i.e., the values in the i-th row of $\mathbf{R}_n$),

$$\lim_{n\to\infty} \mathbf{r}_i(n) = \boldsymbol{\pi}, \qquad \forall i, j \in \mathscr{S} \tag{3.32}$$

A corresponding theorem is not true if we consider infinite state spaces.

Finite Markov chains can be shown to have at least one stationary distribution. However, if at least one of the transition probabilities is zero, the behavior of the process may depend on the topology of the transition graph:

- The stationary distribution may be not unique.
- The limiting distributions $\lim_{n\to\infty} \mathbf{r}_i(n)$ may depend on i.

Example 3.9 The MC in example 3.7 has at least two limiting distributions: $\boldsymbol{\pi}_A = (0,1,0)^\intercal$ and $\boldsymbol{\pi}_B = (0,0,1)^\intercal$. Also, any convex combination of them $\boldsymbol{\pi} = \alpha\boldsymbol{\pi}_A + (1-\alpha)\boldsymbol{\pi}_B$, with $0 \leq \alpha \leq 1$ is a stationary distribution.

The limiting distributions depend on the initial state: is is easy to verify that

$$\lim_{n\to\infty} \mathbf{r}_2(n) = (0,0,1) \tag{3.33}$$

$$\lim_{n\to\infty} \mathbf{r}_1(n) = (0,1,0) \tag{3.34}$$

$$\lim_{n\to\infty} \mathbf{r}_0(n) = \left(0, \frac{1}{3}, \frac{2}{3}\right) \tag{3.35}$$

3.4.4 Computing the stationary distribution

In general, the stationary distributions are all the solutions of the set of equations (3.31) with the probabilistic constraint $\mathbb{1}^\intercal \boldsymbol{\pi} = 1$, that is

$$\begin{pmatrix} \mathbf{P}^\intercal - \mathbf{I} \\ \mathbb{1}^\intercal \end{pmatrix} \boldsymbol{\pi} = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix} \tag{3.36}$$

The system of linear equations given by (3.36) is consistent (it has at least one solution) but overdetermined, because the left matrix has $m+1$ rows but only m columns. In fact, using (3.22), it is easy to see that

$$\mathbb{1}_m^\intercal (\mathbf{P}^\intercal - \mathbf{I}) = \mathbf{0} \tag{3.37}$$

so that the rank of $\mathbf{P}^\intercal - \mathbf{I}$ is at most $m-1$. Thus, we can remove one of the rows in the matrix (any one of them) to solve the system.

3.5 Stationary Processes

[2]

Stationarity refers to the time invariance of some statistical properties of a stochastic process. We have already seen some forms of time invariance in the previous sections:

- The homogeneity of Markov chain, X_n, which implies the time invariance of the conditional probabilities: $P(X_n|X_m) = P(X_{n+k}|X_{m+k})$, for any $k \geq 0$, $n \geq m \geq 0$.
- The time invariance of IID processes, which implies the invariance of the single-variable marginal distributions, $P(X_{n+k}) = P(X_n)$ for any $h > 0$
- The time invariance property of a SRW, which implies that $P(X_{n+h} - X_h) = P(X_n)$.

In this section we introduce two forms of stationarity which are particularly interesting in statistical signal processing applications. In its more restrictive form, stationarity implies the invariance of all joint distributions.

3.5.1 Strict-Sense Stationarity

Definition 3.8 (Strict-Sense Stationarity) A stochastic process is said to be Strict-Sense Stationary (or simply SSS) if all joint distributions are invariant to a time shift, that is, for any $N > 0$, any sampling times $n_0, \ldots, n_{N-1}$ (in the time domain of the process) and any time shift $m > 0$,

$$P(X_{n_0}, X_{n_1}, \ldots, X_{n_{N-1}}) = P(X_{n_0+m}, X_{n_1+m}, \ldots, X_{n_{N-1}+m}) \tag{3.38}$$

Thus, if a stochastic process X_n is SSS, the process $Y_n = X_{n+m}$ defines the same probability measure over the signal space.

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Example 3.10 (Bernoulli process)
A Bernoulli process, $\mathscr{B}(p)$ given by (3.7) is IID, thus, its joint probability mass function has the form

$$P_{X_{n_0},\ldots,X_{n_{N-1}}}(x_0,\ldots,x_{N-1}) = \prod_{i=0}^{N-1} p^{x_i}(1-p)^{1-x_i} = p^{\Sigma_{i=0}^{N-1} x_i}(1-p)^{N-\Sigma_{i=0}^{N-1} x_i} \tag{3.39}$$

which does not depend on $n_0,\ldots,n_{N-1}$. Thus, the Bernoulli process is SSS.

In general, all IID processes are SSS. On the contrary, independent but not identically distributed processes are not SSS. This is the case of non-constant deterministic process, as the following example shows

Example 3.11 (Deterministic process)
Consider the deterministic process given by $X_n = 2n$ (with probability 1), that is, for each time n, X_n is a discrete random variable with PMF

$$P_{X_n}(k) = \delta[k-2n] \tag{3.40}$$

Since $P_{X_n}(k)$ depends on n, X_n is not SSS.

In general, strict stationarity is difficult to check and overly restrictive, and weaker forms of stationarity are more interesting. In the following, we will focus in the invariance of first and second order statistics[3]

3.5.2 First and Second-Order Statistics

In order to define wide-sense stationarity, we will first define some first and second-order statistics of stochastic processes. We will assume, in general, stochastic processes which can take complex values:

Definition 3.9 (First and second order statistics) Given stochastic process X_n we define the following statistics:

- **Mean**:
$$\mu_X[n] = \mathbb{E}\{X_n\} \tag{3.41}$$
- **Autocorrelation**:
$$r_X[n_1,n_2] = \mathbb{E}\{X_{n_1}X_{n_2}^*\} \tag{3.42}$$
- **Cross correlation**
$$r_{XY}[n_1,n_2] = \mathbb{E}\{X_{n_1}Y_{n_2}^*\} \tag{3.43}$$

Given stochastic processes X_n and Y_n with means $\mu_X[n]$ and $\mu_Y[n]$, we define the following joint statistics:

[3] In this document, first and second order statistics refer to the expected value of polynomial functions of a random variable up to degree 2, that, it, the mean, the correlation and other related statistics, like the covariance. This should not be confused with the so called "order statistics".

- **Cross covariance** between X_n and Y_n

$$c_{XY}[n_1,n_2] = \mathbb{E}\{(X_{n_1} - \mu_X[n_1])(Y_{n_2} - \mu_Y[n_2])^*\} \tag{3.44}$$

- **Autocovariance**

$$c_X[n_1,n_2] = \mathbb{E}\{(X_{n_1} - \mu_X[n_1])(X_{n_2} - \mu_X[n_2])^*\} \tag{3.45}$$

Note that $c_X[n,n]$ is the variance of X_n. It is easy to verify that the autocovariance and the autocorrelation are related by

$$c_X[n_1,n_2] = r_X[n_1,n_2] - \mu_X[n_1]\mu_X^*[n_2] \tag{3.46}$$

Example 3.12 (Bernoulli process)
For a Bernoulli process, $\mathscr{B}(p)$, using (3.7) we get

$$\mu_X[n] = \mathbb{E}\{X_n\} = p, \tag{3.47}$$

also

$$\begin{aligned} r_X[n_1,n_2] = \mathbb{E}\{X_{n_1}X_{n_2}\} &= \begin{bmatrix} \mathbb{E}\{X_{n_1}^2\} & \text{if } n_1 = n_2 \\ \mathbb{E}\{X_{n_1}\}\mathbb{E}\{X_{n_2}\} & \text{if } n_1 \neq n_2 \end{bmatrix} = \begin{bmatrix} p & \text{if } n_1 = n_2 \\ p^2 & \text{if } n_1 \neq n_2 \end{bmatrix} \\ &= p^2 + p(1-p)\delta[n_1 - n_2] \end{aligned} \tag{3.48}$$

and $c_X[n_1,n_2] = r_X[n_1,n_2]$.

Example 3.13 (Deterministic process)
Consider the deterministic process in Example 3.11, we have

$$\mu_X[n] = \mathbb{E}\{X_n\} = \mathbb{E}\{2n\} = 2n, \tag{3.49}$$

$$r_X[n_1,n_2] = \mathbb{E}\{X_{n_1}X_{n_2}\} = \mathbb{E}\{4n_1n_2\} = 4n_1n_2 \tag{3.50}$$

and, using (3.46)

$$c_X[n_1,n_2] = 0 \tag{3.51}$$

Example 3.14 Consider the stochastic process X_n given by

$$X_n = (1+n^2)Y \tag{3.52}$$

where Y is a real Gaussian random variable with zero mean and unit variance.
The mean and the autocorrelation of the process can be easily calculated as

$$\mu_X[n] = \mathbb{E}\{Y\} \cdot (1+n^2) = 0 \tag{3.53}$$

$$r_X(n,m) = \mathbb{E}\{X_nX_m\} = (1+n^2)\cdot(1+m^2)\cdot\mathbb{E}\left\{Y^2\right\} = (1+n^2)\cdot(1+m^2) \tag{3.54}$$

Since X_m has zero mean, the autocovariance is equal to the autocorrelation, and the variance is

$$\sigma_X^2[n] = r_n[n,n] = \left(1+n^2\right)^2 \tag{3.55}$$

Example 3.15 (Simple Random Walk)
For the SRW X_n given by (3.8), the value of the process at any time n depends on the value of a set of n random variables $(Y_1, \ldots Y_n)$. We have seen that the process is zero-mean with variance n. The autocorrelation (equal to the autocovariance) can be computed as

$$\begin{aligned} r_X[n_1,n_2] = \mathbb{E}\{X_{n_1}X_{n_2}\} &= \sum_{k=1}^{n_1}\sum_{\ell=0}^{n_2}\mathbb{E}\{Y_kY_\ell\} = \sum_{k=1}^{n_1}\sum_{\ell=0}^{n_2}\delta[k-\ell] \\ &= \min(n_1,n_2) \end{aligned} \tag{3.56}$$

3.5.3 Wide-Sense Stationarity

We are ready to define a weaker form of stationarity:

Definition 3.10 (Wide-Sense Stationarity) A two-sided stationary process is Wide-Sense Stationary (or simply WSS) if its mean and autocorrelation functions are invariant to a time shift, that is, for any integers n, n_1, n_2 and any time shift $m \in \mathbb{Z}$,

$$\mu_X[n] = \mu_X[n+m] \tag{3.57}$$

and

$$r_X[n_1,n_2] = r_X[n_1+m, n_2+m] \tag{3.58}$$

Thus, if a process is WSS, its mean is constant and its autocorrelation only depends on the difference between n_1 and n_2. For this reason, we will typically use the notation

$$\mu_X = \mu_X[n] \tag{3.59}$$

$$r_X[n] = r_X[m+n,m] \tag{3.60}$$

Also, combining (3.59) and (3.60) with (3.46), it is straighforward to see that the covariance function is also invariant to a time shift, and we can write

$$c_X[n] = c_X[m+n,m] \tag{3.61}$$

Example 3.16 Let's go back to the examples from the previous section.

- **Bernoulli process**: The Bernoulli process X_n from Example 3.10 is SSS, since their joint probability function, (3.39), does not depend on the time variables. Consequently, it is also WSS, as can be seen by observing that the average in (3.47) is constant and the autocorrelation in (3.48) depends on the time difference only.
- **Deterministic process**: For the deterministic process $X_n = 2n$ in Example 3.11, we have $\mu_X[n] = 2n$, which depends on n. Thus, it is not WSS.

- The process X_n in Example 3.14 has zero mean (thus, the mean is invariant to a time shift), but the autocorrelation $r_X[n_1,n_2]$ (3.54) cannot be expressed as function of the time difference n_1-n_2 only. Thus, the process is not WSS.
- **Simple Random Walk**: The SRW from Example 3.15 is not stationary since its autocorrelation (3.56) is not invariant to a time shift: for any $m>0$

$$\begin{aligned} r_X[n_1+m,n_2+m] &= \min(n_1+m,n_2+m) = \min(n_1,n_2)+m = r_X(n_1,n_2)+m \\ &\neq r_X[n_1,n_2] \end{aligned} \tag{3.62}$$

3.5.3.1 Properties of stationary processes

Theorem 3.2 *The autocorrelation WSS process X_n satisfies the following properties:*

1. ***Hermiticity**: $r_X[n] = r_X^*[-n]$.*
2. *$r_X[0] = \mathbb{E}\{|X_n|^2\}$.*
3. *It is **positive-definite**: for any integer $N>0$ and any coefficients $c_n \in \mathbb{C}, n = 0,\ldots,N-1$,*

$$\sum_{m=0}^{N-1}\sum_{n=0}^{N-1} c_m c_n^* r_X[n-m] \geq 0 \tag{3.63}$$

4. ***Maximality** at the origin: $|r_X[n]| \leq r_X[0]$.*

Proof The hermiticity and property 2 are a direct consequence of the definition. To prove the positive definiteness, note that, for any coefficients c_n, we have

$$0 \leq \mathbb{E}\left\{\left|\sum_{n=0}^{N-1} c_n X_n\right|^2\right\} = \sum_{m=0}^{N-1}\sum_{n=0}^{N-1} c_m c_n^* r_X[n-m] \tag{3.64}$$

The maximality at the origin is a consequence of the positive definiteness: since (3.63) is true for any complex coefficients, we can take $c_0=1$, $c_1=c_2=\ldots=c_{N-2}=0$ to get

$$(1+|c_{N-1}X_{N-1}|^2)r_X[0] + c_{N-1}r_X[N-1] + c_{N-1}^* r_x^*[n] \geq 0 \tag{3.65}$$

and, taking $c_{N-1} = -\frac{r_X^*[N-1]}{|r_X[N-1]|}$ (which implies $|c_{N-1}|^2=1$), we get

$$2r_X[0] - 2\,|r_X[N-1]|| \geq 0 \tag{3.66}$$

Since this is true for any $N>0$, we get $|r_X[n]|| \leq r_X[0]$, for any n, which completes the proof. □

3.5.3.2 WSS and SSS processes

Note that stationarity in the strict sense implies the wide sense stationarity, but the opposite is not generally true: there are WSS processes that are not SSS, as the following example shows:

Example 3.17 Let X_n be a stochastic independent process such that, if n is even,

$$X_n \sim \begin{cases} P\{X_n = 1\} = P\{X_n = -1\} = \frac{1}{2}, & \text{if } n \text{ is even} \\ \mathcal{N}(0,1), & \text{if } n \text{ is odd} \end{cases} \tag{3.67}$$

Since $P(X_n)$ changes from n even to n odd, the process is not SSS. However, since the process has zero mean and unit variance for all n, the autocorrelation is

$$r_X(n_1,n_2) = \delta[n_2 - n_1] \tag{3.68}$$

which depends only on the time difference. Thus, the process is WSS.

Also, note that all IID processes are SSS, with autocovariance

$$c_X[n] = \mathbb{E}\{(X_n - \mu_X)(X_n - \mu_X)^*\} = \sigma_X^2 \delta[n] \tag{3.69}$$

If the process is zero mean, we have

$$r_X[n] = \sigma_X^2 \delta[n] \tag{3.70}$$

In general, a SP whose autocovariance is a delta function is said to be *white*. All zero-mean IID processes are white, but the opposite is not true in general: some white processes may be non IID.

3.5.3.3 Jointly stationary processes.

The two-side stationary processes X_n and Y_n are said to be jointly stationary *in the strict sense* if their joint statistical properties do not vary with a displacement in time, that is, if, for any values of N and N', any value of m and any instants $n_0, \ldots, n_{N-1}$ and $n'_0, \ldots, n'_{N'-1}$,

$$P(X_{n_0}, \ldots, X_{n_{N-1}}, Y_{n'_0}, \ldots, Y_{n'_{N'-1}}) = P(X_{n_0+m}, \ldots, X_{n_{N-1}+m}, Y_{n'_0+m}, \ldots, Y_{n'_{N'-1}+m}) \tag{3.71}$$

Likewise, we will say that X_n and Y_n are jointly wide-sense stationary if both are WSS and for any integer values of n_1, n_2 and m we have

$$r_{XY}[n_1,n_2] = r_{XY}[n_1+m, n_2+m] \tag{3.72}$$

which allows us to describe the cross-correlation function using a single variable

$$r_{XY}[n] = r_{XY}[m+n, m] \tag{3.73}$$

3.5.4 Linear systems with stochastic inputs

Let X_n be a stochastic process that is the input to a linear time-invariant system with impulse response $h[n]$. The output process Y_n is given by

$$Y_n = X_n * h[n] \tag{3.74}$$

In general, the statistical characterization of Y_n is not easy, except in particular cases (such as the Gaussian), but we can find some useful expressions for some first and second order moments.

For example, the mean of the output process is

$$\begin{aligned}\mu_Y[n] =& \mathbb{E}\{Y_n\} = \mathbb{E}\{h[n] * X_n\} = h[n] * \mathbb{E}\{X_n\} \\ =& h[n] * \mu_X[n],\end{aligned} \tag{3.75}$$

the cross-correlation of the input and output is

$$\begin{aligned}r_{YX}[n_1,n_2] =& \mathbb{E}\{Y_{n_1}X^*_{n_2}\} = \mathbb{E}\left\{X^*_{n_2}\sum_{k=-\infty}^{\infty} X_k h[n_1-k]\right\} \\ =& \sum_{k=-\infty}^{\infty} h[n_1-k]r_X[k,n_2] \\ =& h[n_1] * r_X[n_1,n_2]\end{aligned} \tag{3.76}$$

and the autocorrelation of the output is

$$\begin{aligned}r_Y[n_1,n_2] =& \mathbb{E}\{Y_{n_1}Y^*_{n_2}\} = \mathbb{E}\left\{Y_{n_1}\sum_{k=-\infty}^{\infty} h^*[k]X^*_{n_2-k}\right\} \\ =& \sum_{k=-\infty}^{\infty} h^*[k]r_{YX}[n_1,n_2-k] \\ =& r_{YX}[n_1,n_2] * h^*[n_2]\end{aligned} \tag{3.77}$$

Combining (3.76) and (3.77), we get

$$r_Y[n_1,n_2] = h[n_1] * r_X[n_1,n_2] * h^*[n_2] \tag{3.78}$$

In the above equation, there are two convolution operations. We must perform the first on the variable n_1 and the second on the variable n_2. Equations (3.75) and (3.78) show that both the mean and the autocorrelation of the output depend exclusively on the mean and the autocorrelation of the input, respectively, as well as on the impulse response of the system.

3.5.4.1 Stationary processes.

We can use the general expressions for the output mean and the output autocorrelation to derive specific expression for WSS processes. If X_n is stationary, $\mu_X[n] = \mu_X$, and

$$\mu_Y = \mu_X \sum_{n=-\infty}^{\infty} h[n] \tag{3.79}$$

Also, by the properties of the convolution operator, for any $m \in \mathbb{Z}$ we can write

$$\begin{aligned} r_Y[n_1+m, n_2+m] &= h[n_1] * r_X[n_1+m, n_2+m] * h^*[n_2] = h[n_1] * r_X[n_1, n_2] * h^*[n_2] \\ &= r_Y[n_1, n_2] \end{aligned} \tag{3.80}$$

and, therefore Y_n is also stationary. In such case, we can express the autocorrelation as a function of a single variable. By calling $n = n_1 - n_2$, the cross-correlation of the input and output is

$$\begin{aligned} r_{YX}[n_1, n_2] =& h[n_1] * r_X[n_1 - n_2] = \sum_{k=-\infty}^{\infty} h[n_1 - k] r_X[k - n_2] \\ &= \sum_{k=-\infty}^{\infty} h[n_2 + n - k] r_X[k - n_2] = \sum_{k=-\infty}^{\infty} h[n-k] r_X[k] \\ =& h[n] * r_X[n] \end{aligned} \tag{3.81}$$

Finally, the autocorrelation of the output is

$$\begin{aligned} r_Y[n] =& r_Y[n_1, n_2] = r_{YX}[n_1, n_2] * h^*[n_2] \\ &= \sum_{k=-\infty}^{\infty} h^*[k] r_{YX}[n_1 - n_2 + k] = \sum_{k=-\infty}^{\infty} h^*[k] \sum_{\ell=-\infty}^{\infty} h[n + k - \ell] r_X[\ell] \\ &= \sum_{\ell=-\infty}^{\infty} r_X[\ell] \sum_{k=-\infty}^{\infty} h^*[k] h[n + k - \ell] = r_X[n] * h[n] * h^*[-n] \\ =& r_X[n] * r_h[n] \end{aligned} \tag{3.82}$$

where

$$r_h[n] = h[n] * h^*[-n] \tag{3.83}$$

Function $r_h[n]$ is usually called the *deterministic autocorrelation* of $h[n]$.

Example 3.18 Let X_n be a unit variance white process. The stationary process Y_n resulting from passing X_n through a linear and invariant and causal filter, given by the difference equation

$$Y_n = 0.6 \cdot Y_{n-1} + X_n \tag{3.84}$$

is stationary. According to (3.75), since the impulse response of this system is $h[n] = 0.6^n u[n]$, the process mean will be,

$$\mu_Y[n] = h[n] * \mu_X[n] = 0 \tag{3.85}$$

and its autocorrelation

$$r_Y[n] = r_X[n] * h[n] * h[-n] = \frac{0.6^{|n|}}{0.64} \tag{3.86}$$

Example 3.19 (Low-Pass Filtering of White Noise)
Let the input X_n be a zero-mean white noise process with autocorrelation:

$$r_X[n] = \delta[n]$$

This process is passed through an LTI system with impulse response:

$$h[n] = 2^{-n}u[n]$$

From Eq. (3.79), since the input mean μ_X is 0:

$$\mu_Y = 0 \cdot \sum_{n=0}^{\infty} 2^{-n} = 0$$

Also, since the deterministic autocorrelation is

$$r_h[n] = h[n] * h[-n] = \frac{2^{-|n|}}{1-(1/2)^2} = \frac{4}{3} \cdot 2^{-|n|},$$

we can apply Eq. (3.82) to find that the autocorrelation of the output process, Y_n, is

$$\begin{aligned} r_Y[n] &= r_X[n] * r_h[n] = \delta[n] * \left(\frac{4}{3} \cdot 2^{-|n|}\right) \\ &= \frac{4}{3} \cdot 2^{-|n|} \end{aligned}$$

The output variance (the power of the signal) is $r_Y[0] = \frac{4}{3}$. Note that the output samples are correlated; for example, the correlation between adjacent samples ($n = 1$) is $r_Y[1] = \frac{2}{3}$.

3.5.5 Gaussian processes

Definition 3.11 (Gaussian Process) A real SP X_n is a Gaussian Process (GP) if, for any $N > 0$ and any arbitrary set of N time instants $\{n_0, \ldots, n_{N-1}\}$, the joint distribution

$$P(X_{n_0}, X_{n_1}, \ldots, X_{n_{N-1}}) \tag{3.87}$$

is Gaussian.

Gaussian processes have several interesting properties:

1. Since the parameters of the (multidimensional) Gaussian distribution are the mean and the covariance matrix, any GP is completely characterized by its mean and its autocorrelation function.
2. Thus, if a GP is WSS, it is SSS.
3. Since any linear combination of Gaussian random variables is Gaussian, if a GP is the input to a linear time-invariant system, the output is also GP characterized by the output mean and the output autocorrelation driven by Eqs. (3.75) and (3.98).

3.5.6 Power Spectral Density

The power spectral density (PSD) is the fundamental tool used to extend the Fourier analysis of deterministic signals to stochastic processes.

While the realizations (individual outcomes) of a stochastic process are signals, we cannot simply define the Fourier Transform of the process by transforming its realizations. For a stationary process, the realizations are typically signals of finite average power but infinite energy. Consequently, these signals do not satisfy the Dirichlet conditions for the existence of a standard Fourier Transform.

To resolve this, we can compute the Fourier Transform of a truncated process (one limited in time). Let X_n be a stochastic process with mean $\mu_X[n]$ and autocorrelation $r_X[n_1, n_2]$. We define the truncated process $X_{N,n}$ as:

$$X_{N,n} = \begin{cases} X_n, & \text{if } |n| \leq N \\ 0, & \text{if } |n| > N \end{cases} \tag{3.88}$$

By restricting the signal to the interval $[-N, N]$, we create a signal with finite energy. This allows us to perform Fourier Analysis on $X_{N,n}$ and subsequently define the power spectrum by taking the limit as $N \to \infty$.

The Fourier Transform of $X_{N,n}$, given by

$$X_N\left(e^{j\omega}\right) = \sum_{n=-\infty}^{\infty} X_{N,n} e^{-j\omega n} = \sum_{n=-N}^{N} X_n e^{-j\omega n} \tag{3.89}$$

is, in turn, a stochastic process of mean

$$\mathbb{E}\{X_N\left(e^{j\omega}\right)\} = \sum_{n=-N}^{N} \mu_X[n] e^{-j\omega n} \tag{3.90}$$

and squared mean

$$\mathbb{E}\left\{\left|X_N\left(e^{j\omega}\right)\right|^2\right\} = \mathbb{E}\left\{\left|\sum_{n=-N}^{N} X_n e^{-j\omega n}\right|^2\right\} \tag{3.91}$$

If the process is stationary, the mean quadratic value grows indefinitely with N until it becomes infinity, but we can avoid this effect by normalizing (3.91) with respect to the length of the integration interval, $2N$. We define the *power spectral density* or PSD of a stochastic process as the limit of the expression above.

Definition 3.12 (Power Spectral Density) The power spectal density (PSD) of a stochastic process X_n is defined as

$$S_X(e^{j\omega}) = \lim_{N\to\infty} \frac{1}{2N+1} \mathbb{E}\{|X_N(e^{j\omega})|^2\} \tag{3.92}$$

The expression in (3.92) is the basis of some spectral estimation methods that we will discuss in later chapters, but we can find an alternative expression for stationary processes,

that allows to compute the PSD using the autocorrelation, using a classical result from Wiener and Khinchin that we state without proof.

Theorem 3.3 (Wiener-Khinchin theorem)
If X_n is WSS and $r_X[n]$ is absolutely summable,

$$S_X\left(e^{j\omega}\right) = \sum_{n=-\infty}^{\infty} r_X[n]e^{-j\omega n} \tag{3.93}$$

Therefore, the power spectral density of a stationary process X_n is equal to the Fourier Transform of its autocorrelation function and, conversely

$$r_X[n] = \frac{1}{2\pi}\int_{-\infty}^{\infty} S_X(e^{j\omega})e^{j\omega n}d\omega \tag{3.94}$$

Example 3.20 White processes
If X_n is a white WSS process with autocorrelation $r_X[k] = \sigma_X^2\delta[k]$, its power spectral density is constant at all frequencies.

$$S_X(e^{j\omega}) = \sigma_X^2 \tag{3.95}$$

3.5.6.1 Properties of the PSD

The power spectral density has several interesting properties:

- **Non-negativty**: for any $\omega \in \mathbb{R}$,
$$S_X\left(e^{j\omega}\right) \geq 0 \tag{3.96}$$
This is a direct consequence of the definition, as the PSD is the limit of positive functions.
- **Symmetry**: for any real stochastic process, the PSD is even function.
$$S_X\left(e^{j\omega}\right) = S_X\left(e^{-j\omega}\right) \geq 0 \tag{3.97}$$
This is a consequence of the Wiener-Khinchin Theorem, and the symmetry property of the autocorrelation function.

3.5.6.2 Power spectral density at the output of linear systems

It is interesting to analyze the power spectral density of stochastic processes that pass through linear filters. Starting from the expression for the output autocorrelation obtained in (3.78), we can write

$$r_Y[n] = r_X[n] * h[n] * h^*[-n] \tag{3.98}$$

and applying the convolution property to (3.98), we get

$$S_Y\left(e^{j\omega}\right) = S_X\left(e^{j\omega}\right)\left|H\left(e^{j\omega}\right)\right|^2 \tag{3.99}$$

3.5.7 Ergodicity

If the expected value of a random variable X is unknown, it can be estimated as the sample average of K independent realizations $x_0, \ldots, x_{K-1}$

$$\mathbb{E}\{X\} \approx \frac{1}{K} \sum_{k=0}^{K-1} x_k \tag{3.100}$$

This estimation is supported by the (weak and strong) laws of large numbers, that guarantee, under quite general conditions, the convergence of the sample average to the mean as the number of samples goes to infinity.

We can estimate the mean of a stochastic process in the same way, by averaging multiple realizations. However, in many practical applications, this is not possible because only a single sample of the process is available.

If a stochastic process X_n is WSS, the mean is constant, and we can try to estimate it as the average of all samples from a single realization. Thus, we may wonder if time averages of the SP converge to the mean. A process satisfying this property is called mean-ergodic.

Definition 3.13 (Mean-ergodicity)
A WSS process X_n with mean μ_X is mean-ergodic if the time average

$$S_N = \frac{1}{2N+1} \sum_{n=-N}^{N} X_n \tag{3.101}$$

converges in squared mean to the mean, μ_X, that is

$$\lim_{N\to\infty} \mathbb{E}\left\{ |S_N - \mu_X|^2 \right\} = 0 \tag{3.102}$$

Note that S_N is itself a random variable with the same mean as the process

$$\mathbb{E}\{S_N\} = \frac{1}{2N+1} \sum_{n=-N}^{N} \mathbb{E}\{X_n\} = \mu_X \tag{3.103}$$

and the expectation in (3.102) is the variance. Expanding the expression of this variance, it is straightforward to obtain the following condition for ergodicity:

Theorem 3.4 *A WSS process X_n with mean μ is mean-ergodic iff*

$$\lim_{N\to\infty} \frac{1}{(2N+1)^2} \sum_{n=-N}^{N} \sum_{m=-N}^{N} c_X[n-m] = 0 \tag{3.104}$$

Proof Noting that

$$\begin{aligned}\mathbb{E}\{|S_N-\mu_X|^2\} =& \mathbb{E}\left\{\left|\frac{1}{2N+1}\sum_{n=-N}^{N}(X_n-\mu_X)\right|^2\right\} = \\ =& \frac{1}{(2N+1)^2}\sum_{n=-N}^{N}\sum_{m=-N}^{N}\mathbb{E}\{(X_n-\mu_X)(X_m^*-\mu_X^*)\} \\ =& \frac{1}{(2N+1)^2}\sum_{n=-N}^{N}\sum_{m=-N}^{N}c_X[n-m] \end{aligned} \tag{3.105}$$

the proof is completed. □

We can use the above theorem to state a sufficient condition for ergodicity, that is satisfied by many WSS processes of practical interest:

Theorem 3.5 *If the covariance function of a WSS process X_n is absolutely summable, that is*

$$\sum_{n=-\infty}^{\infty}|c_X[n]|<\infty \tag{3.106}$$

the process is mean-ergodic.

Proof Since $c_X[n]$ is absolutely summable, $B=\sum_{n=-\infty}^{\infty}|c_X[n]|$ is finite, and we can use B to upper bound the variance in (3.105) as

$$\left|\mathbb{E}\{|S_N-\mu_X|^2\}\right| \leq \frac{1}{(2N+1)^2}\sum_{n=-N}^{N}\sum_{m=-N}^{N}|c_X[n-m]| \leq \frac{1}{(2N+1)^2}\sum_{n=-N}^{N}B = \frac{B}{2N+1} \tag{3.107}$$

Since the upper bound converges to zero for large N, the process in mean-ergodic. □

Note that all white processes (and, in particular, IID processes) satisfy (3.106). Thus, they are mean-ergodic.

Example 3.21 Consider the stationary process given by the causal recursive filter $X_n = 0.8X_{n-1}+Z_n$, where the input, Z_n, is a Gaussian IID process of unit variance. It is easy to see that the impulse response of the filter is $h[n]=0.8^n\cdot u[n]$, where $u[n]$ is the step function. Thus, X_n is a zero-mean stationary process with autocovariance given by (3.82), which is equal to

$$c_X[n] = \frac{0.8^{|n|}}{0.36} \tag{3.108}$$

Since the autocovariance vanishes for $|k|>1$,the process is mean-ergodic. Figure 3.4 represents the first samples (from $n=0$ to $n=79$) of 5 realizations. The average of 1000 realizations is represented in the lower part. The 80 sample average of each realization is shown on the right margin. As a consequence of stationarity, the average signal in the bottom approaches a constant signal around the mean, which is zero. As a consequence of the ergodicity, the averages of each signal also approaches the mean.

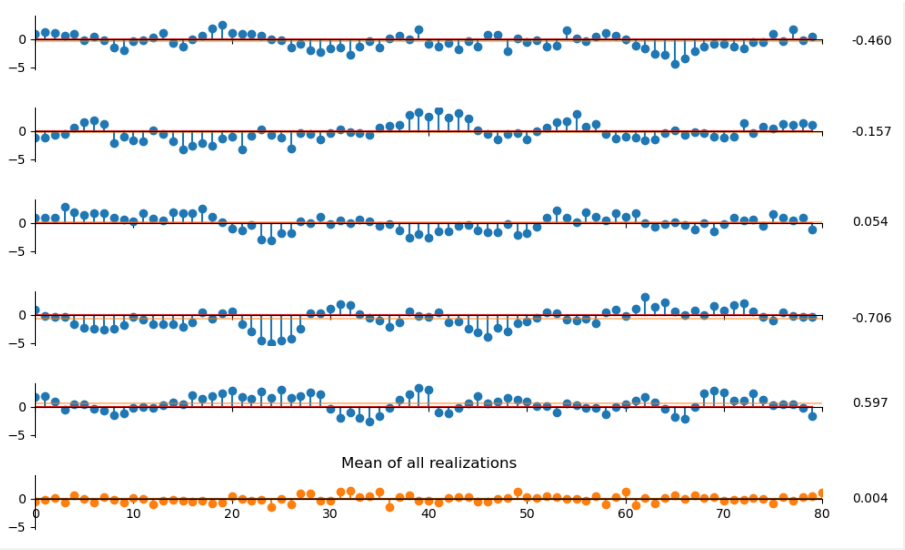


Fig. 3.4 5 realizations of the stochastic process of the example 3.21 (top), the average of 1000 realizations (bottom) and the averages of the represented samples (right). Because of the stationarity, the average signal (at the bottom) approaches a to a zero signal, which is mean of the process. Because of the ergodicity, time averages also approximate the mean.

3.5.7.1 Ergodicity in the autocorrelation

In a similar way, ergodicity is defined in autocorrelation. A WSS process X_n is autocorrelation-ergodic if

$$r_N = \frac{1}{2N+1} \sum_{m=-N}^{N} X_{m+n} X_m^* \tag{3.109}$$

converges to $r_X[n]$ in squared mean. In general, X_n is autocorrelation-ergodic if, for every value of m, the process $Z_n[m] = X_{m+n} X_m^*$ is mean-ergodic.

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Chapter 4
Linear Filtering

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4.1 Introduction

A common challenge in estimation involves determining the coefficients of a linear filter with M coefficients based solely on the observation of its inputs and outputs. This task, along with related ones, falls under the generic term 'linear filtering.' In this chapter, we will demonstrate how the techniques described in Chapter 1 can be applied to design ML, MAP, MAD and MMSE estimators for the coefficients of the aforementioned filter. This will also include the estimation of future filter outputs, provided that the corresponding inputs are known.

4.2 The filtering problem

Assume that a finite impulse response filter (FIR) $s[n]$, with $s[n] = 0$, for n other than $0, 1, \ldots, M-1$ is used to filter a signal[1] $u[n]$. The result is then added to a certain Gaussian noise $\varepsilon[n]$, which is an IID zero-mean stochastic process with variance σ_ε^2 and independent of $s[n]$, giving rise to an observation $x[n]$ (see Figure 4.1). That is, the corresponding entries are

$$\begin{aligned} x[n] &= u[n] * s[n] + \varepsilon[n] \\ &= u[n]s[0] + u[n-1]s[1] + \ldots + u[n-M+1]s[M-1] + \varepsilon[n]. \end{aligned} \tag{4.1}$$

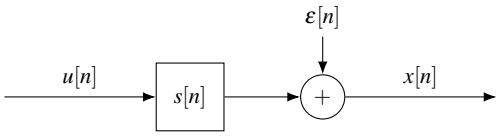


Fig. 4.1 The signal model used in this chapter.

For simplicity, we will assume that the input signal starts at $n = 0$, that is, $u[n] = 0$ for $n < 0$. Also, we will assume that it is known; thus, a satistical model for $u[n]$ is unnecessary for our analysis).

The filtering problem consists in estimating the filter $s[n]$ from the input signal and a finite set of N samples from the output, $x[0], \ldots, x[N-1]$. Also, we will tackle the output prediction problem: predicting unobserved values of the output for a given input signal $u[n]$.

Our goal is to solve the filtering problem using the tools from estimation theory developed in Chapter 1. To do so, we will first represent the target variables (the filter coefficients) and the observations (the output signal) in vector form, and the relation between them using a vector equation. Thus, we will join the nonzero coefficients in an M-dimensional vector

[1] Although the notation $u[n]$ is frequently used to refer to the step function, in this chapter it is used to denote an arbitrary input signal.

$$\mathbf{s} = \begin{bmatrix} s[0] \\ s[1] \\ \vdots \\ s[M-1] \end{bmatrix}_{M\times 1} . \tag{4.2}$$

Also, we will represent any M-length window of consecutive input values in vector form as

$$\mathbf{u}[n] = \begin{bmatrix} u[n] \\ u[n-1] \\ \vdots \\ u[n-M+1] \end{bmatrix}_{M\times 1} , \tag{4.3}$$

we can write

$$x[n] = \mathbf{u}[n]^\top \mathbf{s} + \varepsilon[n]. \tag{4.4}$$

Note that, for any n, $\mathbf{u}[n]$ contains only the input values that are relevant to compute $x[n]$.

Taking $n = 0, 1, \ldots, N-1$ in Eq. (4.4), we get a system of N linear equations relating the filter coefficients and the observations. We can join all of them into a single matrix equation by defining the observation vector

$$\mathbf{x} = \begin{bmatrix} x[0] \\ x[1] \\ \vdots \\ x[N-1] \end{bmatrix}_{N\times 1} , \tag{4.5}$$

the input matrix

$$\begin{aligned} \mathbf{U} &= [\mathbf{u}[0] \;\; \mathbf{u}[1] \;\; \ldots \;\; \mathbf{u}[M-1] \;\; \ldots \;\; \mathbf{u}[N-1]] \\ &= \begin{bmatrix} u[0] & u[1] & \ldots & u[M-1] & \ldots & u[N-1] \\ 0 & u[0] & \ldots & u[M-2] & \ldots & u[N-2] \\ \vdots & \vdots & \ddots & \vdots & \ldots & \vdots \\ 0 & 0 & \ldots & u[0] & \ldots & u[N-M] \end{bmatrix}_{M\times N} , \end{aligned} \tag{4.6}$$

and the noise vector

$$\boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon[0] \\ \varepsilon[1] \\ \vdots \\ \varepsilon[N-1] \end{bmatrix}_{N\times 1} , \tag{4.7}$$

Using (4.5), (4.6) (4.7), we can write the signal model (4.4) as

$$\boxed{\mathbf{x} = \mathbf{U}^\top \mathbf{s} + \boldsymbol{\varepsilon}} \tag{4.8}$$

The filtering problem reduces to the problem of estimating $\mathbf{s}$ given $\mathbf{x}$ knowing (4.8) and the noise statistics.

4.3 ML solution

Eq. (4.4) shows that, given $\mathbf{s}$, $x[n]$ is Gaussian with mean

$$\mathbb{E}\{x[n] \mid \mathbf{s}\} = \mathbf{u}[n]^\top \mathbf{s} + \mathbb{E}\{\varepsilon[n]\} = \mathbf{u}[n]^\top \mathbf{s} \tag{4.9}$$

and variance

$$\mathbb{E}\{(x[n] - \mathbf{u}[n]^\top \mathbf{s})^2 \mid \mathbf{s}\} = \sigma_\varepsilon^2 \tag{4.10}$$

that is, the likelihood function is

$$p(x[n] \mid \mathbf{s}) = \mathcal{N}(x[n] \mid \mathbf{u}[n]^\top \mathbf{s}, \sigma_\varepsilon^2), \tag{4.11}$$

where the notation $\mathcal{N}(x \mid \mu, v)$ is used to refer to the *normal* (Gaussian) pdf of a random variable with mean μ and variance v, evaluated at x.

Given $\mathbf{s}$, $x[n]$ depends on $\varepsilon[n]$ only, which is an IID process. Thus, all samples from $x[n]$ are independent given $\mathbf{s}$, that is,

$$p_{\mathbf{X}|\mathbf{S}}(\mathbf{x} \mid \mathbf{s}) = \prod_{n=0}^{N-1} \mathcal{N}(x[n] \mid \mathbf{u}[n]^\top \mathbf{s}, \sigma_\varepsilon^2) = \mathcal{N}(\mathbf{x} \mid \mathbf{U}^\top \mathbf{s}, \sigma_\varepsilon^2 \mathbf{I}). \tag{4.12}$$

The value of $\mathbf{s}$ that maximizes $p(\mathbf{x}|\mathbf{s})$ is

$$\begin{aligned}
\hat{\mathbf{s}}_{\mathrm{ML}} &= \operatorname*{argmax}_{\mathbf{s}} p_{\mathbf{X}|\mathbf{S}}(\mathbf{x} \mid \mathbf{s}) = \operatorname*{argmax}_{\mathbf{s}} \log p_{\mathbf{X}|\mathbf{S}}(\mathbf{x} \mid \mathbf{s}) \\
&= \operatorname*{argmin}_{\mathbf{s}} \frac{1}{2}(\mathbf{x} - \mathbf{U}^\top \mathbf{s})^\top (\sigma_\varepsilon^2 \mathbf{I})^{-1} (\mathbf{x} - \mathbf{U}^\top \mathbf{s}) + \frac{1}{2} \log |\sigma_\varepsilon^2 \mathbf{I}| + \frac{N}{2} \log(2\pi) \\
&= \operatorname*{argmin}_{\mathbf{s}} ||\mathbf{x} - \mathbf{U}^\top \mathbf{s}||^2
\end{aligned} \tag{4.13}$$

This minimum can be easily obtained by taking the gradient with respect to $\mathbf{s}$, equalizing to zero and clearing, leading to

$$\boxed{\hat{s}_{\mathrm{ML}} = (\mathbf{U}\mathbf{U}^\top)^{-1}\mathbf{U}\mathbf{x}.} \tag{4.14}$$

4.4 Bayesian Solution

To obtain a Bayesian estimator of $\mathbf{s}$ it is necessary to know its prior probability distribution, $p(\mathbf{s})$. A common choice for this distribution is

$$p_{\mathbf{S}}(\mathbf{s}) = \mathcal{N}(\mathbf{s}|\mathbf{0}, \sigma_s^2 \mathbf{I}), \tag{4.15}$$

since it accommodates any set of real coefficients and assumes they have a zero mean and a dispersion determined by σ_s^2. It is also possible to set $\sigma_s^2 \to \infty$ to approach a uniform

distribution. In any case, the use of this prior distribution enables the analytic derivation of the posterior distribution.

Given the likelihood, $p(\mathbf{x}|\mathbf{s})$, and the prior distribution $p(\mathbf{s})$, the posterior distribution $p(\mathbf{s}|\mathbf{x})$ can be obtained. While it is feasible to directly apply Bayes' theorem and simplify the expression as much as possible, this process can be quite tedious. Instead, we will achieve the result in two steps.

First we will determine the joint pdf of $\mathbf{s}$ and $\mathbf{x}$. A simple way to do this is to observe that

$$\begin{bmatrix}\mathbf{s}\\ \mathbf{x}\end{bmatrix} = \begin{bmatrix}\mathbf{I}\\ \mathbf{U}^\top\end{bmatrix}\mathbf{s} + \begin{bmatrix}\mathbf{0}\\ \boldsymbol{\varepsilon}\end{bmatrix} \tag{4.16}$$

that is, vector $[\mathbf{s}^\top\ \mathbf{x}^\top]^\top$ is a linear combination of two Gaussian random vectors and, thus, is jointly Gaussian:

$$p_{\mathbf{S},\mathbf{X}}(\mathbf{s},\mathbf{x}) = \mathscr{N}\left(\begin{bmatrix}\mathbf{s}\\ \mathbf{x}\end{bmatrix} \middle| \begin{bmatrix}\mathbf{m}_\mathbf{S}\\ \mathbf{m}_\mathbf{X}\end{bmatrix}, \begin{bmatrix}\mathbf{V}_\mathbf{S} & \mathbf{V}_\mathbf{SX}\\ \mathbf{V}_\mathbf{SX}^\top & \mathbf{V}_\mathbf{X}\end{bmatrix}\right) \tag{4.17}$$

where

$$\mathbf{m}_\mathbf{S} = \mathbb{E}\{\mathbf{s}\} = \mathbf{0} \tag{4.18}$$

$$\mathbf{m}_\mathbf{X} = \mathbb{E}\{\mathbf{x}\} = \mathbf{U}^\top\mathbb{E}\{\mathbf{s}\} + \mathbb{E}\{\boldsymbol{\varepsilon}\} = \mathbf{0} \tag{4.19}$$

$$\mathbf{V}_\mathbf{S} = \mathrm{Var}\{\mathbf{s}\} = \sigma_s^2\mathbf{I} \tag{4.20}$$

$$\mathbf{V}_\mathbf{SX} = \mathbb{E}\{\mathbf{s}\mathbf{x}^\top\} = \mathbb{E}\{\mathbf{s}\mathbf{s}^\top\}\mathbf{U} + \mathbb{E}\{\mathbf{s}\boldsymbol{\varepsilon}^\top\} = \sigma_s^2\mathbf{U} \tag{4.21}$$

$$\mathbf{V}_\mathbf{X} = \mathbb{E}\{\mathbf{x}\mathbf{x}^\top\} = \mathbf{U}^\top\mathbb{E}\{\mathbf{s}\mathbf{s}^\top\}\mathbf{U} + \mathbb{E}\{\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^\top\} = \sigma_s^2\mathbf{U}^\top\mathbf{U} + \sigma_\varepsilon^2\mathbf{I} \tag{4.22}$$

Therefore,

$$p_{\mathbf{S},\mathbf{X}}(\mathbf{s},\mathbf{x}) = \mathscr{N}\left(\begin{bmatrix}\mathbf{s}\\ \mathbf{x}\end{bmatrix} \middle| \begin{bmatrix}\mathbf{0}\\ \mathbf{0}\end{bmatrix}, \begin{bmatrix}\sigma_s^2\mathbf{I} & \sigma_s^2\mathbf{U}\\ \sigma_s^2\mathbf{U}^\top & \sigma_s^2\mathbf{U}^\top\mathbf{U} + \sigma_\varepsilon^2\mathbf{I}\end{bmatrix}\right) \tag{4.23}$$

From previous chapter, we know that if $p_{\mathbf{S},\mathbf{X}}(\mathbf{s},\mathbf{x})$ is Gaussian, the conditional distribution $p_{\mathbf{S}|\mathbf{X}}(\mathbf{s} \mid \mathbf{x})$ is also Gaussian, with mean and covariance

$$\begin{aligned}\mathbf{m}_{\mathbf{S}|\mathbf{X}} &= \mathbf{m}_\mathbf{S} + \mathbf{V}_\mathbf{SX}{\mathbf{V}_\mathbf{X}}^{-1}(\mathbf{x} - \mathbf{m}_\mathbf{X})\\ &= \mathbf{U}\left(\mathbf{U}^\top\mathbf{U} + \tfrac{\sigma_\varepsilon^2}{\sigma_s^2}\mathbf{I}\right)^{-1}\mathbf{x}\end{aligned} \tag{4.24}$$

$$\begin{aligned}\mathbf{V}_{\mathbf{S}|\mathbf{X}} &= \mathbf{V}_\mathbf{S} - \mathbf{V}_\mathbf{SX}{\mathbf{V}_\mathbf{X}}^{-1}\mathbf{V}_\mathbf{SX}^\top\\ &= \sigma_s^2\mathbf{I} - \sigma_s^2\mathbf{U}\left(\mathbf{U}^\top\mathbf{U} + \tfrac{\sigma_\varepsilon^2}{\sigma_s^2}\mathbf{I}\right)^{-1}\mathbf{U}^\top\end{aligned} \tag{4.25}$$

The above expressions involve the computation of the inverse of an $N \times N$ matrix, which may be not feasible for large signal records (the computational cost is $\mathscr{O}(N^3)$. However, using the *matrix inversion lemma* (see the Appendix in Sec. 4.7), we can obtain the alternative expressions

$$\mathbf{m}_{\mathbf{S}|\mathbf{X}} = \mathbf{P}\mathbf{U}\mathbf{x} \tag{4.26}$$

$$\mathbf{V}_{\mathbf{S}|\mathbf{X}} = \sigma_\varepsilon^2 \mathbf{P} \tag{4.27}$$

where

$$\boxed{\mathbf{P} = (\mathbf{U}\mathbf{U}^\top + \tfrac{\sigma_\varepsilon^2}{\sigma_s^2}\mathbf{I})^{-1}.} \tag{4.28}$$

Eq. (4.28) involves the inversion of a matrix $M \times M$, which is usually much smaller than $N \times N$.

Using these expressions, the MMSE, MAP and MAD estimates of $\mathbf{s}$ are:

$$\boxed{\hat{\mathbf{s}}_{\text{MSE}} = \hat{\mathbf{s}}_{\text{MAP}} = \hat{\mathbf{s}}_{\text{MAD}} = \mathbf{P}\mathbf{U}\mathbf{x}} \tag{4.29}$$

Also, note that taking $\sigma_\varepsilon^2 \to 0$ (negligible noise) and/or $\sigma_s^2 \to \infty$ (which can be interpreted as assuming an infinitely wide uniform prior) these Bayesian solutions become equivalent to the ML in (4.14).

4.4.1 Probabilistic prediction of the filter output

Once we have resolved several estimators of filter $\mathbf{s}$, we now begin to consider the problem of predicting a new output $x[k]$ at some time $k > N$. Continuing with the Bayesian perspective, we will obtain the posterior pdf of the target variable, $x[k]$, in light of the outputs already observed, $\mathbf{x}$. That is, we aim to calculate $p(x[k] \mid \mathbf{x})$.

First, it should be noted that $\mathbf{x}, x[k]$ and $\mathbf{s}$ are jointly Gaussian. This follows from Eq. (4.23), which can be extended to any arbitrary number of outputs, including $x[k]$. This necessarily implies that $\mathbf{x}$ and $x[k]$ are jointly Gaussian (when marginalizing $\mathbf{s}$) and finally that $p(x[k]|\mathbf{x})$ must be Gaussian. Given that

$$x[k] = \mathbf{u}[k]^\top \mathbf{s} + \varepsilon[k] \tag{4.30}$$

is a linear transformation of $\mathbf{s}$ with independent white noise, we can easily compute the posterior mean and variance, respectively, as follows:

$$\mathbb{E}\{x[k] \mid \mathbf{x}\} = \mathbf{u}[k]^\top \mathbb{E}\{\mathbf{s} \mid \mathbf{x}\} + \mathbb{E}\{\varepsilon[k] \mid \mathbf{x}\} = \mathbf{u}[k]^\top \hat{\mathbf{s}}_{\text{MSE}} \tag{4.31}$$

$$\begin{aligned}
\text{Var}\{x[k] \mid \mathbf{x}\} &= \mathbb{E}\left\{ \left(\mathbf{u}[k]^\top (\mathbf{s} - \hat{\mathbf{s}}_{\text{MSE}}) + \varepsilon[k] \right)^2 \mid \mathbf{x} \right\} \\
&= \mathbf{u}[k]^\top \mathbb{E}\{ (\mathbf{s} - \hat{\mathbf{s}}_{\text{MSE}}) (\mathbf{s} - \hat{\mathbf{s}}_{\text{MSE}})^\top \mid \mathbf{x} \} \mathbf{u}[k] + \mathbb{E}\{\varepsilon[k]^2 \mid \mathbf{x}\} \\
&= \mathbf{u}[k]^\top \mathbf{V}_{\mathbf{S}|\mathbf{X}} \mathbf{u}[k] + \sigma_\varepsilon^2 \\
&= \sigma_\varepsilon^2 \mathbf{u}[k]^\top \mathbf{P}\mathbf{u}[k] + \sigma_\varepsilon^2
\end{aligned} \tag{4.32}$$

Therefore, the Bayesian prediction of the output is succinctly given by

$$\boxed{\hat{x}_{\mathrm{MSE}}[k] = \hat{x}_{\mathrm{MAP}}[k] = \hat{x}_{\mathrm{MAD}}[k] = \mathbf{u}[k]^{\top}\hat{\mathbf{s}}_{\mathrm{MSE}}.} \tag{4.33}$$

Eq. (4.33) shows that the optimal prediction of the output at any time $k > N$ can be computed by passing the input signal $u[n]$ though the filter with the coefficients in the Bayesian estimation of $\mathbf{s}$ (see Fig. 4.2).

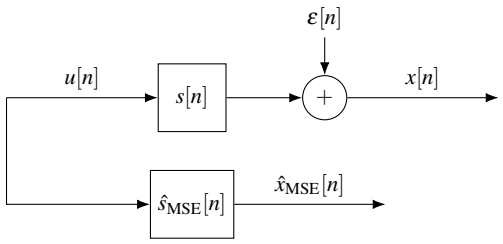


Fig. 4.2 The best prediction of future values of the output (knowing the input) can be computed by passing the input signal through a filter with the coefficients of the Bayesian estimation of $\mathbf{s}$.

4.5 Online calculus

It is possible to obtain the above solutions online, that is, as new input-output pairs are obtained. While complete calculations could be repeated each time a new sample arrives, there are often more efficient ways to do this.

Note that estimating $\mathbf{s}$ using Eqs. (4.14) or (4.29) requires inverting an $M \times M$ matrix. This has a cost $\mathscr{O}(M^3)$, that is, if we double the size of the filter, M we multiply its computational cost by eight. Suppose now that you want to estimate $\mathbf{s}$ as new input-output pairs are received, that is, we are given first $\{u[0], x[0]\}$, then $\{u[1], x[1]\}$ and so on. In this case, we could reuse the results of the previous estimate to calculate the new updated estimate of $\mathbf{s}$, thus reducing the cost $\mathscr{O}(M^3)$ that would have a *naive* method that simply recalculates everything again every time a sample arrives.

4.5.1 Bayesian solution

$\hat{\mathbf{s}}_{\mathrm{MSE}}$ can be obtained exactly as more samples are available (i.e. as N increases) without redoing all calculations, by reusing the previous solution. To do this, it is defined

$$\mathbf{P}_N = (\mathbf{U}\mathbf{U}^{\top} + \frac{\sigma_\varepsilon^2}{\sigma_s^2}\mathbf{I})^{-1}, \tag{4.34}$$

$$\mathbf{r}_N = \mathbf{U}\mathbf{x} \tag{4.35}$$

and the following recursive calculation is used (the first equation corresponds to the direct application of the matrix inversion lemma to the $\mathbf{P}$ update):

$$\begin{aligned}\mathbf{P}_{N+1} &= \mathbf{P}_N - \frac{\mathbf{P}_N\mathbf{u}[N+1]\mathbf{u}[N+1]^\top\mathbf{P}_N}{1+\mathbf{u}[N+1]^\top\mathbf{P}_N\mathbf{u}[N+1]}\\ \mathbf{r}_{N+1} &= \mathbf{r}_N + \mathbf{u}[N+1]x[N+1]\\ \mathbf{s}_{N+1} &= \mathbf{P}_{N+1}\mathbf{r}_{N+1},\end{aligned}$$

which only has a cost $\mathscr{O}(M^2)$ per step (as opposed to applying the complete original equation at each step, which would cost $\mathscr{O}(M^3)$). This algorithm is called *recursive least squares* (RLS).

4.5.2 ML solution

An online approximation to $\hat{\mathbf{s}}_{\text{ML}}$ with computational cost $\mathscr{O}(M)$ can be obtained just by noting that

$$\hat{\mathbf{s}}_{\text{ML}} = \underset{\mathbf{s}}{\text{argmax}}\, p(\mathbf{x}|\mathbf{s}) = \underset{\mathbf{s}}{\text{argmin}}\, ||\mathbf{x}-\mathbf{U}^\top\mathbf{s}||^2 \tag{4.36}$$

and then use stochastic gradient to minimize $||\mathbf{x}-\mathbf{U}^\top\mathbf{s}||^2$.

Notice that

$$||\mathbf{x}-\mathbf{U}^\top\mathbf{s}||^2 = \sum_{n=0}^{N-1}(x[n]-\mathbf{u}[n]^\top\mathbf{s})^2, \tag{4.37}$$

so a gradient descent method would calculate the gradient of that expression and iteratively shift the estimate of the minimum in the opposite direction of the gradient in each step. A descent by stochastic gradient performs the same operation, but considering only one of the additions of the mentioned sum in each step. So, the updating of coefficients that must be iterated to perform the minimization is in this case

$$\hat{\mathbf{s}}_{n+1} = \hat{\mathbf{s}}_n + \mu\left(x[n]-\mathbf{u}[n]^\top\hat{\mathbf{s}}_n\right)\mathbf{u}[n], \tag{4.38}$$

where μ is an adaptation step that should be "small enough". This algorithm is called *least mean squares* (LMS).

4.6 Problems

4.1 Consider the sequence

$$u[1]\ldots u[7] \equiv 0.7,\ -0.1,\ 0.7,\ -0.2,\ -0.1,\ 1.5,\ -1.1$$

which is fed as input to a linear filter of three coefficients, $\mathbf{s} = [s_1, s_2, s_3]^\top$. The following elements of the output sequence are known, (corrupted with Gaussian noise of variance 0.25):

$$x[1]\ldots x[6] \equiv -0.60,\ 1.13,\ 0.57,\ 0.42,\ 1.25,\ -2.58$$

a) What is the ML estimate of $\mathbf{s}$ given the data?
b) Use the obtained filter to predict $x[7]$, $\hat{x}_{\text{ML}}$.

c) Calculate the MSE, MAP and MAD estimates of **s** assuming that the prior pdf of its components is $s_i \sim \mathcal{N}(0,1)$ and that they are independent.
d) Get the MSE estimate of $x[7]$, $\hat{x}_{\text{MSE}}$.
e) Calculate the MSE in prediction b). (That is, the mean of $(\hat{x}_{\text{ML}} - x[7])^2$ given the data).
f) Calculate the MSE in prediction d). (That is, the mean of $(\hat{x}_{\text{MMSE}} - x[6])^2$ given the data)

4.7 Appendix: the matrix inversion lemma

In this section we apply the matrix inversion lemma (also known as the Woodbury matrix identity) to obtain Eqs. (4.26). This lemma states that, for any matrices **U**, **C**, **V** and any non-singular matrix **A**,

$$(\mathbf{A}+\mathbf{VCU})^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{V}\left(\mathbf{C}^{-1}+\mathbf{UA}^{-1}\mathbf{V}\right)^{-1}\mathbf{UA}^{-1} \tag{4.39}$$

Proving this equality is not difficult: multiplying the right-hand side of the equality by $\mathbf{A}+\mathbf{UCV}$, it is easy to see that the result is the identity matrix.

Taking

$$\mathbf{A} = \frac{\sigma_\varepsilon^2}{\sigma_s^2}\mathbf{I} \tag{4.40}$$

$$\mathbf{C} = \mathbf{I} \tag{4.41}$$

$$\mathbf{V} = \mathbf{U}^\top \tag{4.42}$$

we get

$$\begin{aligned}\left(\mathbf{U}^\top\mathbf{U}+\frac{\sigma_\varepsilon^2}{\sigma_s^2}\mathbf{I}\right)^{-1} &= \frac{\sigma_s^2}{\sigma_\varepsilon^2}\mathbf{I} - \left(\frac{\sigma_s^2}{\sigma_\varepsilon^2}\right)^2\mathbf{U}^\top\left(\mathbf{I}+\frac{\sigma_s^2}{\sigma_\varepsilon^2}\mathbf{U}\mathbf{U}^\top\right)^{-1}\mathbf{U} \\ &= \frac{\sigma_s^2}{\sigma_\varepsilon^2}\mathbf{I} - \frac{\sigma_s^2}{\sigma_\varepsilon^2}\mathbf{U}^\top\left(\mathbf{U}\mathbf{U}^\top+\frac{\sigma_\varepsilon^2}{\sigma_s^2}\mathbf{I}\right)^{-1}\mathbf{U} \\ &= \frac{\sigma_s^2}{\sigma_\varepsilon^2}\mathbf{I} - \frac{\sigma_s^2}{\sigma_\varepsilon^2}\mathbf{U}^\top\mathbf{P}\mathbf{U}\end{aligned} \tag{4.43}$$

Where **P** is given by (4.28). Note that, by definition,

$$\left(\mathbf{U}\mathbf{U}^\top+\frac{\sigma_\varepsilon^2}{\sigma_s^2}\mathbf{I}\right)\mathbf{P} = \mathbf{P}\left(\mathbf{U}\mathbf{U}^\top+\frac{\sigma_\varepsilon^2}{\sigma_s^2}\mathbf{I}\right) = \mathbf{I} \tag{4.44}$$

so that

$$\mathbf{U}\mathbf{U}^\top\mathbf{P} = \mathbf{P}\mathbf{U}\mathbf{U}^\top = \mathbf{I} - \frac{\sigma_\varepsilon^2}{\sigma_s^2}\mathbf{P} \tag{4.45}$$

We will use these equalities below. Using (4.43) into (4.24), we get

$$\begin{aligned}
\mathbf{m}_{\mathbf{S}|\mathbf{X}} &= \mathbf{U}\left(\mathbf{U}^\top\mathbf{U} + \frac{\sigma_\varepsilon^2}{\sigma_s^2}\mathbf{I}\right)^{-1}\mathbf{x} \\
&= \frac{\sigma_s^2}{\sigma_\varepsilon^2}\mathbf{U}\mathbf{x} - \frac{\sigma_s^2}{\sigma_\varepsilon^2}\mathbf{U}\mathbf{U}^\top\mathbf{P}\mathbf{U}\mathbf{x} \\
&= \frac{\sigma_s^2}{\sigma_\varepsilon^2}\mathbf{U}\mathbf{x} - \frac{\sigma_s^2}{\sigma_\varepsilon^2}\left(\mathbf{I} - \frac{\sigma_\varepsilon^2}{\sigma_s^2}\mathbf{P}\right)\mathbf{U}\mathbf{x} \\
&= \frac{\sigma_s^2}{\sigma_\varepsilon^2}\mathbf{U}\mathbf{x} - \frac{\sigma_s^2}{\sigma_\varepsilon^2}\mathbf{U}\mathbf{x} + \mathbf{P}\mathbf{U}\mathbf{x} \\
&= \mathbf{P}\mathbf{U}\mathbf{x} \qquad (4.46)
\end{aligned}$$

$$\begin{aligned}
\mathbf{V}_{\mathbf{S}|\mathbf{X}} &= \sigma_s^2\mathbf{I} - \sigma_s^2\mathbf{U}\left(\mathbf{U}^\top\mathbf{U} + \frac{\sigma_\varepsilon^2}{\sigma_s^2}\mathbf{I}\right)^{-1}\mathbf{U}^\top \\
&= \sigma_s^2\left[\mathbf{I} - \mathbf{U}\left(\frac{\sigma_s^2}{\sigma_\varepsilon^2}\mathbf{I} - \frac{\sigma_s^2}{\sigma_\varepsilon^2}\mathbf{U}^\top\mathbf{P}\mathbf{U}\right)\mathbf{U}^\top\right] \\
&= \sigma_s^2\left[\mathbf{I} - \frac{\sigma_s^2}{\sigma_\varepsilon^2}\mathbf{U}\mathbf{U}^\top + \frac{\sigma_s^2}{\sigma_\varepsilon^2}\mathbf{U}\mathbf{U}^\top\mathbf{P}\mathbf{U}\mathbf{U}^\top\right] \\
&= \sigma_s^2\left[\mathbf{I} - \frac{\sigma_s^2}{\sigma_\varepsilon^2}\mathbf{U}\mathbf{U}^\top + \frac{\sigma_s^2}{\sigma_\varepsilon^2}\left(\mathbf{I} - \frac{\sigma_\varepsilon^2}{\sigma_s^2}\mathbf{P}\right)\mathbf{U}\mathbf{U}^\top\right] \\
&= \sigma_\varepsilon^2\mathbf{P} \qquad (4.47)
\end{aligned}$$

Chapter 5
Spectral Estimation

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5.1 Introduction

This chapter studies a very important estimation problem, which is that of estimating the power spectral density (PSD) of a stationary process. We will consider two families of estimators: 1) classical (or non-parametric) and 2) parametric estimators, which are based on a model for the PSD.

Computing the estimate of $S_x(e^{j\omega})$, which we will denote by $\hat{S}_x(e^{j\omega})$, from an arbitrarily large number of realizations of a stationary process (see Figure 5.1) would be a (relatively) easy task. Of course, this is an idealized scenario as we do not have access to all realizations and, even more, we also do not have access to all time samples of the same realization. Thus, the objective in this chapter is to compute $\hat{S}_x(e^{j\omega})$ from N samples of a single realization of the process $x[n]$.

The spectral estimation problem is defined only for wide-sense stationary (WSS) processes for which the mean function is time-independent, that is, $\mu_x = \mu_x[n] = \mathbb{E}[x[n]]$, and the auto-correlation function depends only on the time difference, i.e., $r_x[m] = r_x[n, n-m] = \mathbb{E}[x[n]x^*[n-m]]$. For non-stationary processes, the usual practice is to apply the estimators to small windows, since on a local scale we can assume that non-stationary processes are WSS. For instance, this is typically done when analyzing speech signals, which are usually described using non-stationary processes. Moreover, since only one realization is available, the process must be ergodic such that expectations can be substituted by time averages.

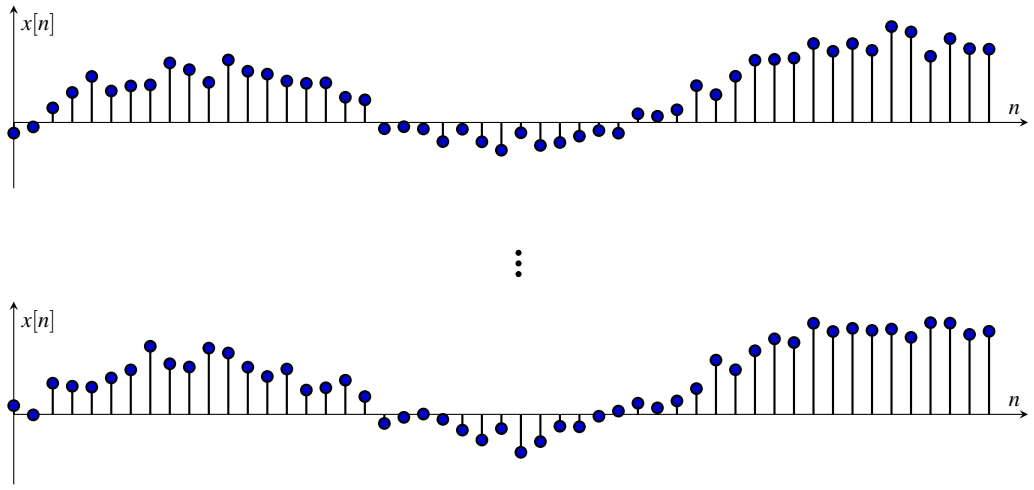


Fig. 5.1 Realizations of a discrete stochastic process

5.2 Preliminaries: Spectral analysis of deterministic signals

Before going into the spectral analysis of stochastic processes, it is convenient to study the case of deterministic signals, which will help us to understand the concept of spectral

resolution. Thus, the problem is to compute the Fourier transform of the deterministic signal $x[n]$. However, this relatively "simple" task has two problems. First, we do not have access to the whole signal $x[n]$, but only to a finite record thereof.

$$x_w[n] = \begin{cases} x[n], & n = 0, \ldots, N-1, \\ 0, & \text{otherwise.} \end{cases}$$

Defining now the window

$$w_{R,N}[n] = \begin{cases} 1, & n = 0, \ldots, N-1, \\ 0, & \text{otherwise,} \end{cases}$$

we may rewrite $x_w[n] = w_{R,N}[n]x[n]$, which allows us to compute the Fourier transform of $x_w[n]$ as[1]

$$X_w(e^{j\omega}) = \mathscr{F}(x_w[n]) = \sum_{n=0}^{N-1} x_w[n]e^{-j\omega N} = \frac{1}{2\pi} W_{R,N}(e^{j\omega}) \circledast X(e^{j\omega}), \tag{5.1}$$

where $\circledast$ denotes the circular convolution. So, the Fourier transform of the windowed signal, $x_w[n]$, is related to that of $x[n]$ through the Fourier transform of the window $w_{R,N}[n]$, which is given by

$$W_{R,N}(e^{j\omega}) = e^{-j\omega(N-1)/2} \frac{\sin\left(\frac{\omega N}{2}\right)}{\sin\left(\frac{\omega}{2}\right)} = e^{-j\omega(N-1)/2} P_N(e^{j\omega}),$$

and its amplitude $|P_N(e^{j\omega})|$ is depicted in Figure 5.2. As this figure shows, the width of the main lobe is $4\pi/N$.

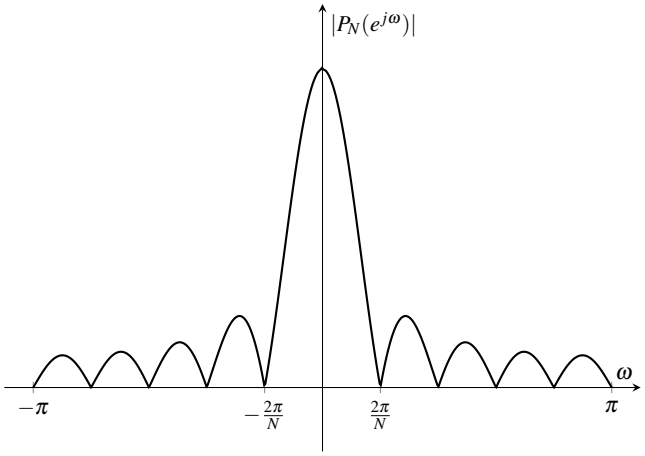


Fig. 5.2 Module of the Fourier transform of the rectangular window

[1] In the "Signals and Systems" parlance, this Fourier transform is named Discrete Time Fourier Transform (DTFT).

The second issue is that the DTFT in (5.1) is a function of a continuous variable. Hence it cannot be computed nor stored in a computer. The solution is simple and consists in discretizing the spectrum, which yields the Discrete Fourier Transform (DFT). Thus, we are only able to compute $X_w(e^{j\omega_k})$, with $\omega_k = 2\pi k/N$ and $k = 0,\ldots,N-1$. The DFT is typically computed using the fast Fourier transform (FFT) algorithm.

The aforementioned procedure based on the DFT/FFT gets only N samples of the spectrum for length-N signals, but we can get more samples by zero-padding the signals, i.e., by simply adding $N_{\text{fft}} - N$ zeros after the N samples. This procedure increases the number of frequencies but it does not increase the resolution as it does not modify the window.

Example 5.1 (Spectral analysis of a complex exponential)

This example considers the spectral analysis of a finite record of a complex exponential, i.e., $x[n] = e^{j\omega_0 n}, n = 0,\ldots,N-1$. Using the DTFT of a complex exponential, given by

$$X(e^{j\omega}) = 2\pi\delta(\omega - \omega_0),$$

and $W_N(e^{j\omega})$, $X_w(e^{j\omega})$ becomes

$$X_w(e^{j\omega}) = e^{-j(\omega-\omega_0)(N-1)/2}\frac{\sin\left(\frac{(\omega-\omega_0)N}{2}\right)}{\sin\left(\frac{\omega-\omega_0}{2}\right)} = e^{-j(\omega-\omega_0)(N-1)/2}P_N(\omega-\omega_0),$$

and its magnitude squared is

$$|X_w(e^{j\omega})|^2 = \left|P_N\left(e^{j(\omega-\omega_0)}\right)\right|^2.$$

Figure 5.3 plots, in logarithmic scale, $|X_w(e^{j\omega})|^2$ and $|X_w(e^{j\omega_k})|^2$ for two different values of N_{fft}.

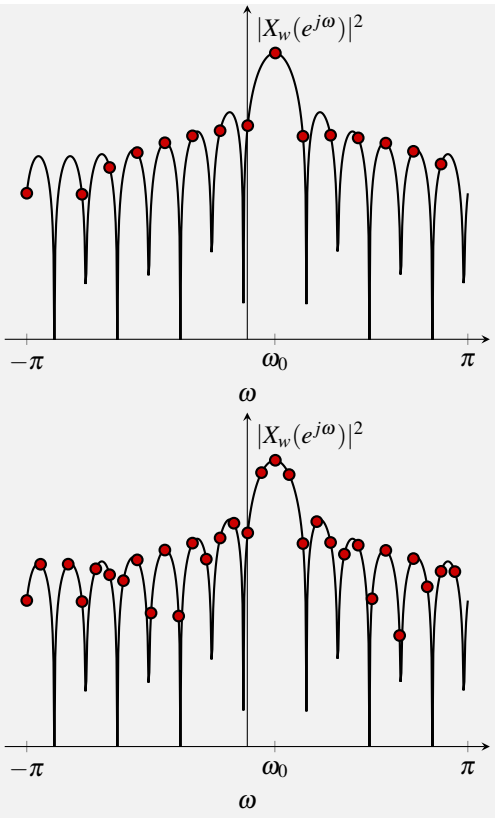


Fig. 5.3 Fourier transform (in logarithmic scale) of a windowed complex exponential

As we have seen in this example, the spectral analysis of deterministic signals depends on two factors. First, the number of available samples, which determines the window and, therefore, the shape of the windowed spectrum. As we have seen in Figure 5.3, the rectangular window has a narrow main lobe at the expense of high secondary lobes. This effect could be reduced by pre-multiplying $x_w[n]$ by a different window, which would reduce the height of the secondary lobes, but it would widen the main lobe. Second, the number of frequencies where the estimated PSD is evaluated.

Example 5.2 (Spectral analysis of two complex exponentials)

This example considers the spectral analysis of a finite record of the sum of two complex exponential, i.e., $x[n] = e^{j\omega_0 n} + e^{j\omega_1 n}, n = 0,\ldots,N-1$, which will help us to understand the concept of resolution. Using (5.1), we have

$$X_w(e^{j\omega}) = e^{-j(\omega-\omega_0)(N-1)/2} P_N(\omega-\omega_0) + e^{-j(\omega-\omega_1)(N-1)/2} P_N(\omega-\omega_1).$$

and its magnitude squared is

$$|X_w(e^{j\omega})|^2 = \left|P_N\left(e^{j(\omega-\omega_0)}\right)\right|^2 + \left|P_N\left(e^{j(\omega-\omega_1)}\right)\right|^2 + 2\cos\left(\frac{(\omega_0-\omega_1)(N-1)}{2}\right) P_N\left(e^{j(\omega-\omega_0)}\right) P_N\left(e^{j(\omega-\omega_1)}\right).$$

Figure 5.4 plots, in logarithmic scale, $|X_w(e^{j\omega})|^2$ for two different separations between the frequencies of the exponential components. As we can see in this figure, for small frequency separations, it is impossible to identify in the spectrum the two complex exponentials.

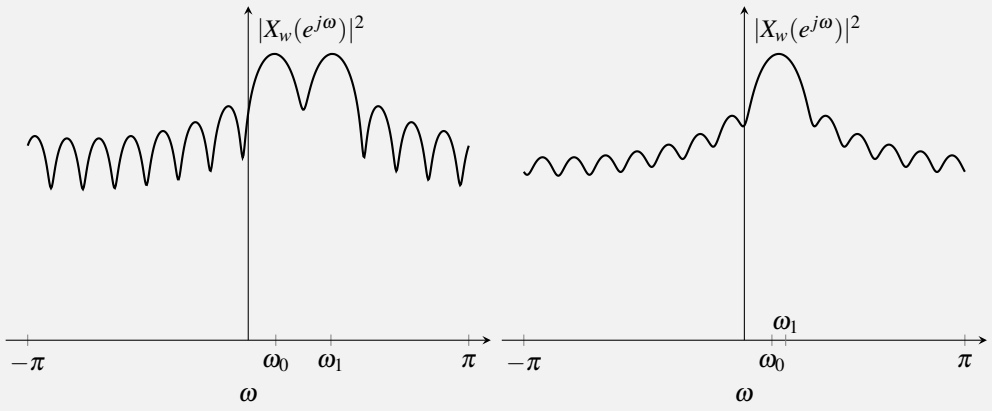


Fig. 5.4 Fourier transform (in logarithmic scale) of the sum of two complex exponentials

5.3 Non-parametric methods in spectral estimation

In this section, we turn our attention to the case of stochastic signals and, in particular, to the development of non-parametric spectral estimation methods. We will therefore study the periodogram and variations thereof.

Before proceeding, let us note that throughout this section, we will only consider DTFTs. However, we have to keep in mind that, in practice, we can only compute the DFT (using the FFT algorithm), as we have seen in Section 5.2.

Remind that the power spectral density, or power spectrum, of a stochastic process $x[n]$, is defined as

$$S_x(e^{j\omega}) = \lim_{N\to\infty} \frac{1}{2N-1} \mathbb{E}\left[\left|\sum_{n=-N+1}^{N-1} x[n]e^{-j\omega n}\right|^2\right]. \tag{5.2}$$

If the process is WSS and the autocorrelation is absolutely summable (the usual case in practice), this definition is equivalent to the Fourier transform of the autocorrelation, i.e.,

$$S_x(e^{j\omega}) = \mathscr{F}(r_x[m]), \tag{5.3}$$

where

$$r_x[m] = \mathbb{E}[x[n]x^*[n-m]], \tag{5.4}$$

These alternative expressions for the power spectrum motivate two different strategies for spectral estimation:

1. Drop the limit in (5.2) and estimate the expectation from an finite sample.
2. Estimate the autocorrelation function and compute its Fourier transform.

Both strategies are closely related, as we will see in the following sections.

5.3.1 The periodogram and the correlogram

The **periodogram**, which is a term coined by Arthur Schuster in 1898, is obtained from on (5.2) by simply dropping the expectation and considering a finite number of samples, i.e.,

$$\hat{S}_x^p(e^{j\omega}) = \frac{1}{N}\left|\sum_{n=0}^{N-1} x[n]e^{-j\omega n}\right|^2 = \frac{1}{N}\left|X_N(e^{j\omega})\right|^2, \tag{5.5}$$

where $X_N(e^{j\omega}) = \mathscr{F}(x_N[n])$ is the Fourier transform of the truncated version of $x[n]$

$$x_N[n] = \begin{cases} x[n], & 0 \le n \le N-1 \\ 0, & \text{otherwise} \end{cases} \tag{5.6}$$

The **correlogram**, based on (5.3), is defined as

$$\hat{S}_x^c(e^{j\omega}) = \mathscr{F}(\hat{r}_x[m]),$$

where

$$\hat{r}_x[m] = \frac{1}{N}\sum_{n=m}^{N-1} x[n]x^*[n-m], \quad m = 0,\ldots,N-1, \tag{5.7}$$

Despite the differences in names and definitions, the correlogram and the periodogram are identical estimators: rewriting $\hat{r}_x[m]$ as

$$\hat{r}_x[m] = \frac{1}{N}\sum_{n=m}^{N-1} x[n]x^*[n-m] = \frac{1}{N}\sum_{n=-\infty}^{\infty} x_N[n]x_N^*[n-m] = \frac{1}{N}x_N[n] * x_N^*[-n],$$

and taking its Fourier transform yields

$$\hat{S}_x^c(e^{j\omega}) = \mathscr{F}(\hat{r}_x[m]) = \frac{1}{N}\mathscr{F}\left(x_N[n] * x_N^*[-n]\right).$$

Finally, applying the properties of the Fourier transform, $\hat{S}_x^c(e^{j\omega})$ simplifies to

$$\hat{S}_x^c(e^{j\omega}) = \frac{1}{N}\mathscr{F}\left(x_N[n]\right)\mathscr{F}\left(x_N^*[-n]\right) = \frac{1}{N}X_N(e^{j\omega})X_N^*(e^{j\omega}) = \frac{1}{N}\left|X_N(e^{j\omega})\right|^2 = \hat{S}_x^p(e^{j\omega}),$$

which is the periodogram in (5.5).

5.3.1.1 Bias and variance of the periodogram

To understand why we need more refined estimators of the power spectral density, now we shall perform the statistical analysis of the periodogram (or correlogram), i.e., we will compute its bias and variance as we would do with any other estimator.

First, note that the autocorrelation estimate in (5.3.1) is biased:

$$\begin{aligned}\mathbb{E}\left[\hat{r}_x[m]\right] &= \frac{1}{N}\sum_{n=m}^{N-1}\mathbb{E}\left[x[n]x^*[n-m]\right] = \frac{1}{N}\sum_{n=m}^{N-1} r_x[m] \\ &= \frac{N-|m|}{N} r_x[m], \end{aligned} \tag{5.8}$$

Note that, for $m = 0$ or any m such that $r_x[m] = 0$, the autocorrelation estimate is unbiased. However,for any other values, $\mathbb{E}\left[\hat{r}_x[m]\right] \neq r_x[m]$, and the relative value of the bias increases with m.

Seemingly, we can easily remove this bias by replacing the factor $1/N$ in (5.7) by $1/(N-|m|)$. In this way, we would get unbiased estimates of the autocorrelation values for any m. Unfortunately, it can be show that the resulting function is, in general, not a feasible autocorrelation function (i.e. its Fourier transform may take negative values at some frequencies).

As a consequence of the autocorrelation bias, the periodogram is also biased. To see it, it is useful to define the triangular, or Barlett window, as

$$w_{T,N}[m] = \begin{cases} \frac{N-|m|}{N}, & |m| \leq N-1, \\ 0, & \text{otherwise}, \end{cases}$$

which is depicted in Figure 5.5.

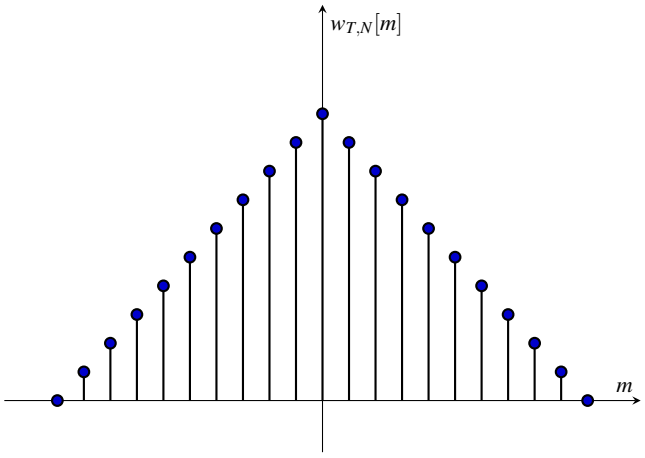


Fig. 5.5 Triangular window

The mean of the autocorrelation in (5.8) can then be written as

$$\mathbb{E}\left[\hat{r}_x[m]\right] = \frac{N-|m|}{N} r_x[m] = w_{T,N}[m] r_x[m], \tag{5.9}$$

Using (5.9), the bias of the periodogram becomes

$$\mathbb{E}\left[\hat{S}_x^p(e^{j\omega})\right] = \mathscr{F}\left(w_{T,N}[m] r_x[m]\right) = \frac{1}{2\pi} W_{T,N}(e^{j\omega}) \circledast S_x(e^{j\omega}), \tag{5.10}$$

where

$$\begin{aligned} W_{T,N}(e^{j\omega}) &= \mathscr{F}\left(w_{T,N}[m]\right) = \frac{1}{N}\mathscr{F}\left(w_{R,N}[m] * w_{R,N}[-m]\right) = |W_{R,N}(e^{j\omega})|^2 \\ &= \frac{1}{N}\frac{\sin^2\left(\frac{\omega N}{2}\right)}{\sin^2\left(\frac{\omega}{2}\right)}, \end{aligned}$$

is the Fourier transform of the triangular window and is depicted in Figure 5.6. Comparing Figures 5.2 and 5.6, it can be seen that the level of secondary lobes is smaller for the triangular window. By analogy with Example 5.2, we can say that the bias of the periodogram is related with its resolution.

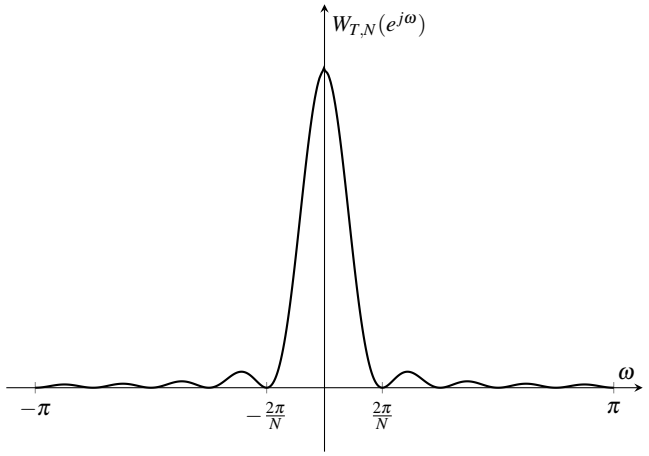


Fig. 5.6 Fourier transform of the triangular window

Note that the width of the main lobe is inversely proportional to N. Therefore, by increasing the sample size we can narrow the main lobe and reduce the bias. This is because the periodogram is asymptotically unbiased: the bias reduces to zero as N grows.

$$\lim_{N\to\infty} \hat{S}_x^p(e^{j\omega}) = S_x(e^{j\omega}).$$

As a special case, note that the autocorrelation function of a white process is zero for all $m \neq 0$, therefore, the estimate of its autocorrelation is unbiased, and so the periodogram.

The analysis of the variance of the periodogram is cumbersome and can only be done in particular cases. For white noise, it can be shown that

$$Var\left(\hat{S}_x^p(e^{j\omega})\right) = S_x^2(e^{j\omega}),$$

and in general we can say that

$$Var\left(\hat{S}_x^p(e^{j\omega})\right) \lesssim S_x^2(e^{j\omega}),$$

where $\lesssim$ denotes approximately proportional to. This expression tells us that the variance does not decrease for larger data records. That is, the periodogram is not a consistent estimate of the PSD.

5.3.2 The Blackman-Tukey estimator

One of the reasons for the behavior of the periodogram variance is the poor quality of the estimate $\hat{r}_x[m]$ for values of m close to N. This problem is what the Blackman-Tukey (BT) estimator tries to improve. The idea is to ignore or weight the samples of $\hat{r}_x[m]$ for m close to N. Thus, the BT estimator is

$$\hat{S}_x^{BT}(e^{j\omega}) = \mathscr{F}(w_M[m]\hat{r}_x[m]) = \sum_{m=-N+1}^{N-1} w_M[m]\hat{r}_x[m]e^{-j\omega m}, \tag{5.11}$$

where $w[m]$ is a window that must fulfill

$$w_M[m] = \begin{cases} f(|m|), & |m| \leq M-1, \\ 0, & \text{otherwise}, \end{cases}$$

where $f(|m|)$ is a monotonically decreasing function of $|m|$ and $M \leq N$. This window ignores the lags of the estimated auto-correlation for $|m| > M-1$ and weights the lags for large m. The choice of the window is critical to achieve good performance, but, in any case, it must guarantee that $\hat{S}_x^{BT}(e^{j\omega}) \geq 0$.

Using the properties of the Fourier transform, we may rewrite $\hat{S}_x^{BT}(e^{j\omega})$ as

$$\hat{S}_x^{BT}(e^{j\omega}) = \frac{1}{2\pi} W_M(e^{j\omega}) \circledast \hat{S}_x^p(e^{j\omega}) = \frac{1}{2\pi}\int_{-\pi}^{\pi} W_M(e^{j\psi})\hat{S}_x^p(e^{j(\omega-\psi)})d\psi,$$

where $W_M(e^{j\omega}) = \mathscr{F}(w_M[n])$. Then, the Blackman-Tukey estimator is locally smoothing the periodogram, which reduces its variance. However, there is no free lunch and we will show that this variance reduction translates into lower resolution (or larger bias). Concretely, the bias of the BT estimator is

$$\mathbb{E}\left[\hat{S}_x^{BT}(e^{j\omega})\right] = \frac{1}{2\pi} W_M(e^{j\omega}) \circledast \mathbb{E}\left[\hat{S}_x^p(e^{j\omega})\right] = \frac{1}{2\pi} W_M(e^{j\omega}) \circledast W_{T,N}(e^{j\omega}) \circledast S_x(e^{j\omega}).$$

Finally, since $w_M[n]$ is shorter than $w_{T,N}[n]$, it can be shown that $W_M(e^{j\omega})$ is wider than $W_{T,M}(e^{j\omega})$, which translates into a lower resolution. This behavior is depicted in Figure 5.7 for $w_M[n] = w_{T,M}[n]$. Note that the y-axis is in logarithmic scale.

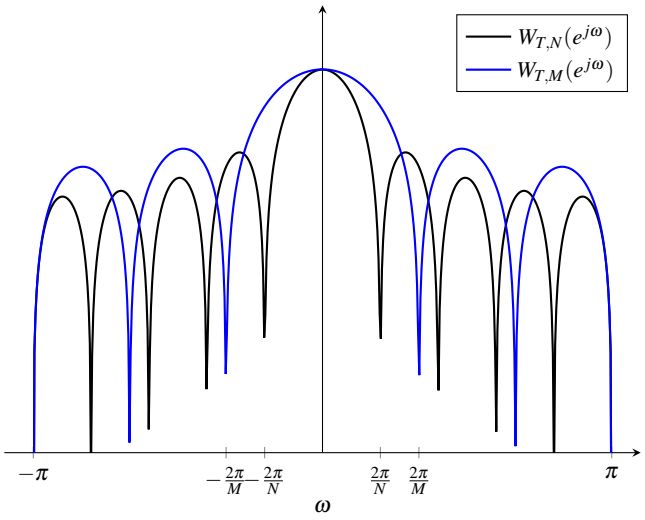


Fig. 5.7 Fourier transform (in logarithmic scale) of two triangular windows of different lengths

5.3.3 Estimators based on the averaged periodogram

The Blackman-Tukey estimator yields a smaller variance than that of the periodogram because, as we have seen, it smooths the periodogram. An alternative to reduce the variance is to average several periodograms. However, the question is: How do we obtain such periodograms? The answer is easy and consists in dividing the N observations into windows of length $M < N$.

The Barlett method is one of the possible estimators based on the averaged periodogram. First, it divides the N observations into L non-overlapping windows of length M as

$$x_l[n] = x[(l-1)M+n], \tag{5.12}$$

where $n = 0,\ldots,M-1$, and $l = 1,\ldots,L$, and computes the periodogram of each window, that is,

$$\hat{S}^p_{x,l}(e^{j\omega}) = \frac{1}{M}\left|\sum_{n=0}^{M-1} x_l[n]e^{-j\omega n}\right|^2 = \frac{1}{M}\left|X_l(e^{j\omega})\right|^2. \tag{5.13}$$

Then, the Barlett estimator is given by simply averaging the individual periodograms

$$\hat{S}^B_x(e^{j\omega}) = \frac{1}{L}\sum_{l=1}^{L}\hat{S}^p_{x,l}(e^{j\omega}). \tag{5.14}$$

Although it is out of the scope of these notes, we must point out that $\hat{S}^B_x(e^{j\omega})$ is somehow related to $\hat{S}^{BT}_x(e^{j\omega})$.

There are two further improvements of the periodogram. The first one is based on the Barlett estimator but substituting the individual periodograms by Blackman-Tukey esti-

mates. The second one is based on dividing the N observations into L overlapping windows. The combination of both improvements is known as the Welch method.

One final question remains: What happens to the bias and variance of these methods. Regarding the bias, it is going to be higher (lower resolution) than that of the periodogram since $M < N$, as also happened to the Blackman-Tukey estimate. As for the variance, it is going to be reduced by a factor of L, the number of windows. That is,

$$Var\left(\hat{S}_x^{ap}(e^{j\omega})\right) \approx \frac{1}{L} Var\left(\hat{S}_x^{p}(e^{j\omega})\right),$$

where $\hat{S}_x^{ap}(e^{j\omega})$ is any averaged periodogram (either Barlett or Welch methods) and $\approx$ is due to the non-independence between the windows. It would be an equality when the windows are independent, i.e., the Barlett method.

5.4 Parametric methods in spectral estimation

The problem of non-parametric methods is that they estimate an infinite number of parameters (the PSD at each frequency) from a sequence of N observations. Clearly, this is an ill-posed problem since there are (many) more parameters to estimate than observations. To overcome this issue, we could postulate a parametric model for the PSD and estimate only the parameters of such model using the N observations. For instance, the model could be $S_x(e^{j\omega}) = a + b\cos^2(\omega)$ and, hence, we only have to estimate a and b.

Parametric approaches, as described above, can provide a significant performance boost if the signal fits the postulated model, otherwise the performance could be even worse than that of non-parametric methods. It is therefore of the utmost importance to select the proper model.

In this chapter we will analyze autorregresive (AR) models, which are particularly amenable for parametric estimation.

5.4.1 Auto-Regressive (AR) models

We say that the stochastic process $x[n]$ follows an auto-regressive model of order p (or, simply, an AR(p) model, if it is the output of a causal, linear and time-invariant filter driven by the recursive relation

$$x[n] = u[n] - \sum_{k=1}^{p} a_k x[n-k], \tag{5.15}$$

when the input, $u[n]$ is an IID Gaussian process with zero mean and variance $\sigma^2 > 0$.

The impulse response of the filter can be represented by means of the recursive relation (5.15), replacing $u[n]$ by $\delta[n]$

$$h[n] = \delta[n] - \sum_{k=1}^{p} a_k h[n-k],$$

Applying the Fourier transform to this equation, we get,

$$H(e^{j\omega}) = 1 - \sum_{k=1}^{p} a_k e^{-j\omega k} H(e^{j\omega}),$$

therefore

$$H(e^{j\omega}) = \frac{1}{1 + \sum_{k=1}^{p} a_k e^{-j\omega k}},$$

5.4.1.1 Power Spectum

Since $x[n]$ is the output of filter $h[n]$ for input $u[n]$,

$$x[n] = u[n] * h[n] \tag{5.16}$$

therefore, the power spectrum is

$$S_x(e^{j\omega}) = S_u(e^{j\omega})|H(e^{j\omega})|^2 = \frac{\sigma^2}{\left|1 + \sum_{k=1}^{p} a_k e^{-j\omega k}\right|^2}. \tag{5.17}$$

According to Weierstrass theorem, for large values of p, the PSD model in (5.17) can approximate arbitrarily close any continuous PSD. Hence, there is a strong interest in this kind of models.

5.4.2 Auto-correlation

Using (5.15), we can express the auto-correlation function of the AR(p) process $x[n]$ as

$$\begin{aligned} r_x[m] &= \mathbb{E}\{x[n]x^*[n-m]\} \\ &= \mathbb{E}\left\{\left(u[n] - \sum_{k=1}^{p} a_k x[n-k]\right) x^*[n-m]\right\} \\ &= \mathbb{E}\{u[n]x^*[n-m]\} - \sum_{k=1}^{p} a_k r_x[m-k] \end{aligned} \tag{5.18}$$

We can develop this expression by analyzing the cases $m > 0$, $m = 0$ and $m < 0$ separately.

- For $m > 0$, since the filter is causal, $x^*[n-m]$ does not depend on $u[n]$, therefore,

$$\mathbb{E}\{u[n]x^*[n-m]\} = \mathbb{E}\{u[n]\}\mathbb{E}\{x^*[n-m]\} = 0, \qquad m > 0 \tag{5.19}$$

therefore, (5.18) simplifies to

$$r_x[m] = -\sum_{k=1}^{p} a_k r_x[m-k], \qquad m > 0$$

- For $m = 0$, we get,

$$\begin{aligned}\mathbb{E}\{u[n]x^*[n-m]\} &= \mathbb{E}\{u[n]x^*[n]\} \\ &= \mathbb{E}\left\{u[n]\left(u^*[n] - \sum_{k=1}^{p} a_k^* x^*[n-k]\right)\right\} \\ &= \mathbb{E}\{u[n]u^*[n]\} = \sigma^2 \end{aligned} \tag{5.20}$$

  therefore

$$r_x[0] = \sigma^2 - \sum_{k=1}^{p} a_k r_x[m-k].$$

- Finally, for $m < 0$, since the autocorrelation function is Hermitian, we have $r_x[m] = r_x^*[-m]$.

  Joining the cases $m > 0$, $m = 0$ and $m < 0$, we get

$$r_x[m] = \begin{cases} -\sum_{k=1}^{p} a_k r_x[m-k] + \sigma^2, & m = 0, \\ -\sum_{k=1}^{p} a_k r_x[m-k], & m > 0, \\ r_x^*[-m], & m < 0. \end{cases} \tag{5.21}$$

We see in (5.21) that the relationship between the model parameters (σ^2, $a_1, \ldots, a_p$) and the auto-correlation is linear. which simplifies the estimation of such parameters. The estimation procedure consists in substituting the theoretical auto-correlation by an estimate and then solving a linear system of equations. The PSD estimate is obtained by substituting the estimated parameters in the corresponding model.

5.4.3 Parameter estimation from the autocorrelation

Since we need to obtain $p+1$ parameters, i.e., $a_1, \ldots, a_p$ and σ^2, we need $p+1$ equations from (5.21), which are known as the Yule-Walker equations:

$$\begin{aligned} r_x[0] &= -a_1 r_x[-1] - a_2 r_x[-2] + \ldots - a_p r_x[-p] + \sigma^2, \\ r_x[1] &= -a_1 r_x[0] - a_2 r_x[-1] + \ldots - a_p r_x[-p+1], \\ &\vdots \\ r_x[p] &= -a_1 r_x[p-1] - a_2 r_x[p-2] + \ldots - a_p r_x[0]. \end{aligned}$$

The last p equations depend only on $a_1, \ldots, a_p$. We can write them in matrix form by using $r_x[-m] = r_x^*[m]$ and defining

$$\mathbf{r}_x = \left[r_x[1]\ r_x[2]\ \cdots\ r_x[p]\right]^T,$$

and

$$\mathbf{R}_x = \begin{bmatrix} r_x[0] & r_x^*[1] & \cdots & r_x^*[p-1] \\ r_x[1] & r_x[0] & \cdots & r_x^*[p-2] \\ \vdots & \vdots & \ddots & \vdots \\ r_x[p-1] & r_x[p-2] & \cdots & r_x[0] \end{bmatrix},$$

to arrive at

$$\begin{bmatrix} \hat{a}_1 \\ \hat{a}_2 \\ \vdots \\ \hat{a}_p \end{bmatrix} = -\mathbf{R}_x^{-1}\mathbf{r}_x,$$

The matrix $\mathbf{R}_x$ has a special structure, namely, it is constant along diagonals, and is therefore known as Toeplitz. This fact is important for solving the system of equations (computing the matrix inverse) as it reduces the complexity from $\mathscr{O}(p^3)$ to $\mathscr{O}(p^2)$. The remaining parameter to be estimated, the variance, is easily obtained as

$$\hat{\sigma}^2 = r_x[0] + \hat{a}_1 r_x^*[1] + \hat{a}_2 r_x^*[2] + \ldots + \hat{a}_p r_x^*[p].$$

Finally, as we have already seen before, in practical scenarios the auto-correlation function is not available and must therefore be replaced by an estimate. The overall procedure consist on three steps:

1. Estimate the autocorrelation function
2. Compute the model parameters solving the Yule-Walker equations
3. Estimate the power spectrum using the parameter estimates.

5.4.4 Maximum Likelihood estimation

We can avoid the estimation of the autocorrelation function by applying estimation theory to compute the model parameters from the signal observations. Assume we have recorded a set of N observations from the process, $\{x[n], 0 \leq n \leq N-1\}$. Our goal is to estimate the model parameters based on these observations. To do so, the following vector notation will be useful:

$$\mathbf{a} = (a_1, a_2, \ldots, a_{p-1})^\top \tag{5.22}$$

$$\mathbf{x} = (x[p], x[p+1], \ldots, x[N-1])^\top \tag{5.23}$$

$$\mathbf{x}_n = (x[n-1], x[n-2], \ldots, x[n-p])^\top \tag{5.24}$$

Note that $\mathbf{x}_n$ contains all values from the signal record that multiply the coefficients to determine $x[n]$, in such a way that the signal model (5.15) can be expressed as

$$x[n] = u[n] - \mathbf{x}_n^\top \mathbf{a}, \tag{5.25}$$

Also, note that vector $\mathbf{x}$ contains all observations from the process starting for $n > p$. We have excluded the observations for $n < p$ because the maximization of the complete likelihood including these values (i.e. $p(\mathbf{x}, \mathbf{x}_p \mid \mathbf{a})$) becomes much more complex. To avoid these complications, we will take $\mathbf{x}_p$ as fixed data, maximizing the likelihood for the rest of observations, that is, we will compute the estimate

$$\hat{\mathbf{a}}_{\mathrm{ML}} = \underset{\mathbf{a}}{\operatorname{argmax}}\, p(\mathbf{x} \mid \mathbf{x}_p, \mathbf{a}) \tag{5.26}$$

To do so, using (5.25), we factorize the likelihood as

$$\begin{aligned} p(\mathbf{x} \mid \mathbf{x}_p, \mathbf{a}) =& p(x[N-1], x[N-2], \ldots, x[p] \mid \mathbf{x}_p, \mathbf{a}) \\ =& p(x[N-1] \mid x[N-2], \ldots, x[p], \mathbf{x}_p, \mathbf{a}) \cdot p(x[N-2] \mid x[N-3], \ldots, x[p], \mathbf{x}_p \mathbf{a}) \\ & \cdot \ldots \cdot p(x[p] \mid \mathbf{x}_p, \mathbf{a}) \\ =& \prod_{n=p}^{N-1} p(x[n] \mid \mathbf{x}_n, \mathbf{a}) \end{aligned} \tag{5.27}$$

Since the input process is IID, zero-mean, Gaussian, we can write

$$p(x[n] \mid \mathbf{x}_n, \mathbf{a}) = \mathcal{N}\left(x[n] - \mathbf{x}_n^\top \mathbf{a}, \sigma^2\right) \tag{5.28}$$

therefore

$$\begin{aligned} p(\mathbf{x} \mid \mathbf{x}_p, \mathbf{a}) =& \prod_{n=p}^{N-1} \mathcal{N}\left(x[n] - \mathbf{x}_n^\top \mathbf{a}, \sigma^2\right) \\ =& \frac{1}{\sigma^{N-p}(2\pi)^{(N-p)/2}} \exp\left(-\frac{1}{2\sigma^2} \sum_{n=p}^{N-1} \left(x[n] - \mathbf{x}_n^\top \mathbf{a}\right)^2\right) \end{aligned} \tag{5.29}$$

Therefore, the ML estimate is the solution of a least squares problem

$$\hat{\mathbf{a}}_{\mathrm{ML}} = \underset{\mathbf{a}}{\operatorname{argmin}} \sum_{n=p}^{N-1} \left(x[n] - \mathbf{x}_n^\top \mathbf{a}\right)^2 \tag{5.30}$$

which can be expressed in matrix form by defining the observation matrix

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}_p^\top \\ \mathbf{x}_{p+1}^\top \\ \vdots \\ \mathbf{x}_{N-1}^\top \end{pmatrix} \tag{5.31}$$

to arrive at

$$\hat{\mathbf{a}}_{\mathrm{ML}} = \underset{\mathbf{a}}{\operatorname{argmin}} \|\mathbf{x} - \mathbf{X}\mathbf{a}\|^2 = \left(\mathbf{X}\mathbf{X}^\top\right)^{-1} \mathbf{X}^\top \mathbf{x} \tag{5.32}$$

The same approach can be applied to the estimation of the variance, σ^2:

$$\hat{\sigma}^2_{\mathrm{ML}} = \underset{\sigma}{\operatorname{argmax}} \frac{1}{\sigma^{N-p}(2\pi)^{(N-p)/2}} \exp\left(-\frac{1}{2\sigma^2}\sum_{n=p}^{N-1}\left(x[n]-\mathbf{x}_n^\top \mathbf{a}\right)^2\right)$$
$$= \frac{1}{N-p}\sum_{n=p}^{N-1}\left(x[n]-\mathbf{x}_n^\top \mathbf{a}\right)^2 \tag{5.33}$$

5.4.5 Signal prediction

The estimation of the parameters and an AR process can be done by simply solving a system of equations. Moreover, from the expression of an AR model

$$x[n] = u[n] - \sum_{k=1}^{p} a_k x[n-k],$$

we note that they can be used to predict future samples by ignoring the input, i.e.,

$$x[n] = -\sum_{k=1}^{p} \hat{a}_k x[n-k],$$

where the coefficients of the model have been replaced by some estimates. That is, from a record of N samples, $x[0],\ldots,x[N-1]$, we can estimate the model parameters and, afterwards, we can predict $x[N], x[N+1], \ldots$.